



ADLM STUDIO · EVERY PRODUCT

# The complete ADLM user guide

Every ADLM product in one book: getting started, the Installer Hub, QUIV, HERON, Rate Gen, SERVIQ for Revit MEP, Time Pro, the ADLM Cloud, the sample projects and the QS handbook.

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[adlmstudio.net/guides](https://adlmstudio.net/guides)

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# Getting started with ADLM Studio

Create your account, buy a licence, install your software and find your way around your account, billing, seats and support.

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This guide is for everyone who uses ADLM Studio software, whether you have just heard of us or you are renewing for the third year. It walks you through creating your account, paying for a product, installing it, and looking after your licence, devices, invoices and support requests on [adlmstudio.net](https://adlmstudio.net). At the end there is a table of every ADLM product with a link to its own guide.

## Before you start

You need:

- An email address you can open now. We send a six-digit code to it before your account can be used.
- A WhatsApp number, with the country code (for example +2348012345678). It is required when you sign up.
- A web browser on a phone, tablet or computer. Everything in this guide works in any modern browser.
- For the desktop products (QUIV, HERON, RateGen, Revit MEP, CIVIQ and Time Pro), a Windows 64-bit computer. These are installed with the [ADLM Installer Hub](#).

One ADLM account signs you in to every product, on every machine. The same email and password work on the website, in the Installer Hub and inside Revit, PlanSwift and the other host programs.

### TIP

Keep this guide open on your phone while you work on your computer. The steps are short so you can follow them side by side.

## Create your account

### Sign up with your email

1. Open [adlmstudio.net](https://adlmstudio.net) and click **Sign up** in the top bar. On a phone, open the menu first.
2. Enter your **First name** and **Last name**.
3. Enter your **WhatsApp number** (e.g. +2348012345678).
4. Enter your **Email**.
5. Enter a **Password**. It must be at least 8 characters and contain at least one letter and one number.
6. Click **Sign up**.

You are taken straight to the **Confirm your email** screen.

### NOTE

If the site says the user already exists, an account with that email (or the same name before the @) is already registered. Click **Login** instead and use **Forgot password?** if you need to.

## Confirm your email address

Your account cannot be used, and you cannot buy anything, until your email address is confirmed.

1. Open the email from ADLM Studio. It contains a six-digit code. It can take a minute to arrive, so check your spam or promotions folder too.
2. Type the six digits into the box on the **Confirm your email** screen.
3. Click **Confirm**.

The code works for 30 minutes. On the same screen you can:

- Click **Send a new code** if it has expired or never arrived.
- Click **Wrong address? Change it** if you mistyped your email, enter **The right email address**, then click **Send the code there**.
- Click **Sign out** to leave and come back later.

Until you confirm, a yellow bar across the top of every page reads "Confirm your email address to use your account." Click **Enter the code** in that bar to come back to this screen.

### IMPORTANT

The desktop products also refuse to sign in until your email is confirmed. They tell you to sign in at [adlmstudio.net](https://adlmstudio.net), enter the code, and then try again.

## Sign up with Google, Microsoft or Autodesk

Under the sign-up form you may see buttons such as **Continue with Google**, **Continue with Microsoft** or **Continue with Autodesk**. Only the ones we have switched on are shown. Clicking one creates your ADLM account from that provider's account (or signs you in, if you already have one with the same email).

### IMPORTANT

Google, Microsoft and Autodesk sign-in work on the website only. The Windows software always asks for an email and a password. After a social sign-in you are taken to your Profile and asked to **Set a password for the desktop software**. Do it straight away, otherwise you cannot open QUIV, HERON, RateGen, the Revit MEP plugin, Time Pro or CIVIQ. Your social sign-in keeps working on the website.

## Sign in

1. Click **Login** in the top bar.
2. Enter your **Email or username** and your **Password**. Click the eye icon to show or hide what you typed.
3. Click **Sign in**.

If you connected Google, Microsoft or Autodesk, you can click the matching **Continue with** button instead.

## Forgot your password

1. On the sign-in page, click **Forgot password?**.
2. Enter your **Email or username** and click **Send code**.
3. Check your email for a six-digit code. It works for 10 minutes.
4. Type the code into **Enter 6-digit code**.

5. Type a **New password** (at least 8 characters, with a letter and a number).
6. Click **Set new password**.
7. Sign in with the new password.

If the code does not arrive, click **Didn't get a code? Resend**.

## The codes we send you

ADLM Studio uses short six-digit codes in a few places. Each one is sent to you only, and each has a short life:

| Code                    | When you get it                                                                    | Valid for        |
|-------------------------|------------------------------------------------------------------------------------|------------------|
| Email confirmation      | After you sign up, or when you change the address                                  | 30 minutes       |
| Password reset          | After you click <b>Forgot password?</b>                                            | 10 minutes       |
| Destructive-action code | Before deleting a project or locking a contract, if you turned this on (see below) | About 10 minutes |

### TIP

Never share a code with anyone. ADLM support does not need your codes or your password to help you.

## Extra protection for deleting projects

On your Profile, under **Security & Admin**, you can tick **Require an email code for destructive actions**. When it is on, we email you a six-digit code before you can delete a project or lock or unlock a contract. Untick it to turn it off.

## Your profile and WhatsApp number

There are two places to edit your details. Both save to the same account.

### On the Profile page

Click **Profile** in the top bar. Under **Account details** you can:

1. Click **Upload new photo** to set your picture.
2. Edit your **Username**, **First Name** and **Last Name**.
3. Enter your **Firm name** and **Location**.
4. Choose your **Pricing location (State)**. Your RateGen prices, and the prices in projects you price afterwards, use the zone your state belongs to. Click **View Prices** to see them.
5. Click **Save**.

### NOTE

If your name has been printed on a course certificate, it is locked. The page says so. Contact support if it needs correcting.

## In Account settings

Open your account area at [/manage](#) and click **Account settings** in the left-hand menu (see "The Manage area" below). The **Your profile** panel has **First name**, **Last name**, **Phone** and an optional **Display name**. Click **Save profile** when you are done.

- **Work email** cannot be changed here. The email is the account itself. To change it, contact support.
- **Phone** is your WhatsApp number. Enter it with the country code, for example +234 803 000 0000.
- **Rates and currency** lets you pick **Where you work** (your state). The **Geopolitical zone** fills itself in. **Currency** switches how prices are shown to you between NGN and USD. Click **Save location**.
- **The firm** holds your **Practice name** and **Address**. These are printed on invoices raised from now on. Click **Save firm details**.

## Verifying your WhatsApp number

If you see a **Verify on WhatsApp** button under **Phone** in Account settings:

1. Make sure the number in **Phone** is right, and click **Save profile** first. A changed number must be saved before it can be verified.
2. Click **Verify on WhatsApp**.
3. Open WhatsApp on your phone. A six-digit code arrives from ADLM Studio.
4. Type it into the **6-digit code** box and click **Confirm**.

When it works, the field reads **Verified on WhatsApp**. Click **Send again** if the code does not arrive. Changing your number later removes the verified mark until you verify the new one.

### NOTE

WhatsApp verification is being switched on gradually. If you do not see the button, there is nothing you need to do.

## Buying a product or subscription

You can buy from any product page or from the checkout page.

### From a product page

1. Click **Products** in the top bar and open a product.
2. Under **Choose your term**, pick how long you want. Longer terms show the percentage you save.
3. Click **Get started now**. You are taken to checkout with the product already chosen.

If a product shows **Notify me when available** instead, it is not on sale yet. Click it to be told when it is.

### At checkout

Click **Purchase** in the top bar to open the **Subscribe** page. You must be signed in, with your email confirmed.

1. At the top, choose your **Currency**: NGN or USD.
2. Choose **Purchase for: Personal** (one seat, one computer) or **Organization** (several seats for a firm).
3. In the **Products** list on the left, click a product.

4. Set the **Duration**. For monthly products, tap **1 mo**, **6 mo** or **12 mo**, or use the minus and plus buttons. Six and 12 months unlock better pricing.
5. If you chose **Organization**, set **Seats / users**. Some products have a minimum number of users for organisation licences; the page tells you.
6. If the product offers it, tick **First-time install** to add our one-time installation fee.
7. If you pay in NGN, you can add **Extra project storage** in blocks of 10 project slots.
8. Check the total at the bottom of the box and click **Add to order**.
9. Repeat for any other products. A green dot in the list marks products already in your order. To change one, click it, adjust it and click **Update order**. To take it out, click **Remove**.

Your order builds up in the **Summary** on the right.

## Buying for an organisation

When **Purchase for** is set to **Organization**, an **Organization details** box appears:

1. Enter the **Organization name**. It is required.
2. Optionally add an **Organization email** and **Phone**.
3. Prices are shown per seat. The seats you choose on each product are the number of users or computers your organisation needs.

In the same box you can tick **Need physical training for your office?**. Choose a **Training location**, and the location, duration and **Training cost** are shown. You can also tick **Also install BIM Softwares on office computers?**. The training date is agreed with you after your purchase is approved.

## Bundles and multi-month discounts

- Most products are cheaper per month when you buy 6 or 12 months at once. The saving is shown on the product page and is applied automatically at checkout.
- You can put several products in one order and pay for them together. Each appears as its own line in the **Summary**.

## Coupons

1. In the **Summary**, find **Discount coupon**.
2. Type your code in **Enter coupon code**.
3. Click **Apply**.

When the code is accepted, the summary shows **Applied:** with the code and the discount, and the **Discount** line is taken off your **Subtotal**. Any VAT is then added and the final **Total** shown. Add products to your order before applying a coupon.

### TIP

When a promotion is running, a coupon banner appears across the top of the public pages with the code.

## Auto-renewal

If you are paying in NGN, the **Summary** shows **Auto-renew my subscription**. Tick it before you pay. It only applies when you pay by card: we save the card and charge it at the price at that time shortly before your licence expires, so your access carries on without a new order. If a renewal is declined, you get an email and it is tried again before your access ends.

You can switch auto-renew on or off for each product, or remove the saved card, at any time (see "Saved card and auto-renewal").

## Paying

Click **Pay**. The **Complete your payment** box opens.

### Paying by card (NGN orders)

1. Click **Pay with Card (Paystack)**.
2. Enter your card details in the Paystack window. Your bank may ask for an OTP.
3. When the payment succeeds you see **Payment successful! Your access has been activated.** and you are taken to your account area.

Your licence is active as soon as the card payment is confirmed. If you were debited but the page could not confirm it, wait a few minutes: the payment is confirmed automatically, and your dashboard updates.

#### NOTE

Card payments are charged in Naira. Cards issued in Nigeria clear reliably. If you are paying from outside Nigeria, the site may suggest bank transfer instead. If you do hold a Nigerian card, click **I have a Nigerian card. Let me pay by card anyway**. On a USD order, click **Switch to NGN & pay by card** to pay by card.

### Paying by bank transfer

1. In the **Complete your payment** box, note the **Account number**, **Account name** and **Bank**.
2. Make the transfer from your bank app.
3. Click **I have paid**.

A bank transfer is checked by our team before your licence is switched on. Your order shows **Awaiting admin approval** until then. If we sent you an order link by email, open it and click **Upload your receipt** to attach a photo or PDF of your transfer slip (up to 8MB). That speeds up approval.

## Quotations

If you need a figure for your firm's approval before you buy, build a quotation. You do not need to be signed in.

1. Open [/quote](#) , or on a phone choose **Get Quotation** from the menu.
2. Choose the **Currency** and **Billing** (Yearly or Monthly).
3. Under **Select Software Products**, click each product you need.
4. For each selected product, set **Number of PCs / Users**.
5. If you want training, tick **I need physical training at my office** and choose a location.
6. Read **Your Quotation** and the **Estimated Total**.
7. Click **Print Quote** to print or save it as a PDF, or click **Email Quote**, enter **Your name** and **Email address**, and click **Send**.
8. When you are ready, click **Proceed to Purchase**.

#### NOTE

A quotation is an estimate. The final price is the one shown at checkout.

## Invoices and receipts

- **Orders.** Every purchase is listed on your Profile under **Orders, Invoices & Installations**, in the **Orders** tab, with its status: **Paid**, **Approved**, **Rejected** or **Awaiting admin approval**. Once an order is approved, click **Print / Download Receipt** to open the receipt, then print it or save it as a PDF.
- **Invoices.** Invoices raised for you by ADLM Studio are in the **Invoices** tab. Click **View Invoice** to open one, or **Download PDF** to save it. When an invoice is paid, **Download Receipt** appears beside it.
- **In the Manage area.** **Billing & invoices** lists the same invoices with **PDF** and, for paid ones, **Receipt** links.

### TIP

If ADLM sent an invoice to your email before you had an account, sign up with that same email. The invoice is linked to your new account automatically.

## Installing your software

Once you have an active subscription, your dashboard shows a blue **Installer Hub** banner.

1. Click **Download Installer Hub**.
2. Run the downloaded file on your Windows computer.
3. Sign in to the Hub with your ADLM email and password.
4. Install the products your licence covers.

**Download User Guide** and **Watch Setup Guide** sit beside the download button. The full walkthrough is in the [ADLM Installer Hub guide](#). You can also download the Hub from **Downloads** in the Manage area.

### IMPORTANT

Close Revit and PlanSwift before installing a plugin. If the host program is open during the install, the ADLM panel does not appear until you close it and start it again.

If you bought a **First-time install**, the **Installations** tab on your Profile shows **Installation pending** with **Next steps**. You can install it yourself with the Installer Hub, or download AnyDesk and send your AnyDesk address to support so we can install it for you remotely. The status changes to **Installation complete** when it is done.

## Your dashboard

Click **Dashboard** in the top bar. The top of the page shows four counters: **Active Products**, **Active Subscriptions**, **Tutorials Watched** and **Total Orders**.

Below them are three tabs:

- **My Products** lists the products on your account. Click **View** to open a product's page.
- **Subscriptions** shows each licence with its status (**active**, **expiring soon** in the last 7 days, or **expired**), whether it is a personal or organisation licence, the billing interval and the **Expires** date. Click **Open** to go to that product's projects on the website. Click **Manage** to go to checkout with the product ready to renew or extend.
- **Learning** shows the **User guides** for the products you own (illustrated PDFs), your **Online Courses** with progress, and your **Physical Trainings**.

**Orders & Billing** opens your Profile, where orders, invoices and installations live.

On the right, **Quick Links** take you to your Revit projects, the **Learning Center**, **User Guides** and **Contact Support**.

#### NOTE

If we ever need everyone to reinstall, a red **Action required: reinstall** box appears on the dashboard with links to download the Hub and watch the setup video. Follow it before opening your products again.

## The Manage area

ADLM Studio also has an account area with a menu down the left-hand side. You arrive there after paying by card, from your receipts and invoices, or by opening `/manage`. On a phone, the menu folds away behind the menu button.

The menu has three groups:

- **Work: Overview, My tools** (QUIV, HERON, RateGen, Revit MEP, CIVIQ) and **Projects**. These are your project workspaces, covered in the [ADLM Cloud guide](#). Tools you do not own are greyed out with an option to add them.
- **Learn: My learning, Assignments, Certificates, Lessons & events** and **Guides & docs**.
- **Manage: Overview, Products & seats, Team, Billing & invoices, Downloads** and **Support**.

At the bottom are **Account settings** and **Sign out**.

### Overview

The Manage **Overview** greets you by name and shows **Products active**, **Seats in use**, **Next charge** and **Courses running**. Below that:

- **Your products** lists each licence. Click **All products** for the full list.
- **Needs your attention** lists anything you should act on, such as an expiring licence, with a link to fix it. When all is well it says "Nothing needs you. Everything is active and installed."
- **ADLM Installer Hub** has the download button.
- **Recent activity** and **Your learning** show what has happened lately.

### Products & seats

The **Products** page lists everything ADLM makes. Use the filter buttons at the top to narrow the list.

- For products you own: click **Download** to go to your installers, or **Seats** (or **Assign the spare seat**) to see who holds them.
- For products you do not own: click **What it does** to read about it, or **Add to subscription** to buy it.
- **Talk to us about seats** opens a support ticket.

### Team

See "Organisation seats and team members" below.

## Billing & invoices

- **Next charge** shows each product, its seats and what your next renewal comes to, with any VAT and the **Total due**. Switch between **Monthly** and **Yearly** to compare. Use the minus and plus buttons to see what a different number of seats would cost, then click **Buy the change** to go to checkout, or **Reset** to undo. Nothing changes until you pay at checkout.
- **Yearly costs less** appears when paying yearly would save you money. Click **Switch to yearly** (or **Switch back to monthly**). It takes effect at your next renewal.
- **Invoices** lists every invoice with **PDF** and **Receipt** links.
- **Payment method** shows your saved card, if any.
- **Billing details** shows who invoices are addressed to. Click **Edit details** to change them in Account settings. Past invoices are not reissued.
- **Cancelling** explains that a licence runs to the end of the period you paid for. Your rate library and project data stay in your account whether or not a licence is active. Click **Talk to us first** if you want to cancel.

## Downloads

- **Download the Installer Hub** gets the Hub. It installs only what your licences cover.
- **Your products** lists each licensed product. Click **Installer** to download a single product's package for a machine that cannot run the Hub.
- **Not in the subscription** lists products you could add.
- **Before you install** has short answers to the most common installation problems.

## Guides & docs

The illustrated PDF user guides. For each guide you can preview it or download it. **Still stuck?** lets you ask Ada or open a support ticket.

## Account settings

Your profile, rates and currency, firm details and password are covered above and below. This page also has:

- **Machines signed in:** every computer that has activated a licence, with the products on it and when it was last used. See "Devices and licence limits".
- **Connected accounts:** connect or **Disconnect** Google, Microsoft or Autodesk sign-in.
- **Appearance:** choose a theme for this device. It saves straight away.
- **Notifications:** switch on or off the emails you get. See "What's New and release emails".
- **Your data:** **Activity log** opens your project activity on the Profile page, and **Ask for an export** opens a support ticket.

## Saved card and auto-renewal

When you pay by card with auto-renew ticked, the card is saved for renewals. To manage it, open your Profile and find **Billing & auto-renewal**.

- The card shows as its type and last four digits, with the expiry date.
- Each subscription has its own **Auto-renew** tick box. Untick it to stop that product renewing. The line shows how many months each renewal buys.
- If a renewal failed, the line says **Last renewal attempt failed** with the reason.

- To stop all renewals, click **Remove card**, then **Yes, remove**. Auto-renew is switched off for every product.

If you have no saved card, pay any order by card and it is saved automatically for renewals. To use a different card, pay your next renewal with the new one.

## Organisation seats and team members

### How seats work

- A **seat** is one licence for one computer. A seat activates against a machine, not a person.
- A **Personal** licence has one seat.
- An **Organization** licence has as many seats as you bought for each product.

To see your seats, open **Team** in the Manage area (the page is called **Team & seats**). It shows:

- **Seats per product**: how many seats each product has and how many are in use.
- **Machines**: each computer holding a seat, the products on it and when it was last used, with a **Free the seat** button.
- **Free activations**: **Seats owned**, **In use** and **Free right now**.

To add seats, click **Buy more seats** (or **Buy more**). You are taken to checkout. Installing on a new machine costs nothing extra while you have a free seat.

### Adding team members

Adding colleagues to one shared account is not available yet. Today, each person signs in with their own ADLM account, and a licence is bought against that account. If you need several people under one bill, click **Talk to us about a practice account** on the Team page and we set it up for you by hand.

#### TIP

To work with a colleague on a single project, you do not need to share an account. Projects can be shared with a link; see the [ADLM Cloud guide](#).

## Devices and licence limits

Each seat can be active on one computer at a time. If you sign in to a product on a computer when all your seats are already in use, the product refuses and tells you which computer holds the seat. The message reads like this:

- "This QUIV licence is already in use on another computer..." (one seat), or
- "All 3 seats on this QUIV licence are in use..." (several seats)

followed by the name of the computer and when it was last used.

### Moving a licence to a new computer

You can do this yourself, without a ticket:

1. On any device, sign in at [adlmstudio.net](https://adlmstudio.net) and open **Team** in the Manage area (or **Account settings**, under **Machines signed in**).

2. Find the old computer in the list.
3. Click **Free the seat** (on the Team page) or **Sign out** (in Account settings).
4. Confirm. The seat is freed at once. Nothing on the old computer is deleted; it simply stops being licensed at its next check-in.
5. Sign in to the product on the new computer.

#### IMPORTANT

Free the seat first, then install on the new machine. Reinstalling without freeing the seat uses up an activation that support then has to release for you.

## Things that count, and things that do not

- A rebuilt, reformatted or reimaged computer counts as a new device. Free the old entry first.
- Switching between Wi-Fi, ethernet or a VPN does **not** use up a seat. Activation is tied to the device, not the network.
- Changing your password signs your other machines out at their next check-in, so they will ask for the new password.

## Changing your password

You can change your password in either of these places:

### On your Profile

1. Click **Profile** in the top bar.
2. Scroll to **Change your password**.
3. Enter your **Current password**.
4. Enter a **New password** (at least 8 characters, with a letter and a number).
5. Enter it again in **Confirm password**.
6. Click **Change password**.

### In Account settings

1. Open **Account settings** in the Manage area.
2. In the **Password** panel, enter your **Current password** and **New password**.
3. Click **Change password**.

The new password is the one the Windows products sign in with. Your other machines will ask for it at their next check-in.

If you signed up with Google or Microsoft and have never set a password, the Profile card is called **Set a password for the desktop software** instead, and only asks for **Password** and **Confirm password**.

#### TIP

If you have forgotten your current password, sign out and use **Forgot password?** on the sign-in page.

## What's New and release emails

### What's New

Click **What's New** in the top bar to see what has changed in every ADLM product. Each product has a card; click **View updates** to read its release notes, newest first. Each product's guide on this site also has a "What's new" section.

### Emails we send

- **Account email:** receipts, licence activations, renewal notices and support replies. These are part of your account and always arrive.
- **Product update emails:** when a new version ships for a product you are licensed for.
- **New video emails:** when ADLM Studio publishes a video on YouTube.
- **Marketing email:** news and offers.

### Choosing what you get

Open **Account settings** in the Manage area and use the switches under **Notifications**:

| Switch             | What it covers                                                |
|--------------------|---------------------------------------------------------------|
| Product updates    | When a build ships for something you are licensed for         |
| Billing            | Invoices, renewals and anything that changes what you pay     |
| Seats and machines | When a seat is used, freed, or a machine signs in             |
| Courses and events | New courses, training dates and free lessons (off by default) |
| New videos         | An email when ADLM Studio publishes a video on YouTube        |

### Unsubscribing from an email

Every marketing, product-update and new-video email has an unsubscribe link at the bottom. You do not need to sign in.

1. Click the unsubscribe link in the email.
2. A page opens asking you to confirm.
3. Click the button: **Stop marketing email**, **Stop product update emails** or the matching button for video emails.
4. The page says **Done**.

#### NOTE

Unsubscribing does not stop receipts, licence activations, renewal notices or support replies.

# Getting support

## Ask Ada first

The **Ask Ada** button sits in the corner of every page. Ada is ADLM's assistant: she answers questions about products, prices and training from ADLM's own information, and once you are signed in she can also answer about your own subscriptions and projects. She can hand you to a person on WhatsApp whenever you prefer.

## Open a support ticket

There are two ways to raise a ticket. Both need you to be signed in.

### From the Support page

1. Go to `/support` , or click **Contact Support** under **Quick Links** on your dashboard.
2. Click **Raise a Technical Ticket**. The **Request technical help** page opens.
3. Enter a **Title**, for example "Can't open RateGen after update".
4. Describe the problem in **What's happening?**, including any error message.
5. Choose a **Category**: **Technical issue**, **Account / login**, **Billing / subscription**, **Feature request** or **General question**.
6. Choose **Which product / software?**.
7. Optionally click **Add images** to attach up to 5 screenshots, 2MB each.
8. If you would like us to fix it on your computer, install AnyDesk and paste your **AnyDesk address**.
9. Click **Submit request**.

Your tickets and their status are listed under **Your requests** on the same page. We reply by email.

### From the Manage area

1. Open **Support** in the Manage area and click **Open a ticket**.
2. Choose **Which product** and, if it is about one computer, **Which machine**.
3. Enter a **Summary** and fill in **What happens, and what you expected**.
4. Optionally add your **AnyDesk address**.
5. Click **Send to support**.

A ticket opened here already carries your licences and machines, so we do not need to ask for them. Your tickets appear under **Your tickets**, with any reply from support.

#### TIP

The two things that shorten a ticket most are the exact wording of any message you saw, and what you had just done when it appeared. A screenshot helps.

## WhatsApp

For anything that stops you working, WhatsApp is the quickest way to reach us: **+234 810 650 3524**. Click **WhatsApp Live Support** on the Support page, or **WhatsApp** in the Manage area's Support page, to open a chat.

| Contact  | Details                |
|----------|------------------------|
| Email    | admin@adlmstudio.net   |
| WhatsApp | +234 810 650 3524      |
| Hours    | Mon to Fri, 9 to 6 WAT |
| Where    | Lagos, Nigeria         |

## Network check

If the website keeps saying "Failed to fetch" or will not load your account, open `/network-check`. It sends a few tiny test requests and shows which ones got through. Nothing on your account changes.

1. Sign in first, so the checks that use your sign-in are included.
2. Open `/network-check` and wait a few seconds.
3. Click **Copy results**, or **Send to ADLM on WhatsApp**, and quote the **Reference code** shown at the top.

## Learning, courses and trainings

### Free lessons and paid courses

Click **Learn** in the top bar. The page has two parts:

- **Free Lessons:** the ADLM Studio YouTube channel, shelved by software. Use the buttons at the top to filter by software, and click a title to watch.
- **Paid Courses:** in-depth courses with certificates. Open a course to see what it covers and how to enrol.

Courses you are enrolled in appear on your dashboard's **Learning** tab and under **My learning** in the Manage area, with your progress, modules and assignments. When you finish a course, **Download Certificate** appears. Your certificates are also under **Certificates** in the Manage area.

#### NOTE

Your name is locked once it has been printed on a certificate, so check your first and last name on your Profile before you finish a course.

### Trainings and events

Click **Trainings** in the top bar to see ADLM's online sessions, office trainings and conferences. Firms can also add physical training at their office when buying an organisation licence (see "Buying for an organisation"). Approved physical trainings appear on your dashboard under **Physical Trainings**, with **View location** and **Download Calendar (.ics)** buttons.

## ADLM products and their guides

| Product            | What it does                                                           | Where it runs     | Guide                               |
|--------------------|------------------------------------------------------------------------|-------------------|-------------------------------------|
| ADLM Installer Hub | Installs, updates and licenses every ADLM desktop product from one app | Windows 10 and 11 | <a href="#">Installer Hub guide</a> |

| Product        | What it does                                                       | Where it runs              | Guide                            |
|----------------|--------------------------------------------------------------------|----------------------------|----------------------------------|
| QUIV for Revit | Quantity takeoff and estimating inside Autodesk Revit              | Revit add-in (Windows)     | <a href="#">QUIV guide</a>       |
| HERON          | Takeoff-to-budget estimating for PlanSwift                         | PlanSwift add-in (Windows) | <a href="#">HERON guide</a>      |
| RateGen        | Rate build-ups and market pricing for quantity surveyors           | Windows and the website    | <a href="#">RateGen guide</a>    |
| ADLM MEP       | MEP quantity takeoff for Autodesk Revit                            | Revit add-in (Windows)     | <a href="#">MEP guide</a>        |
| ADLM Time Pro  | Site productivity tracking and programme durations                 | Windows 10 and 11          | <a href="#">Time Pro guide</a>   |
| ADLM Cloud     | Your projects, budgets, valuations, reports and billing on the web | Any browser                | <a href="#">ADLM Cloud guide</a> |

## What's new in October 2026

Recent changes to your account and the website:

- **Card payments and auto-renewal.** Pay by card for instant activation, and let subscriptions renew themselves. Manage the saved card and auto-renew per product from your Profile.
- **Downloadable receipts.** Every paid invoice has a receipt PDF.
- **Email confirmation.** New accounts confirm their email with a six-digit code before they are used, and you can correct a mistyped address on the same screen.
- **Free a seat yourself.** Move a licence to a new computer from **Team** or **Account settings** without raising a ticket.
- **Set a password after social sign-in.** Accounts created with Google or Microsoft are asked to set a password for the desktop software.
- **Ada.** The Ask Ada assistant replaces the old help bot.
- **The Manage area.** A new account area with a left-hand menu for products, seats, billing, downloads, support and settings.
- **Upload your transfer receipt.** Attach a photo or PDF of a bank transfer to your order so it is approved faster.
- **Network check.** A one-click test at `/network-check` for "Failed to fetch" problems.

## Troubleshooting

### I never received my confirmation code

Wait a minute and check your spam or promotions folder. On the **Confirm your email** screen, click **Send a new code**. If the address is wrong, click **Wrong address? Change it**. Each code works for 30 minutes.

## The desktop software says my email is not confirmed

Sign in at [adlmstudio.net](https://adlmstudio.net), enter the six-digit code on the **Confirm your email** screen (click **Send a new code** if needed), then sign in to the product again.

## I signed up with Google and the software will not accept my sign-in

The Windows products need an email and a password. Open your Profile and use **Set a password for the desktop software**, then sign in to the product with your email and that password.

## My card was debited but my licence is not active

Card payments are usually confirmed within moments. If the page could not confirm it, the payment is confirmed automatically shortly afterwards. Refresh your dashboard after a few minutes. If it is still not active, open a ticket with the **Billing / subscription** category or message us on WhatsApp.

## My card was declined

Cards issued outside Nigeria are often declined because the charge is in Naira. Pay by bank transfer instead, then click **I have paid** and upload your receipt.

## My order says Awaiting admin approval

Bank transfers are checked by our team before the licence is switched on. Make sure you clicked **I have paid**, and upload your transfer receipt on the order page if you have the link. Contact support on WhatsApp if it has been more than a working day.

## My coupon was not accepted

Add at least one product to your order first, then apply the code. Check the spelling. Coupons can expire or apply only to certain products; the message under the box says why.

## A product says the licence is in use on another computer

All your seats are taken. Free the old computer's seat from **Team** or **Account settings** in the Manage area, then sign in again. If you need more computers at once, buy more seats.

## I replaced my computer and cannot activate

A new or rebuilt computer counts as a new device. Free the old one under **Team** in the Manage area, then sign in on the new one.

## The ADLM panel is missing in Revit or PlanSwift after installing

Revit or PlanSwift was open during the install. Close it and start it again.

## The website says "Failed to fetch"

Run the network check at [/network-check](#) and send us the results with the reference code. It is usually something on your office network blocking part of the website.

## I cannot change my email address

The email is your account. Contact support to change it.

## Frequently asked questions

### Do I need a separate account for each product?

No. One ADLM account signs in to every product, on every machine.

### Can I use one licence on two computers?

Not at the same time. A personal licence has one seat, which is one computer. To use two computers at once, buy an organisation licence with two seats. To move between computers, free the seat on the old one first.

### Can I pay in US dollars?

You can view prices and place an order in USD. Card payments are charged in Naira, so to pay by card switch the order to NGN; otherwise pay by bank transfer.

### What happens when my subscription ends?

The licence stops working at the end of the period you paid for. Your rate library, projects and takeoffs stay in your account, and are there again when you renew. Click **Manage** on your subscription to renew.

### How do I renew?

Turn on auto-renew and pay by card, and it renews itself. Otherwise, click **Manage** on the subscription on your dashboard, or buy the product again at checkout.

### Where are my receipts?

On your Profile under **Orders, Invoices & Installations**, and under **Billing & invoices** in the Manage area.

### Can I add my colleagues to my account?

Not yet. Each person uses their own ADLM account. For several people under one bill, ask us about a practice account from the **Team** page.

### Will unsubscribing stop my renewal notices?

No. Receipts, licence activations, renewal notices and support replies always arrive.

### Who do I contact if I am stuck?

Ask Ada, open a ticket from the Support page, or message us on WhatsApp on +234 810 650 3524, Monday to Friday, 9 to 6 WAT.

# ADLM Installer Hub

Sign in once, then install, update and repair every ADLM product your licence covers.

The ADLM Installer Hub is the Windows app that puts your ADLM software on your computer. You sign in with your ADLM account, and the Hub shows the products your subscription covers, installs them in the right folders, tells you when an update is ready and repairs a broken install. You never copy files or hunt for folders yourself. This guide covers Installer Hub 2.0, the version you download from the ADLM website today.

## Before you start

### What you need

| You need                                                | Why                                                                                                                                                              |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A Windows 10 or Windows 11 PC, 64-bit                   | The Hub does not run on Mac, older Windows or 32-bit Windows.                                                                                                    |
| Administrator rights on that PC                         | The Hub writes plugins into shared program folders, so Windows asks for administrator approval.                                                                  |
| Your ADLM account                                       | The same email (or username) and password you use on <a href="https://adlmstudio.net">adlmstudio.net</a> . There is no separate Hub account.                     |
| An active licence for at least one ADLM desktop product | The Hub download comes with a paid licence. It installs only what your account has bought.                                                                       |
| An internet connection                                  | Needed to sign in, load your products and download them.                                                                                                         |
| The host program, for plugins                           | Revit for QUIV for Revit and the Revit MEP plugin, PlanSwift for ADLM Heron, Civil 3D for CIVIQ, Archicad for QUIV for ArchiCAD. Install the host program first. |

#### NOTE

If your company bought the licences, sign in with the account that made the purchase. Licences belong to the account that bought them, not to whoever uses the computer.

### Where to download the Hub

1. On the computer you want to install on, sign in at [adlmstudio.net](https://adlmstudio.net).
2. Open your [dashboard](#).
3. Click **Download Installer Hub**. (On the newer Downloads page the button reads **Download the Installer Hub**.)
4. Save the setup file. It is a normal Windows installer (.exe).

#### NOTE

The download button only appears when your account has an active licence for a desktop product. If you do not see it, check your subscription on the website, or [contact support](#).

## How many computers

Each licence seat is tied to one computer, the first one you install the product on. You can install, update and repair as often as you like on that computer. To use a seat on a different computer, you first free it from the old one. See [Licences and your computers](#).

## Install the Hub on your PC

You do this once per computer.

1. Double-click the setup file you downloaded.
2. If Windows shows a blue **Windows protected your PC** screen, click **More info**, then **Run anyway**. This appears because the setup file is new to Windows. It is safe to continue if you downloaded it from adlmstudio.net.
3. Windows asks "Do you want to allow this app to make changes to your device?". Click **Yes**.
4. If you want a desktop icon, tick the desktop shortcut box. It is off by default.
5. Click **Install**. Setup chooses the install folder for you.
6. On the last page, leave the launch box ticked and click **Finish** to open the Hub.

The Hub appears in the Start menu as **ADLM Installer Hub**.

### IMPORTANT

If the **Yes** button on the Windows prompt is greyed out, or Windows asks for an administrator password, your Windows account is a standard user. Ask whoever manages the PC to approve it, or to install the Hub for you. The Hub cannot install plugins into program folders without administrator rights.

### TIP

Every time the Hub opens, Windows may ask for administrator approval again. That is expected. Click **Yes**.

## Updating from an older Hub

If you already have Installer Hub 1.x, download the 2.0 setup from the website and run it. It installs over the old version. If the Hub is open, setup offers to close it first. The list of products you have installed on this PC is kept.

The Hub does not update itself. When ADLM releases a new Hub, download the new setup from the website the same way.

## Sign in

The first screen is **Secure Sign In**.

1. In **Email or username**, type your ADLM email or username.
2. In **Password**, type your ADLM website password.
3. Click **Sign in to dashboard**, or press .

The button stays greyed out until both boxes have text in them. After a few seconds the Hub loads your products, checks what is already installed on this PC, compares versions and opens the dashboard.

#### NOTE

In version 2.0 the Hub does not keep you signed in. You sign in each time you open it. Closing the window signs you out.

## If sign-in does not work

Any problem shows in red above the sign-in button.

| What you see                                                                       | What it means and what to do                                                                                                 |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Invalid email, username, or password.                                              | The details do not match. Passwords are case-sensitive. Reset your password on the website, then try again with the new one. |
| We couldn't reach the server. Please check your internet connection and try again. | The Hub could not reach ADLM. Check your internet. On an office network, ask IT to allow the Hub to reach adlmstudio.net.    |
| The request timed out. Please check your connection and try again.                 | The connection is too slow or dropped. Try again on a steadier connection.                                                   |
| Your ADLM account is not allowed to sign in right now.                             | The account is on hold. <a href="#">Contact support</a> .                                                                    |
| The ADLM server is temporarily unavailable. Please try again.                      | A short outage on our side. Wait a few minutes and try again.                                                                |
| Your session has expired. Please sign in again.                                    | You left the Hub open for a long time. Sign in again.                                                                        |

#### TIP

If sign-in fails three times, stop retrying. Use WhatsApp support (see [Getting help](#)) and send the exact red message.

## Switching accounts

Click **Sign out** at the bottom of the sidebar. The Hub returns to the sign-in screen. Products already installed on the PC stay installed.

## Find your way around

The sidebar on the left switches pages. The bar across the top has the search box, **Refresh**, and your name and email.

| Page                       | Use it to                                                                        |
|----------------------------|----------------------------------------------------------------------------------|
| <b>Dashboard</b>           | See a summary of your account and this PC.                                       |
| <b>Installation Center</b> | Install, update and repair your products. You will spend most of your time here. |
| <b>Free Software</b>       | Install or download free ADLM tools.                                             |

| Page                   | Use it to                                                    |
|------------------------|--------------------------------------------------------------|
| <b>Subscriptions</b>   | Check licence status, seats, expiry dates and billing notes. |
| <b>Install History</b> | See what is installed on this PC, which version, and when.   |

At the bottom of the sidebar you will find the **Need help?** card with **Chat on WhatsApp**, **Raise a ticket** and **Download User Guide**, then the theme switch and **Sign out**.

## The Dashboard page

The Dashboard greets you with **Welcome back** and your name, and a line such as "2 active subscription(s), 1 installed, 0 pending, 1 update(s) available." Below it are four counters:

| Counter                         | What it counts                                             |
|---------------------------------|------------------------------------------------------------|
| <b>Subscribed products</b>      | Everything your account has bought, installed here or not. |
| <b>Active subscriptions</b>     | Of those, how many are in force and not expired.           |
| <b>Installed on this device</b> | How many are installed on this PC.                         |
| <b>Total orders</b>             | How many purchases are recorded on your account.           |

## Search and Refresh

Type in the search box to filter products by name. Clear it to see everything again.

Click **Refresh** to read your account again from the server. Use it after you buy or renew something, when a product you expect is missing, or when you have been told an update is out. Refresh also re-checks what is installed on this PC.

## Light and dark mode

Click  **Dark mode** at the bottom of the sidebar to switch the Hub to dark. The button then reads  **Light mode**. The Hub remembers your choice.

## The Installation Center

Click **Installation Center** in the sidebar. Each product your account has bought appears as a card. Products you have not bought are not shown.

If you see "No purchased products are ready for installation on this account yet", your account has no active purchase on record. Click **Refresh**. If it stays empty, check your orders on the website or [contact support](#).

## Reading a product card

Each card shows, from top to bottom:

- The product name and its short key (for example **revit** for QUIV for Revit, **planswift** for ADLM Heron). Quote the key when you contact support.
- A status badge, top right.
- A short description.

- Your licence, for example "Personal license • Seats 1/1".
- The expiry date, for example "Expires 12 Mar 2027", or "No expiry date on record".
- Your billing plan, for example "Monthly plan • Install fee included".
- Where the package comes from and its version.
- What is installed on this PC, for example "Installed on 14 Sep 2026, 10:12 AM. Version 3.1.9 on OFFICE-PC."
- The main button, **Reinstall** (once installed), and **Hide**.

## What the badge and button mean

| Badge                  | Main button                         | Meaning                                                                                              |
|------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Ready</b>           | <b>Install</b>                      | Bought, licence active, package published. Ready to install.                                         |
| <b>Installed</b>       | <b>Installed</b> (greyed out)       | Installed here and up to date. Use <b>Reinstall</b> to repair it.                                    |
| <b>Update</b>          | <b>Update</b>                       | A newer version is out. Click to install it over the old one.                                        |
| <b>Pending</b>         | <b>Install</b>                      | Your purchase is approved but ADLM still has to complete the set-up on our side.                     |
| <b>Package pending</b> | <b>Package pending</b> (greyed out) | Your licence is fine but the installer has not been published yet. Check back later.                 |
| <b>Expired</b>         | <b>Unavailable</b> (greyed out)     | The subscription has run out. Renew on the website, then click <b>Refresh</b> .                      |
| <b>Inactive</b>        | <b>Unavailable</b> (greyed out)     | The subscription is not active, usually a payment or account hold. <a href="#">Contact support</a> . |

While an install runs, the button reads **Installing....**

## Install a product

Most installs finish in a couple of minutes.

1. Close the host program if you are installing a plugin: Revit, PlanSwift, Civil 3D or Archicad. Check the Windows system tray as well.
2. Click **Installation Center** in the sidebar.
3. On the product's card, click **Install**.
4. If the Hub asks you to choose your Revit version, pick it and click **Continue Install** (see below).
5. Wait while the progress panel runs. When it shows **Complete**, the button changes to **Installed**.

## Choosing your Revit version

QUIV for Revit and the Revit MEP plugin need to know which year of Revit you use, because each year loads add-ins from a different folder.

- If the Hub finds only one Revit on your PC, it uses it without asking.
- If it finds more than one, the **Choose Your Revit Version** window opens. Pick the year from **Revit Version** and click **Continue Install**.
- If your Revit year is not in the list, type the four-digit year (for example 2025) into the box.

- Click **Cancel** to stop the install. Nothing is changed.

#### TIP

To add the plugin to a second Revit year, click **Reinstall** on the card and choose the other year. Do this again after each update so every Revit year gets the new version.

## ADLM Heron needs PlanSwift first

ADLM Heron installs into PlanSwift. If PlanSwift is not on the PC, the Hub stops with "PlanSwift is not installed on this machine. Please install PlanSwift before installing the ADLM Plugin." Install PlanSwift, then click **Install** again.

## Watching the progress

During an install, a panel covers the card with a progress bar, the current step, a detail line and a percentage. The steps you may see:

| Step                     | What is happening                                                                   |
|--------------------------|-------------------------------------------------------------------------------------|
| Preparing                | Checking your licence and where the files should go.                                |
| Downloading / Downloaded | Fetching the package from ADLM. The detail line shows how much has arrived.         |
| Cached                   | The package was downloaded before and is being reused, which makes reinstalls fast. |
| Retrying                 | The download dropped. The Hub tries again on its own, up to three times.            |
| Verifying                | Checking the download is complete and has not been tampered with.                   |
| Extracting / Extracted   | Unpacking the files.                                                                |
| Cleaning / Cleanup       | Removing the older version so it cannot clash with the new one.                     |
| Running / Launching      | Running the product's own installer, or starting it.                                |
| Complete                 | Done.                                                                               |

A **Cancel** button is available while it is safe to stop. When you click it, the button reads **Cancelling...** while the Hub finishes the current step safely, then stops. Click **Install** again when you are ready.

## After the install

- **Revit, Civil 3D and Archicad plugins:** start the host program. Plugins load when the program starts, so if it was open, close it and open it again.
- **ADLM Heron:** start PlanSwift. See the [ADLM Heron guide](#) for the first steps.
- **Standalone apps** (ADLM RateGen, ADLM Time Pro): open them from the shortcut the Hub created.

The **Installed on this device** counter goes up by one and the product appears on the **Install History** page.

## Update a product

Updates install over the old version. You do not need to uninstall first.

## How you know an update is waiting

- The Dashboard line says how many updates are available.
- The card's badge changes to **Update**, and the main button reads **Update**.
- The card says "Update available: version ... Click Update to install the newer build."

## To update

1. Close the program you are updating (Revit, PlanSwift, Civil 3D, Archicad or the ADLM app).
2. Click **Update** on the card.
3. Wait for **Complete**, then open the program again.

### NOTE

Updating replaces program files only. Your projects, saved rates, templates and preferences are kept.

### TIP

Told an update is out but you do not see it? Click **Refresh**. The Hub checks the published version against what is installed and the badge changes if an update is ready.

## Repair, hide and remove

### Repair with Reinstall

When a product is installed and up to date, a **Reinstall** button sits beside the main button. It installs the same version again from scratch. Use it when:

- the plugin no longer appears in Revit, PlanSwift, Civil 3D or Archicad;
- antivirus removed or damaged some files;
- a shortcut has gone missing;
- you want the Revit plugin in another Revit year.

Reinstalling does not use another licence seat.

### Hide a card

If your account carries a product you never use on this PC, click **Hide** on its card. The card disappears from view. The licence and anything installed are not touched.

A **Show 1 hidden** (or **Show 2 hidden**, and so on) button then appears next to the search box. Click it to bring every hidden card back. Hidden cards also come back when you click **Refresh** or sign in again.

### Remove (uninstall) a product

Installer Hub 2.0 installs, updates and repairs, but it does not remove products. To remove one:

1. Open Windows **Settings**, then **Apps**, then **Installed apps**.
2. If the product is listed, click it and choose **Uninstall**.
3. If it is not listed (most plugins are copied into place by the Hub and do not appear there), [contact support](#) and we will help you remove it.

### IMPORTANT

Hiding a card is not the same as uninstalling. Hidden products stay on your PC and keep working.

## Uninstall the Hub itself

1. Open Windows **Settings**, then **Apps**, then **Installed apps**.
2. Find **ADLM Installer Hub** and choose **Uninstall**.
3. Windows asks whether to also remove your Installer Hub settings and data (theme and installation history) for all users on this computer. Choose **No** to keep them, or **Yes** to remove them.

Removing the Hub does not remove the products it installed.

## Free Software

Click **Free Software** in the sidebar for free ADLM tools, templates and utilities. You only need to be signed in.

- Items packaged for the Hub show **Install**. They install like a paid product, with the same progress panel, and then read **Installed**.
- Other items show **Download**. The file opens in your browser and saves to your Downloads folder. Open it from there.

## Subscriptions and Install History

### Subscriptions

The **Subscriptions** page lists every product on your account with its licence type and seats, expiry date, billing plan and a status badge. Nothing is installed from here. Use it to check what you own and when it runs out. To renew, go to the website and then click **Refresh** in the Hub.

### Install History

The **Install History** page shows what is installed on this PC, which version, when, and anything still waiting for ADLM to complete. Take a screenshot of this page when you report a problem. It tells us exactly which version you have.

## Licences and your computers

Each licence seat belongs to one computer. When you install a product for the first time, the Hub registers that seat to the PC you installed it on.

### What this means day to day

- Install, update and reinstall as often as you like on that PC. It never uses up another seat.
- You can sign in to the Hub on any computer and see your products. Installing on a new one is what needs a free seat.
- If you have more seats than computers, ADLM can add seats to your plan. Ask support.

## The message you see on a second computer

If every seat on a licence is already in use on other computers, the Hub stops with **Device Not Authorized** and the message "[Product] is bound to another device. Contact admin to revoke the existing device or increase seat count." Nothing is installed and nothing is lost. Free a seat (below), click **Refresh**, then **Install** again.

## Moving to a new computer

1. If you still have the old computer, close the ADLM products on it.
2. On the website, sign in and open [Settings](#). Under **Machines signed in**, find the old computer and click **Sign out**, then confirm. This frees its seat straight away and deletes nothing on that machine.
3. On the new computer, download and install the Hub (see [Install the Hub on your PC](#)).
4. Sign in, go to **Installation Center** and click **Install** on each product.

If you cannot sign the old machine out yourself, use **Chat on WhatsApp** or **Raise a ticket** with your account email and the product name, and ADLM will free it for you.

### TIP

Replacing or reformatting a PC? Free its seat on the website before you wipe it. Reinstalling on the rebuilt PC is then straightforward.

## "Device Registration Notice" after an install

Occasionally the product installs but the Hub cannot update your seat on the server. You see **Device Registration Notice**. The product is installed. If the product then reports a device mismatch when you open it, free the old machine on the website as above, or contact support.

## Working without internet

The Hub needs an internet connection to sign in, load your products and download installers. Installer Hub 2.0 has no offline mode.

- If you lose connection while signed in, **Refresh** fails with a message about the server.
- If the Hub cannot fetch the latest package list from ADLM, it will not install from its built-in list. You see **Server Unreachable** and the install stops, so you never get an out-of-date build that cannot sign in. Reconnect, click **Refresh**, and install again.
- Products you have already installed are not affected by the Hub being offline. How each product behaves without internet is covered in its own guide.

## Getting help

The **Need help?** card at the bottom of the sidebar has three buttons:

- **Chat on WhatsApp** opens WhatsApp with a message to ADLM support already started. This is the fastest route for a stuck install or a seat that needs freeing.
- **Raise a ticket** opens the ADLM [support page](#) in your browser with your account email and Hub version filled in. Best for billing, seat changes, or a problem with screenshots.
- **Download User Guide** opens the illustrated PDF guide.

When you report a problem, include the product name and key (for example **revit**), the exact wording of any message, the step it stopped on, your Revit or PlanSwift version, and a screenshot of the card and the **Install History** page.

## What's new in 2.0

### Installer Hub 2.0 (28 September 2026)

- **"Update available" no longer sticks.** Products could show an update every time you refreshed, even straight after updating. The Hub now recognises the build you have, and only shows **Update** for a genuinely new version.
- **Downloads come from ADLM's own secure storage**, through short-lived private links, instead of a shared file link.
- **A proper Windows installer.** The website download is now a setup file instead of a zip, and setup no longer shows unrelated licence or readme pages.

### Installer Hub 1.0.1 to 1.0.3 (August and September 2026)

- **No more out-of-date installs when the server is down.** If the Hub cannot get the latest package list, it now stops with **Server Unreachable** instead of installing an old build from its built-in list.
- **ADLM Heron's import package no longer lands on the shared desktop** of every user on the PC.
- The Hub installs the latest released build of each product as soon as ADLM publishes it.

### Installer Hub 1.0 (July 2026)

- Sign in once and install every ADLM product from one place.
- **Download User Guide** button in the sidebar.
- Each product card shows its own artwork.
- **Launch** at the end of setup now works.
- Uninstalling the Hub asks before deleting your settings, and keeps them by default.

#### NOTE

A redesigned Installer Hub is being tested by the ADLM team. This guide will be updated when it is released.

## Troubleshooting

### Windows says "Windows protected your PC" when I run setup

This is Windows SmartScreen. It appears for installers that are new to Windows. Click **More info**, then **Run anyway**. Only do this for a setup file you downloaded from [adlmstudio.net](https://adlmstudio.net).

### The Windows prompt says "Unknown publisher"

That is expected for the current setup file. If you downloaded it from your ADLM dashboard, click **Yes**.

## My antivirus blocked or deleted the setup or a plugin

Some security products quarantine newly downloaded files. Signs: setup will not start, an install fails at **Extracting**, or a plugin disappears after working. Restore the file from your antivirus quarantine, or ask IT to allow the ADLM Installer Hub and the ADLM install folders, then use **Reinstall**.

## The install fails with "Access denied" or "file in use"

The program you are installing into is still running. Close Revit, PlanSwift, Civil 3D, Archicad or the ADLM app completely (check the system tray), then click **Install** or **Reinstall** again.

## I installed while Revit was open

The Hub does not stop you, but Revit holds its plugin files open. You may see **Cleanup skipped** with "Revit is running - close Revit and reinstall to remove old version folders." Close Revit, click **Reinstall**, then open Revit again.

## Revit cannot see the plugin

1. Close Revit fully. Plugins load only when Revit starts.
2. Open Revit again.
3. If it is still missing, click **Reinstall** in the Hub and choose the Revit year you actually use.

If you see a message that the add-in was installed but its files do not exist, the wrong Revit year was chosen or a copy failed. Close Revit, click **Reinstall** and pick the correct year.

## I cannot sign in

Check the red message against the table in [If sign-in does not work](#). The most common causes are a mistyped password (passwords are case-sensitive), no internet, and signing in with a personal email when the licence is on the company account.

## The website says "Failed to fetch", or the Hub cannot reach the server even though my internet works

Some office networks, firewalls or security products block or cut short the sign-in details sent with each request. Between July and September 2026 some large company accounts also hit this because of a problem on our side, which was fixed on 29 September 2026.

1. Sign out of the Hub and the website, then sign in again.
2. On the same computer, open [adlmstudio.net/network-check](https://adlmstudio.net/network-check) in your browser. It runs a set of tests and shows what is blocked.
3. If something fails, click **Copy results**, or quote the reference code it shows, and send it to support on WhatsApp.

## My product is not in the Installation Center

1. Click **Refresh**.
2. Look for a **Show ... hidden** button next to the search box. You may have hidden it.
3. Check you are signed in with the account that bought the licence (your name and email show top right).
4. Check the subscription on the website. If it is new, it may still be waiting for ADLM to complete set-up.

## The button says "Package pending" or "Unavailable"

**Package pending** means your licence is fine but the installer is not published yet; check back later.

**Unavailable** means the subscription is expired or inactive; renew on the website, then click **Refresh**.

## "Package integrity check failed"

The download was damaged on the way. Click **Install** again and the Hub fetches a fresh copy. If it keeps happening, contact support.

## "Server Unreachable" when I click Install

The Hub could not get the latest package list from ADLM, so it stopped rather than install an old build. Check your connection, click **Refresh**, and try again.

## The Hub will not start

Right-click the **ADLM Installer Hub** shortcut and choose **Run as administrator**. If Windows refuses, your account does not have administrator rights. Ask whoever manages the PC.

## Downloads are very slow

Packages can be large. Leave the Hub running; it retries a dropped download up to three times. A steadier connection helps.

## The single most useful fix

Close the host program, click **Refresh**, then click **Install** or **Reinstall** again. That solves most problems.

# Frequently asked questions

## Do I need a separate account for the Hub?

No. Use the same email or username and password as the ADLM website.

## Why do I have to sign in every time?

Installer Hub 2.0 does not save your session. Signing in each time keeps your account safe on shared computers.

## Does the Hub keep my password?

No. Your password is sent once to sign you in and then cleared. The Hub keeps only a temporary sign-in while it is open.

## Can I install on more than one computer?

One seat covers one computer. To use a product on a second computer at the same time, you need another seat. To move it, free the old computer on the website first.

## Does reinstalling or updating use up a seat?

No. Install, update and reinstall as often as you need on the same computer.

### **What happens to my installed products if I sign out or close the Hub?**

Nothing. They stay installed and keep working. The Hub is only needed to install, update and repair.

### **Does the Hub update itself?**

No. When ADLM releases a new Hub, download the setup again from your dashboard and run it over the old one.

### **Does the Hub have keyboard shortcuts or a connection signal indicator?**

Not in version 2.0. Use the mouse and the buttons described in this guide.

### **Can my IT team install the Hub for everyone?**

Yes. The Hub uses standard Windows install locations. It needs outbound internet access to adlmstudio.net and the ADLM download service. Each person then signs in with the account that holds the licence.

### **Where are the guides for the products themselves?**

See [QUIV for Revit](#), [ADLM Heron](#), [ADLM RateGen](#) and the other guides on the [guides page](#).



# QUIV for Revit

Measure your Revit model module by module, price it into a bill and a budget, and keep the job on ADLM Cloud.

QUIV is ADLM's quantity take-off and estimating add-in for Autodesk Revit. It reads quantities straight from your model: foundations, frame, walls, roof, doors, windows, finishes and external works. You add only what a quantity surveyor knows that the model does not, such as bar sizes, spacing and filling thicknesses. QUIV turns what you save into a bill of quantities and a budget inside Revit, and saves the job to your ADLM account so you can open it on [adlmstudio.net](https://adlmstudio.net).

It is for quantity surveyors, estimators and engineers who receive Revit models and want a bill of quantities without measuring the drawings again by hand.

## NOTE

This guide covers **QUIV 3.1.11**, the version the Installer Hub gives you today. QUIV 4.0, with a new panel design, is being prepared and is not yet available to customers. See [Coming in 4.0](#).

## Before you start

### What you need

| Item                | Requirement                                                    |
|---------------------|----------------------------------------------------------------|
| Autodesk Revit      | Revit 2024, 2025, 2026 or 2027, on a Windows PC                |
| Your licence        | An active QUIV subscription on your ADLM account               |
| The Installer Hub   | QUIV is installed and updated through the ADLM Installer Hub   |
| Internet            | Needed to sign in and to save to the cloud                     |
| Rate Gen (optional) | Needed to load rates from the ADLM rate library onto your bill |

## Install QUIV

Installing is covered in full in the [Installer Hub guide](#). The short version:

1. Close Revit. Windows will not let the Hub replace add-in files that Revit is holding open.
2. Open the ADLM Installer Hub and sign in with your ADLM account.
3. Find **QUIV for Revit** and click **Install**.
4. When asked, choose the year of Revit you use. If you use more than one year, install once for each.
5. Start Revit. QUIV loads when Revit starts, so a Revit that was already open will not see it until you restart it.

When a new version is published, the Hub offers it as an update. Close Revit, update, then open Revit again. Your saved projects are not affected.

#### NOTE

Your licence is tied to the computer you sign in from. If you change computers, contact ADLM support from [/support](#) rather than buying again.

## Sign in

1. In Revit, open the **ADLM Calculator** tab on the ribbon.
2. Click **ADLM Calculator**.
3. In the **ADLM Sign In** window, enter your **Email** and **Password**. These are the same details you use on [adlmstudio.net](#).
4. Click **Log in**.

QUIV says "Signed in successfully." and opens. It remembers your session, so you normally sign in once on each computer. If you do not have an account yet, use the link under **Don't have an account?**.

#### IMPORTANT

If QUIV says your account has no active licence, your subscription has expired or does not include QUIV. Renew it on [adlmstudio.net](#), then click **ADLM Calculator** again.

## Finding your way around

### The ADLM Calculator tab

QUIV adds one ribbon tab, **ADLM Calculator**, with two buttons on its **ADLM** panel:

| Button                 | What it does                                                                                                                 |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>ADLM Calculator</b> | Opens the QUIV workspace: dashboard, take-off modules, Bill and Budget. Asks you to sign in the first time.                  |
| <b>Model Checker</b>   | Opens the ADLM Model Checker, which tells you whether the model is ready to measure. See <a href="#">The Model Checker</a> . |

### The workspace

QUIV opens as a panel docked inside Revit, so you can keep it beside your model. It has a sidebar on the left and the current page on the right.

The sidebar is grouped the way a bill is written:

| Group                | Modules                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Foundation</b>    | <b>Oversite Qty, Strip Qty, Pad Foundation Qty, Pile Cap Qty, Raft Foundation Qty</b>                                                                      |
| <b>Frames</b>        | <b>Beam Qty, Column Qty, Slab Qty, Steelwork Qty</b>                                                                                                       |
| <b>Architectural</b> | <b>Wall Qty, Roof Qty, Ceiling Qty, Curtain Wall Qty, Door Qty, Window Qty, Staircase Qty, Railing Qty, Finishes Qty, Landscaping Qty, Model Items Qty</b> |
| <b>Database</b>      | <b>BoQ (the Bill) and Budget</b>                                                                                                                           |

**Dashboard** at the top takes you home, and **Log out** at the bottom signs you out. A module can only be opened once it is on your take-off list (see [Starting a project](#)).

## The header buttons

The header across the top has a few small buttons. Hover over one to see what it does:

| Button (tooltip)                    | What it does                                                                                  |
|-------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Pop out to a floating window</b> | Moves QUIV into its own window for more room.                                                 |
| <b>Toggle Dark Mode</b>             | Switches between light and dark. Not shown in Revit 2024.                                     |
| <b>Metric / Imperial</b>            | Switches the rate display between metric and imperial units.                                  |
| <b>Include Hidden Elements</b>      | When on, QUIV measures every element, including hidden ones. When off, only visible elements. |
| <b>Material Constants</b>           | Opens the material constants QUIV uses for mixes and allowances.                              |
| <b>Contact ADLM Support</b>         | Sends a message to ADLM support.                                                              |

Click your name in the header for **View Profile** and **Sign Out**.

### TIP

Docked in a narrow strip, QUIV rearranges itself so nothing is cut off. If it feels cramped, widen the panel or pop it out.

## The Model Checker

Before you measure, the Model Checker tells you whether the model can be measured: what is there, what is missing, what overlaps, and whether reinforcement has been modelled.

### Run a check

1. Open the model you want to measure.
2. On the **ADLM Calculator** tab, click **Model Checker**. The **ADLM Model Checker** window opens.
3. Under **Select Model Type**, choose **Architectural Model** or **Structural Model**.
4. Under **Check Options**, tick the checks you want (see the table below).
5. Click **Run Model Check**.

You will see it work through the model: scanning elements by category, detecting overlapping elements, checking for missing categories, analysing reinforcement, then calculating the readiness score.

| Option                                | What it looks for                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Check for Overlapping Elements</b> | Duplicated elements sitting on top of each other, found by comparing their outlines. Duplicates are the usual reason a quantity comes out double. |
| <b>Check Inter-Model Clashes</b>      | Clashes between your model and the models linked into it.                                                                                         |

| Option                                         | What it looks for                                                                     |
|------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Check Reinforcement</b><br>(InDesign Rebar) | Structural models only. Shows which categories carry modelled rebar and which do not. |

## Read the results

The results screen shows four figures: **Readiness Score**, **Elements Found**, **Issues Found** and **Element Categories**. Below them:

- **Element Categories** lists what was found in the model. A category that is missing has to be modelled first or measured some other way.
- **Reinforcement Analysis** (structural models) shows which categories have modelled rebar. Where there is none, you will enter bar sizes and spacing yourself when you measure.
- **Issues** lists each problem. Click a row to highlight that element in Revit, so you can see the problem and fix it in the model.

## The buttons on the results screen

| Button                    | What it does                                                                                                                                                                                                                                                                    |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Export Excel</b>       | Saves the whole check to a spreadsheet.                                                                                                                                                                                                                                         |
| <b>View QS Query</b>      | Opens a <b>QS Query Report</b> : the questions a quantity surveyor would send the design team about missing or unclear parts of the model. Use <b>Copy to Clipboard</b> or <b>Export PDF</b> to send it.                                                                        |
| <b>Generate QR Code</b>   | Saves the report to the ADLM website and makes a QR code for it. Anyone who scans the code can read the report in a browser, without Revit and without signing in. Use <b>Save QR Code</b> to save the image, or <b>Save Full Report</b> to save the report as a web page file. |
| <b>Proceed to Takeoff</b> | Records your take-off decision in the QS query and closes the checker.                                                                                                                                                                                                          |
| <b>Back to Setup</b>      | Returns to the options to run another check.                                                                                                                                                                                                                                    |

### IMPORTANT

A report published with **Generate QR Code** can be read by anyone who has the link or the code. Check the project name and the category list before you share it.

## Starting a project

### The dashboard

When you open QUIV you land on the dashboard. It greets you by name and shows:

- **Ready for a new project?** with two cards, **Bungalow** and **Multi-Story**.
- **Recent Projects**: the projects you have saved to the cloud, with a search box and the filters **All Projects**, **Takeoff Only** and **Material Only**. Click a project to open it.

## Create a new project

1. On the dashboard, under **Ready for a new project?**, click **Bungalow** or **Multi-Story**.
2. For **Multi-Story**, QUIV asks for the **Foundation type**. Choose **Pad Foundation**, **Pile Foundation** or **Raft Foundation**.
3. QUIV shows "Generating Taking-Off List", then your **Take off List**.
4. Tick the items this job needs. Use **Check All** or **Uncheck All** to start from everything or nothing.
5. Click **Continue**.

The modules you ticked are now open in the sidebar. To change the list later, use the **Edit take off list** button; the confirm button then reads **Update List**.

### TIP

Work down the sidebar from top to bottom. It is the order a bill is written in, so your bill comes out in bill order without sorting.

## Measuring with a module

Click a module in the sidebar to open it. Every module works the same way, from top to bottom:

1. **Model Selection.** Choose the level (or **All Floors** where offered) and the type you are measuring, for example a wall type or a beam type. For some modules you pick the elements in Revit instead.
2. **Model Quantities.** What QUIV read from the model: areas, lengths, volumes and counts. You cannot type over these. To change them, change the model and measure again.
3. **Your inputs.** What only the quantity surveyor knows: thicknesses, depths, bar diameters, numbers of bars, link spacing, lintel treatment. On many modules this section is called **Input Quantities**. Change a value and the results update.
4. **Results.** The quantities that will go into the bill.
5. **View Material Breakdown.** On most modules, this shows the materials behind the results, such as cement in bags, sharp sand and granite in tonnes, blocks, planks and nails.
6. **Save.** Puts the result on the Bill and the Budget.

The first time you save, QUIV asks for a **Project name** in the **Save Takeoff** window. Everything you save afterwards belongs to that project.

If you save a level and type you have already saved, QUIV asks whether to replace the previous figures or add to them. Choose replace when you are correcting a measurement, and add when you are measuring another part of the same thing.

### TIP

QUIV reads the model when you measure, not all the time. If you change the model, open the module again, choose the level and type, and save again.

### NOTE

The type list only shows types that are actually modelled on the level you chose. An empty list nearly always means the wrong level is selected.

## What each module measures

| Module                       | What it measures                                                                                                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oversite Qty</b>          | Hardcore and laterite filling, BRC mesh, DPM, formwork to edges, oversite concrete and blinding.                                                                              |
| <b>Strip Qty</b>             | Excavation, earthwork support, concrete in footing, blockwork in foundation, blinding, surface treatment, levelling and compacting, damp-proof course, backfill and disposal. |
| <b>Pad Foundation Qty</b>    | Excavation, earthwork support, concrete in pads, formwork, blinding and bars.                                                                                                 |
| <b>Pile Cap Qty</b>          | Excavation, concrete in pile caps and piles, formwork, blinding and bars.                                                                                                     |
| <b>Raft Foundation Qty</b>   | Excavation, concrete in slab and beams, hardcore, laterite, DPM, beam formwork and reinforcement.                                                                             |
| <b>Beam Qty</b>              | Concrete in beams, formwork, top and bottom bars and links.                                                                                                                   |
| <b>Column Qty</b>            | Concrete in columns, formwork, main bars and links.                                                                                                                           |
| <b>Slab Qty</b>              | Concrete in slab, formwork to soffit and edges, and reinforcement.                                                                                                            |
| <b>Steelwork Qty</b>         | Structural steel by weight and surface area.                                                                                                                                  |
| <b>Wall Qty</b>              | Blockwork net of openings, rendering, lintel concrete, formwork and reinforcement.                                                                                            |
| <b>Roof Qty</b>              | Roof members such as rafters, purlins, tie beams, king posts, struts, wall plate, fascia and ridge.                                                                           |
| <b>Ceiling Qty</b>           | Ceiling area and noggings.                                                                                                                                                    |
| <b>Curtain Wall Qty</b>      | Curtain walling area and its parts.                                                                                                                                           |
| <b>Door Qty / Window Qty</b> | Doors and windows by type, with area and perimeter.                                                                                                                           |
| <b>Staircase Qty</b>         | Concrete in stairs, formwork and reinforcement.                                                                                                                               |
| <b>Railing Qty</b>           | Railings and balustrades in metres, by level and type.                                                                                                                        |
| <b>Finishes Qty</b>          | Wall paint, floor tiling and ceiling finishes.                                                                                                                                |
| <b>Landscaping Qty</b>       | Paving, turf, fencing and other external works.                                                                                                                               |
| <b>Model Items Qty</b>       | Anything modelled that does not fit a category above.                                                                                                                         |

Steel beams and columns are measured under **Steelwork Qty** only, not as concrete beams or columns.

## Run the whole takeoff

Instead of opening each module yourself, you can hand the whole take-off list to QUIV. It measures every ticked item level by level and type by type, and saves each result to the Bill and the Budget.

1. Start a project and finish your take-off list.
2. Under the list, click **Run the whole takeoff** (in a narrow panel, just **Run**).
3. Read **Hand the takeoff to QUIV**. It explains what happens and lists the items that **Still needs you**, because they start with a pick in Revit.

4. Click **Let QUIV run the whole takeoff**, or **Not now** to go back.

While **QUIV is running the takeoff**, the Revit view switches level by level and highlights each element as it is measured. A progress panel shows what has been saved, how many were skipped and roughly how long is left.

To stop, choose **Stop here and keep what is saved** or **Stop and undo the whole run**. The measurement in progress always finishes first, so a save is never cut in half. When the run ends you can also click **Undo this run** to put the Bill, the Budget and the Take off List back as they were.

#### NOTE

Full runs are limited per day on each account. When you are close to the limit QUIV tells you how many are left. Measuring module by module is never limited.

#### NOTE

A run does not save to the cloud. Check the Bill, then save to the cloud yourself.

## The Bill of Quantities

Click **BoQ** under **Database** in the sidebar. The **Bill of Quantities** lists everything you have saved, with the columns **S/N**, **Takeoff Title**, **Item**, **Category**, **Qty**, **Unit**, **Rate**, **Amount**, **Source** and **Done**.

- **Source** shows where a rate came from.
- **Done** is your own tick, for the lines you have checked or agreed.
- **Search Takeoff Items** filters the list.
- Click **Budget** to switch to the Budget, and **Back to Dashboard** to go home.

The cards at the top show **Measured Total**, **Budget Total**, **Left To Spend** and **Completed**, with **Preliminaries**, **Prime Cost (PC) / Provisional Sum** and **Project Total Cost**.

### Price the bill

| Button                                | What it does                                                                                                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Load RateGen Prices</b>            | Applies rates from the Rate Gen library to matching lines. Needs a Rate Gen licence.                                                                                                        |
| <b>View Rates</b>                     | Opens the <b>RateGen Quantity Rate Viewer</b> . Choose a bill line in the map column, then click <b>Map</b> to write that rate onto it.                                                     |
| <b>From Budget / Manual / RateGen</b> | Chooses where rates come from. <b>From Budget</b> works each rate out from the line's material and labour build-up. <b>Manual / RateGen</b> lets you type a rate or pick one from Rate Gen. |
| <b>Refresh from Cloud</b>             | Re-reads the saved project, so rates and ticks changed on the website show up here.                                                                                                         |

Nothing is priced until you ask. A line nobody has priced shows no rate.

### Export from Revit

| Button                 | What you get                                |
|------------------------|---------------------------------------------|
| <b>Export to Excel</b> | The bill as a spreadsheet, with its totals. |

| Button                 | What you get                                                                                                                                                                                       |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget Workbook</b> | A formula-linked QS workbook with a cover, dashboard, basic prices, budget summary, the bill with rate build-ups, and a material summary. Change one profit margin cell to re-price the whole job. |

Exports print the rate you entered or a Rate Gen rate. Anything not priced shows 0, never a made-up figure.

## Start again

**Reset** clears all take-off items and the attached budget for this model. **Load Latest Local** brings back the most recent copy kept on this PC.

### IMPORTANT

Export the bill first if there is any chance you will want the figures after a **Reset**.

## The Budget

Click **Budget** under **Database**, or **Budget** on the Bill. The **Budget Library** turns the same take-off into what the job will cost you: every material and labour item, what you have bought, and what is left.

- **All**, **Material** and **Labour** filter the list.
- **Takeoff** names the module each line came from.
- **Bought** is yours to tick as you buy. The cards show **All Items Total**, **Purchased** and **Left To Spend**.
- **Load RateGen Prices** and **View Rates** price materials from the Rate Gen library. In the **RateGen Library Viewer**, choose a material in the map column and click **Map**.
- **Clear Prices** zeroes the prices QUIV applied by itself, keeping the quantities and any rate you typed. Save to the cloud afterwards to clear the saved copy too.
- **Export to Excel** saves the budget as a spreadsheet.
- **Bill** takes you back to the Bill of Quantities.

## Save to the cloud

1. Open the Bill (**BoQ** in the sidebar).
2. Click **Save To Cloud**. For a project already on the cloud the button reads **Update To Cloud**.
3. Check the **Project name** and confirm.

If some lines are missing a rate or other details, QUIV tells you before it saves. You can save as often as you like; each save updates the same project.

To open a saved project again, click it under **Recent Projects** on the dashboard. This works on any computer where you sign in with the same account.

### NOTE

You must be online to save to the cloud.

## Linked models

Many jobs come as an architectural model with the structural engineer's model linked into it. QUIV measures the beams, columns and slabs inside a linked structural model.

- **Levels are matched for you.** QUIV pairs the levels of the two models by name where they agree, and by height where they do not. So "00 - GROUND FLOOR" in the engineer's file is recognised as your "Ground floor level". Each linked element is measured on exactly one level.
- **Load the link.** QUIV can only measure a link that is loaded in Revit. If a category comes back empty when you know it is modelled, check the link is loaded in Manage Links.

### IMPORTANT

If the same linked file is placed in your model more than once (copied, mirrored, or linked again when it looked missing), its elements can be measured once for each placement, which doubles the quantities. Open Manage Links in Revit and remove the link instances you do not need before you measure.

## ADLM Cloud: your project on the website

Go to </projects/revit> after signing in on adlmstudio.net. Every project you saved from QUIV is listed there.

Open a project to see its tabs, including **Dashboard**, **Bill of Quantity**, **Budget**, **Valuation**, **3D Model** and **PM Dashboard**. This is where you can also price the bill, plan the budget and value the work.

### Sample projects

At the top of the QUIV projects page you may see **Learning samples**: fully worked duplex projects, one for each foundation type, each measured from its own 3D model. Click the button that reads **Show** and the number of samples, for example "Show 4 samples", then open one to see every tab filled in on a realistic job. Samples are read-only.

### Exports on the website

On an open project, click **Export** to choose a workbook. The choices include:

| Export                                                                                      | What you get                                                                           |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Export bill &amp; budget workbook</b>                                                    | The bill and the budget together, by category (also offered by trade)                  |
| <b>Export generic BoQ (by category)</b>                                                     | A plain bill grouped by category (also offered by trade)                               |
| Elemental BoQ: <b>Bungalow</b> or <b>Multi-storey</b>                                       | A bill grouped by building element                                                     |
| Trade BoQ: <b>Bungalow (Trade format)</b> or <b>Multi-storey (Trade format)</b>             | A bill grouped by work section                                                         |
| Milestone BoQ: <b>Bungalow (Milestone format)</b> or <b>Multi-storey (Milestone format)</b> | One priceable bill per construction stage or storey, as a basis for a payment schedule |

See the [ADLM Cloud guide](#) for the rest of the website, and the [Rate Gen guide](#) for building your own rates.

## What's new in 3.1.11

### 3.1.11 (September 2026)

- Pop-up windows close with `Esc`, and take-off pop-ups fit a narrow docked panel.

### 3.1.10 (September 2026)

- Faster opening of the Bill and the Budget after a take-off that started with foundations.

### 3.1.9 (August 2026)

- Linked structural models are measured, with levels matched by name or by height.
- ♦ **Run the whole takeoff:** QUIV measures every ticked item, splitting every level and every type onto its own bill line.
- A **Railing Qty** module.
- Staircases that measured zero now measure properly; steel members are no longer counted twice.
- Nothing is priced until you ask, and exports no longer print rates nobody set.
- **Clear Prices** removes the prices QUIV applied by itself and keeps the ones you typed.

## Coming in 4.0

QUIV 4.0 is not yet available to customers. When the Installer Hub offers it, QUIV will have a new panel with sign-in inside it, a one-screen new project, one screen per item, and auto take-off results you review and accept. In 4.0, pricing and exports move to ADLM Cloud. This guide will be updated when 4.0 is released.

## Troubleshooting

### There is no ADLM Calculator tab in Revit

QUIV is not loaded. Close Revit, open the Installer Hub, and install or reinstall **QUIV for Revit**, choosing the year of Revit you use. Then start Revit again. Add-ins load only when Revit starts.

### It says my account has no active licence

Your subscription has expired or does not include QUIV. Check your subscription on [adlmstudio.net](https://adlmstudio.net), renew it, then sign in again.

### "Sign in failed. Please verify your credentials."

Check your email and password and try again. Use the same details as on [adlmstudio.net](https://adlmstudio.net).

### "No internet connection detected."

QUIV needs the internet to sign in. Connect, then click **Log in** again.

### My computer changed and now I cannot sign in

Your licence is tied to the computer you first used. Contact ADLM support from [/support](#) to move it.

## A module in the sidebar is greyed out

It is not on your take-off list. Use **Edit take off list**, tick it, and click **Update List**.

## Nothing appears in the type list

Nothing of that kind is modelled on the level you chose. Check the level first: this is nearly always the cause. If the elements are in a linked model, make sure the link is loaded.

## A quantity looks double

Two common causes:

1. **Duplicated elements.** Run the Model Checker with **Check for Overlapping Elements** ticked, then click the issue rows to find the duplicates in Revit.
2. **A link placed twice.** Open Manage Links in Revit and remove link instances you do not need. See [Linked models](#).

## The reinforcement comes out as zero

The rebar is not modelled, which is normal. Enter the bar sizes, numbers and spacing in the module's inputs.

## I changed the model but the quantities did not change

QUIV reads the model when you measure. Open the module, choose the level and type again, and save. When asked, choose to replace the previous figures.

## The rates are all blank

That is expected until you price. Click **Load RateGen Prices** on the Bill, or type rates with **Manual / RateGen**. Loading Rate Gen prices needs a Rate Gen licence.

## The project is not on the website

It was never saved to the cloud, or the save was made while offline. Open the project in QUIV, click **Save To Cloud** or **Update To Cloud** on the Bill, then refresh the projects page in your browser.

## There is no dark mode button

You are using Revit 2024, where QUIV stays in light mode.

## The panel is cramped

Widen the docked panel, or click the **Pop out to a floating window** button.

## Support asks for my log file

QUIV writes a diagnostic log to `%APPDATA%\ADLM\adlm.log`. Paste that address into the Windows File Explorer address bar to find it, and attach the file to your support ticket.

## Frequently asked questions

### Do I need Rate Gen to use QUIV?

No. QUIV measures without it, and you can type your own rates. You need Rate Gen to load rates from the ADLM rate library.

### Can I save before everything is measured?

Yes. Save to the cloud at any time. Each later save updates the same project.

### Is Run the whole takeoff guessing the quantities?

No. It uses the same modules you would use by hand, going through the model level by level and type by type. Check the Bill afterwards as you would check your own work.

### Can I measure a structural model linked into my architectural model?

Yes, for beams, columns and slabs. Make sure the link is loaded. See [Linked models](#).

### Which versions of Revit does QUIV work with?

Revit 2024 to 2027. The Installer Hub asks which year you use and installs the matching version.

### Can I work in imperial units?

Yes. Use **Metric** / **Imperial** in the header to switch the rate display. Decide before you price, so rates and quantities agree.

### Where can I get help?

Raise a ticket or chat with ADLM from [/support](#). With a take-off problem, send the project name, the module, the level and type you chose, the number QUIV gave and the number you expected, and a screenshot of the module page.



# ADLM HERON

Measure PDF drawings in PlanSwift with the ADLM templates, then turn the takeoff into a priced bill, a material and labour budget and a live Excel link.

ADLM HERON (formerly the ADLM PlanSwift Plugin) is a Windows app for quantity surveyors and estimators who take off from 2D drawings in PlanSwift. HERON adds the ADLM template library to PlanSwift, so every slab, wall, finish and service you measure already knows its unit and its formula. HERON then reads your takeoff live out of PlanSwift and gives you a billed takeoff, a material and labour budget priced from your Rate Gen library, a professional Excel bill of quantities and a live link into your own Excel workbooks. You can also save each job to your ADLM Cloud projects and open it on the website. This guide covers HERON 2.9.6.

## Before you start

### What you need

| You need                                  | Notes                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| A Windows PC                              | HERON is a Windows desktop app.                                                                                                                   |
| PlanSwift 10 or 11                        | HERON works alongside PlanSwift. It does not replace it. If you run both versions, the templates go into each one.                                |
| An ADLM account with a HERON subscription | The same email and password you use on the ADLM website.                                                                                          |
| Rate Gen (optional)                       | Pricing from your rate library needs a Rate Gen subscription on the same account. Without it you can still measure, review and export quantities. |
| Microsoft Excel (optional)                | Needed for the Excel export and the Excel Takeoff Link.                                                                                           |
| An internet connection                    | Needed for your first sign-in, for loading rates and for saving to the cloud. Measuring works offline.                                            |

## Installing HERON

HERON is installed from the ADLM Installer Hub, never from a loose download. See [ADLM Installer Hub](#) for the Hub itself.

1. Close PlanSwift completely. The templates are written into PlanSwift's own folders, and Windows locks them while PlanSwift is open.
2. Close Excel too, so the Excel add-in can be registered.
3. Open the Installer Hub and sign in.
4. Find **ADLM Heron** and click **Install** (or **Update** if you already have an older version).
5. Wait for the Hub to report that the install has finished.

The Hub puts four things on your PC:

- The HERON app, with an **ADLM HERON** shortcut on the desktop.
- The ADLM template library, inside every PlanSwift installation it finds.
- A protected copy of the templates, which HERON uses to put them back if PlanSwift ever loses them.

- The Excel add-in (the ADLM Takeoff Link).

#### IMPORTANT

If the Hub reports that access was denied while installing HERON, PlanSwift was still running. Check the notification area beside the clock, close PlanSwift, and click **Install** again.

## One licence, one PC

Your HERON licence is tied to the PC you first sign in on. If you sign in on a second PC, HERON tells you the subscription is bound to a different device. To move your licence to a new computer, contact ADLM support through [/support](#) and ask for the device lock to be reset.

## The order to do things in

You only do steps 1 to 3 once. After that, your day starts at step 4.

1. Install HERON from the Hub with PlanSwift closed.
2. Open HERON and sign in. Signing in is what switches the ADLM templates on.
3. Restart PlanSwift. PlanSwift only reads its template library when it starts.
4. Open or create your job in PlanSwift and scale the drawings.
5. Measure with the ADLM templates, then read, price and export the results in HERON.

## Signing in

1. Open **ADLM HERON** from the desktop shortcut.
2. On the **Sign in to ADLM Plugin** screen, type your email or username in **Username or Email**.
3. Type your password in **Password**. Click the eye button if you want to check what you typed.
4. Click **Sign In**.

HERON checks your subscription and your device, switches the ADLM templates on in PlanSwift, and opens the **Dashboard**. You will see one of these messages:

| Message                                                               | What it means                                                                                                                     |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>ADLM Plugin activated successfully!</b>                            | The templates are in place. Carry on.                                                                                             |
| <b>ADLM templates restored. Please restart PlanSwift to see them.</b> | The templates had gone missing (usually after signing out) and HERON put them back. Close PlanSwift completely and open it again. |
| <b>Welcome, (your name)!</b>                                          | You are signed in. If this is your first sign-in on this PC, restart PlanSwift before you look for the templates.                 |

If sign-in fails, the message tells you why. The common ones are **Invalid username/email or password.**, **Your subscription has expired.**, **Device limit reached for this subscription.** and **Internet required for first sign-in (no valid offline license found)**. See [Troubleshooting](#).

#### TIP

There is no password reset inside HERON. If you have forgotten your password, reset it on the ADLM website, then sign in to HERON with the new one.

## Staying signed in

- HERON remembers you. When you open it again it signs you back in from your saved session.
- A session lasts 10 days. After that you see **Session Expired** and sign in again. This is normal.
- If HERON starts without reaching the server, it signs you in from your saved licence. The header then shows an **Offline session** note. Measuring, reviewing and exporting to Excel all still work. Loading rates, cloud projects and saving to the cloud need a full sign-in: click **Sign Out**, then sign in again while you are connected.

## Signing out

Click **Sign Out** at the bottom of the sidebar. HERON tells you that PlanSwift templates will be hidden on next restart. That is deliberate: the templates belong to your licence. Sign back in and restart PlanSwift to get them back.

### TIP

Do not sign out at the end of each day out of habit. It gains you nothing and costs you a PlanSwift restart the next morning. Just close the window.

## A tour of the HERON window

HERON runs in its own window beside PlanSwift. It never redraws your plans. It reads the job that is open in PlanSwift.

### The sidebar

| Button                   | What it opens                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------|
| <b>Dashboard</b>         | Start a new project, see the units HERON detected and open projects saved to the cloud. |
| <b>Quantity Take Off</b> | Your measured folders and items, their breakdown, and <b>Save / Export</b> .            |
| <b>Budget</b>            | The material and labour schedule behind each item, with your margin.                    |
| <b>Steel Tonnage</b>     | Converts measured steel lengths into tonnes.                                            |
| <b>Connect Excel</b>     | Checks and repairs the Excel add-in and starts the live link.                           |
| <b>Feedback</b>          | Opens WhatsApp with a message to ADLM support.                                          |
| <b>Sign Out</b>          | Signs you out of HERON.                                                                 |

**Quantity Take Off**, **Budget** and **Steel Tonnage** stay greyed out until you start or open a project from the **Dashboard**. The round button on the edge of the sidebar collapses it to icons when you need more room, and you can drag the sidebar edge to resize it.

### The header

Along the top right you will find:

- **The network signal.** Bars show how well you are connected to the ADLM Cloud: **Excellent**, **Good**, **Fair** or **Poor**. It shows **Offline** when you have no internet, and **No server** when your internet works but the ADLM server is not answering. Hover over it for details and click it to check again.

- **The theme button.** Switches between light and dark mode.
- **The gear.** Opens **Settings** (display units, the Takeoff Time Log and your material constants).
- Your name and email.

## The Dashboard

The Dashboard is where every session starts.

- **The top strip** shows the PlanSwift job that is open, the date, your PlanSwift version and the unit system HERON is using, for example **Units: Metric**. HERON reads this from the Scale Units of your PlanSwift pages when you open a project. Check it each time you open a new job.
- **New Project** starts a new takeoff from the job open in PlanSwift. Click **Start New**.
- **Quick Actions** jump straight to **Quantity Take Off** or **Budget**, or **Refresh** the Dashboard.
- **Recent Projects** lists the HERON projects saved to your ADLM Cloud account. Sort them with **Recent**, **A - Z** or **Most Items**, and use the search box to find one by name.

### Opening a saved project

1. Open the matching job in PlanSwift first.
2. On the Dashboard, find the project under **Recent Projects** and click **Open**.
3. If the project was saved from a different PlanSwift job, HERON warns you with **Job Mismatch**. Click **Yes** only if you are sure it is the right job.
4. HERON loads the project and asks whether to build the bill into PlanSwift now. Click **No** for a project you measured in HERON yourself. Click **Yes** only for a bill that came from the website and has no measurements behind it yet (see [Step 2: Build the bill in PlanSwift](#)).
5. HERON opens **Quantity Take Off**. Your next **Save to Cloud** updates this same project.

Click the small arrow on a project card to expand it and see the materials saved with it, with a **Materials total (material + labour)** at the bottom.

## Setting up the job in PlanSwift

Everything HERON reads comes from the job open in PlanSwift: its pages, its folders and its takeoff items. A tidy job gives you a tidy bill.

1. Create a new job in PlanSwift and set its measurement system to metric or imperial. This drives the units your pages report, which HERON then reads.
2. Import your drawings (PDF or image files) into the job.
3. Rename every page you will measure on, for example "Ground Floor Plan" rather than "Page 7".
4. Group pages into folders by drawing set, for example "Admin Struct" and "Admin Arch".
5. Rotate any sideways sheet, and give each separate drawing on a sheet its own page, because the scale is set per page.

#### TIP

When revised drawings arrive mid-job, import the revision as a new page, measure it, then delete the old page. You never lose a takeoff you might want to compare against.

## Scaling your drawings

This is the step that decides whether every number after it is right. A PDF does not know how big the building is until you tell it.

### IMPORTANT

The scale is set for each page, not for the whole job. Every page you measure on needs its own scale.

### Setting the scale

1. Open the page in PlanSwift.
2. Find a long dimension you trust, ideally an overall grid dimension of 6 metres or more. Avoid door and window sizes, which are often drawn roughly.
3. Start PlanSwift's Scale tool.
4. Type the real length in the units shown, for example 4425 for a 4425 mm grid, or 4.425 if you are working in metres.
5. Zoom right in, then click the two ends of that dimension.
6. Repeat on a dimension running the other way. Plots and scans are not always stretched equally in both directions.

### Checking the scale

1. Use PlanSwift's Dimension tool to measure a different dimension on the same page, ideally on the other axis.
2. Compare the result with the figure printed on the drawing.
3. If it is out by more than a few millimetres, the scale is wrong. Scale the page again.

### Common scaling mistakes

| What went wrong                      | What you see, and the fix                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------|
| The page was never scaled            | Quantities look far too big or too small. Scale the page. Existing measurements update to the new scale. |
| Scaled off a wrong dimension         | Everything is out by the same percentage. Check with the Dimension tool and scale again.                 |
| Millimetres typed into a metre field | Quantities out by a factor of 1,000. Scale again, entering the length in the units shown.                |
| Clicked at low zoom                  | A small, steady error across the job. Zoom in until the line is several pixels thick before clicking.    |
| Two scales on one sheet              | Give each part of the sheet its own page, and put the scale in the page name.                            |

### NOTE

Scaling also decides which units HERON uses. HERON reads the page's Scale Units (for example M, MM, FT or IN). If your pages are unscaled, HERON may pick the wrong unit system. You can always set it yourself in **Settings**, under **Display Units**.

#### TIP

If you rescale a page after measuring, go back to HERON's **Quantity Take Off** and click **Reload** so HERON picks up the corrected numbers.

## The ADLM template library

The templates are the heart of HERON. Instead of drawing a shape and deciding later what it is, you pick what you are measuring first, and the template brings its own unit, colour and formula.

In PlanSwift, open the templates panel and choose **Complete ADLM TakeOff Plugin** from the list at the top. The library is grouped by trade:

- SUBSTRUCTURE
- FRAME
- SWIMMING POOL
- STAIRCASE
- BLOCKWORK AND OPENINGS
- FINISHES (wall, floor and ceiling finishes)
- ROOFING
- ELECTRICAL
- PLUMBING
- EXTERNAL WORKS
- RAILING
- HVAC
- FIRE ALARM
- ELV SYSTEMS

Expand a trade to see its folders, and expand a folder to see the templates you measure with. Many templates give you more than one quantity from one measurement. For example, a floor tile area also gives you its skirting, and a wall area gives you its rendering and perimeter.

#### NOTE

If the ADLM library is missing from PlanSwift, see [The ADLM templates are not in PlanSwift](#).

## Measuring with the templates

PlanSwift gives you four kinds of measurement. The template you pick decides which one you use.

| Kind    | Gives you                 | Use it for                                                                                |
|---------|---------------------------|-------------------------------------------------------------------------------------------|
| Area    | m <sup>2</sup> (or sq ft) | Slabs, wall elevations, floor and ceiling finishes, roof coverings, excavation footprints |
| Linear  | m (or ft)                 | Walls on plan, beams, skirting, kerbs, pipe and cable runs, railings                      |
| Segment | m (or ft)                 | Single straight lengths where you want each leg recorded on its own                       |

| Kind  | Gives you | Use it for                                                        |
|-------|-----------|-------------------------------------------------------------------|
| Count | Nr        | Columns, doors, windows, sanitary fittings, light points, sockets |

## Measuring an item

1. Open the page and check it is scaled.
2. In the template library, expand down to the item you want, for example FINISHES, then FLOOR FINISH, then the floor tile template.
3. Start the takeoff from that template (in PlanSwift, click the button that appears beside the template).
4. Name the item after the room or element, for example "Offices" or "Sick bay", and adjust anything the template asks for, such as thickness or height. The name is what appears in your bill, so type it properly.
5. Draw the measurement on the plan, then finish it.
6. Check the running total in PlanSwift's takeoff summary.

### TIP

Measure room by room, not one big shape per floor. Five separately named areas cost no more effort than one, and they give you a bill you can defend line by line.

## Organising the takeoff so it bills well

HERON reads your PlanSwift folders literally: each folder becomes a section of your bill, and item names become descriptions.

- **By storey:** Admin Sub, Admin GF, Admin FF, Admin RF.
- **By block:** prefix everything, for example Admin and Hostel, so you can export one block at a time.
- **By package:** Admin Finishes, Admin Openings, where a trade is let separately.

Some names carry meaning for HERON:

| Name it like this                    | Because                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------|
| 225 Wall Area, 150 Wall Area         | The leading figure is the wall thickness in millimetres, used for blocks, mortar and rendering. |
| Concrete in Column, Concrete in slab | The element type drives the concrete, formwork and reinforcement breakdown.                     |
| UKB 305x165x40, SHS 100x100x5        | Steel section sizes are read by <b>Steel Tonnage</b> and converted to tonnes.                   |

### IMPORTANT

Avoid two items with the same name inside one folder. HERON merges them when it reads the job. If they are different, say so in the name or put them in different folders.

## Quantity Take Off

This is where your PlanSwift measurements become billable quantities. Click **Quantity Take Off** in the sidebar.

The screen has three panels:

- **Take Off Folders:** the folders in your PlanSwift job.
- **Take Off Items:** the items in the selected folder, with **Name**, **Qty** and **Unit**.
- **Take Off Breakdown:** what the selected item resolves into, such as concrete, formwork, reinforcement and other sub-quantities.

Click **Reload** at the top whenever you change something in PlanSwift: a new measurement, a renamed folder or a corrected scale. HERON reads a snapshot of the job, and **Reload** refreshes it.

#### TIP

Reload fixes most surprises. If PlanSwift was restarted while HERON was open, **Reload** also reconnects to it.

## Editing an item's parameters

1. Double-click an item in **Take Off Items**.
2. A detail window opens, with one block for each measured instance of that item (each **Instance** shows its size and parameters).
3. Change the parameters the item offers, such as thickness, depth, height or spacing.
4. Close the window. The breakdown recalculates.

## Adding reinforcement

For concrete items, the detail window has a **Reinforcement** section on each instance.

1. Click **+ Bars** for main and distribution bars, or **+ Links** for stirrups and links.
2. Enter **Dia:** (bar diameter in mm), **Nr:** (number of bars) and **Spc:** (spacing). HERON works out the weight in kg.
3. Use the **×** on a row to remove it.
4. Click **Apply Reinforcement** to keep your changes, or **Cancel** to discard them.

## Roofs

Roof items work from the plan area and the pitch, so you measure the plan, not the slope. If HERON shows **Invalid Roof Area**, or **Pitch value must be greater than zero**., check the roof measurement and its pitch in PlanSwift, then click **Reload**.

## Pricing and saving the bill

When your takeoff is complete, click **Save / Export** at the bottom of **Quantity Take Off**. Nothing is saved yet. HERON opens the **Review Takeoff Before Saving** screen, where you check descriptions, set rates and order the bill.

### What is on the review screen

| Part       | What it does                                                                                                                                           |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status bar | How many items are ready, plus the rate status. With Rate Gen it also shows <b>Rates:</b> matched out of total, and the <b>Total</b> in your currency. |

| Part                                                                   | What it does                                                                                                                             |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name:</b>                                                   | The name the project is saved under in the cloud. Put the client or job name here.                                                       |
| <b>Search Rate:</b>                                                    | Search your Rate Gen library and apply a rate to the selected row (Rate Gen accounts only).                                              |
| <b>Keep edited rates (uncheck to use the latest from your library)</b> | When ticked, the rates you typed come back next time you open this review. When unticked, HERON uses the latest rates from your library. |
| The bill table                                                         | <b>Item, Description, Qty, Unit, Rate and Amount</b> , grouped into sections by folder.                                                  |

## Loading rates

Rates are never loaded on their own. The bill opens with every rate at 0, so you decide when a job is priced.

1. Click **Load rates** at the bottom of the review screen. HERON prices the bill from your Rate Gen library and shows how many items matched.
2. The button then reads **Reload rates**. Click it again whenever you want to pull the latest rates.

Loading rates on the bill also prices the **Budget** behind it, so the two stay in step.

### NOTE

**Load rates** and **Search Rate**: appear only if your account has an active Rate Gen subscription. Without Rate Gen you can still type rates by hand.

## Editing the bill

| To do this                    | Do this                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Change a description          | Click the <b>Description</b> cell and type. Section headings and sub-items are locked.                                                               |
| Set a rate by hand            | Type into the <b>Rate</b> cell. <b>Amount</b> updates when you leave the cell.                                                                       |
| Pick a rate from your library | Select the row, type in <b>Search Rate</b> :, then double-click a result.                                                                            |
| Reorder the bill              | Select a row and click <b>Up</b> or <b>Down</b> , or drag it. The order you set is the order it exports in.                                          |
| Remove a line                 | Click the × at the end of the row. The line comes off this bill and is hidden from the <b>Budget</b> too. Your PlanSwift measurement is not touched. |

### IMPORTANT

A row shaded amber means its rate is lower than the material plus labour cost HERON worked out for it. You can still save. HERON lists those items when you save so you can raise the rates when it suits you.

## Saving to the cloud

1. Check the **Project Name**..
2. Click **Save to Cloud**.

3. If a project with that name already exists on your account and you did not open it from the Dashboard, HERON tells you **Project name already exists**. Choose a different name, or open the existing project from the Dashboard to update it.
4. Wait for **Takeoff saved to cloud!** with the project name. The bill and its material and labour budget are saved together.

While a save is running you can click **Cancel Save**. If the bill saves but the budget does not, HERON tells you so rather than reporting success. Save again once your connection is steady.

Each save creates a new version of the project, so re-saving never wipes out an earlier one. The project then appears under **Recent Projects** on the Dashboard on any PC you sign in from, and on the website.

#### NOTE

If your session is offline, HERON explains that the takeoff cannot be saved to the cloud. Your rows stay in the review window and **Export All** still works. Sign out, sign in again while connected, then save.

## When the contract is locked

A project can be locked on the website once its bill becomes the contract. HERON then shows a red **CONTRACT LOCKED** chip on the review screen and on the **Budget**. You can still save. Your revised quantities and rates are recorded as actuals against the contract, and the contract sum does not change. The save confirmation says so.

## Exporting to Excel

You can export from the review screen at any time, even offline.

- **Export All** writes every folder to one workbook.
- **Export Folder(s)** opens **Select Folders to Export**. Tick the folders you want and click **Export Selected**. Export one folder for a single worksheet, or several for a combined workbook.

Choose where to save the file. HERON confirms when the workbook is written.

The workbook is laid out as a proper bill of quantities:

- One sheet per folder, with **ITEM**, **DESCRIPTION**, **QTY**, **UNIT**, **RATE** and **AMOUNT** columns.
- QS item lettering (A, B, C and so on), a bold section heading and wrapped descriptions.
- Sub-items shown under their parent line.
- A **GEN SUMMARY** sheet that links to each folder's total.

It is an ordinary .xlsx file, so you can restyle it or paste it into your own template.

#### TIP

After an export, HERON may show a one-line note of how long the takeoff took. Click **Dismiss** to hide it. See [Settings](#) to turn the Takeoff Time Log off.

## The Budget: material and labour

Where the bill answers "what will I charge?", the **Budget** answers "what will it cost me?". Click **Budget** in the sidebar.

1. Pick a folder under **Take-Off Folders** on the left. HERON builds the budget for every item in it: concrete, cement, sand, granite, blocks, reinforcement, formwork, labour and more, worked out from built-in QS recipes and your material constants.
2. Click **Load rates** to price the budget from your Rate Gen library. Until then every price is 0. The button then reads **Reload rates**.
3. Check each item. Under it you see **Material / Labour, Qty, Unit, Price** and **Amount**, with **Material, Labour** and **Total** for the item.

The strip at the top shows the whole project at a glance: **Project cost (material + labour)**, **Take-off value (BoQ)**, **Overhead + profit** and **Margin** as a percentage.

### Working with the budget

| Button or action            | What it does                                                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type in a <b>Price</b> cell | Type a price, or type a material or labour name to search Rate Gen and pick a rate. Totals and margins update as you type.                                 |
| <b>Recompute</b>            | Rebuilds the budget, for example after you change quantities in PlanSwift.                                                                                 |
| <b>BoQ rate ← budget</b>    | Sets each item's bill rate from its budget: (material + labour) divided by quantity, plus the overhead and profit percentage from your material constants. |
| <b>Export</b>               | Saves the budget as a linked Excel workbook (see below).                                                                                                   |
| × on an item                | Hides the item from the budget. Your PlanSwift takeoff is not changed.                                                                                     |
| <b>Restore hidden</b>       | Appears when you have hidden items. Brings them back.                                                                                                      |

An item whose material and labour cost is higher than its rate shows **Over budget**. Raise the rate on the review screen, or use **BoQ rate ← budget**.

### The budget workbook

**Export** writes a workbook where everything except the measured quantities and your typed prices is a live formula:

- A **Basic Prices** sheet, the one place each material price is typed, plus an overhead and profit percentage cell.
- A budget sheet for each folder, with material and labour totals and the rate they imply.
- A bill sheet for each folder whose rates link to that budget.
- A **BUDGET SUMMARY** sheet with material, labour, budget, overhead and profit, and contract total for each folder.

Change a basic price in Excel and every rate and margin that depends on it recalculates.

#### NOTE

The budget saves to the cloud together with the bill when you click **Save to Cloud** on the review screen. There is no separate budget save.

## Steel Tonnage

Steel is measured in metres but bought in tonnes. **Steel Tonnage** reads the section size from your item names and does the conversion.

1. Click **Steel Tonnage** in the sidebar.
2. Set **Connection/fittings %** (default 7.5) for cleats, plates, bolts and welds, and **Waste %** (default 0) for offcuts.
3. Click **Scan Job**.
4. Check the table: **Item, Trade, Detected Section, kg/m, Length (m), Net (t), Allowance (t) and Gross (t)**.
5. For any row where no section was found, use **Assign Section** and type to search the catalogue (UKB, UKC, PFC, UKA, SHS, RHS and CHS), or type a kg/m yourself. Your choice is kept when you scan again.
6. Read the job totals at the bottom: **NET, ALLOWANCE and GROSS**.

Roof and truss members (rafters, purlins, tie beams, king posts, struts, chords and bracing) are picked up even when their names carry no section size, so you can assign a section to them.

Where a column was measured as a count rather than a length, use the **Count × Ht?** column. The table shows the tonnage both ways, **Gross if metres (t)** and **Gross if count×ht (t)**, so you can use the one that matches how you measured.

Click **Clear** to empty the table and start again.

#### TIP

Name steel items with the full designation, for example UKB 305x165x40. "Steel beam" cannot be read.

## Settings

Click the gear at the top right to open **Settings**.

### Display Units

Choose **Metric (m, m<sup>2</sup>, m<sup>3</sup>, mm, kg)** or **Imperial (ft, sqft, cuft, in, lb)**. HERON normally picks this for you from your PlanSwift pages. Change it here if it picked wrong.

### Takeoff Time Log

HERON records how long each takeoff takes and how many items, folders and bill lines it produced, so ADLM can report the time its tools save. Only timings and counts are sent. Drawing content, item names, quantities and prices are never sent. Untick **Share takeoff timings with ADLM** to turn it off.

### Material Constants

The constants HERON uses to build your breakdowns and budgets: cement per cubic metre, blocks per square metre, mortar ratios, wastage, labour outputs, overhead and profit and so on.

1. Search by name, category or key to find a constant.
2. Edit its **Value**. The **Default** column shows the standard figure.
3. Click **Save**. HERON confirms **Material constants saved and applied**.
4. Click **Reset Defaults** to go back to the standard figures.

Set these once to match how you price, and every breakdown and budget follows.

## The Excel Takeoff Link

If your bill lives in your own spreadsheet, the Excel Takeoff Link pulls your takeoff, materials and rates straight into it and keeps them up to date. No copy and paste, and no re-export when a quantity changes.

### Turning it on

1. Close Excel completely.
2. In HERON, click **Connect Excel** in the sidebar. HERON checks the add-in.
3. If the add-in is missing, switched off or broken, HERON tells you and offers to install or repair it. Click **Yes**.
4. When HERON reports that the Excel bridge is ready, open Excel and open (or create) a workbook.
5. On the **ADLM** tab of the Excel ribbon, click **Takeoff Link**. The **ADLM Takeoff Link** pane opens.

#### NOTE

HERON must be open, signed in and showing your project for the pane to find data. If the pane says no data was found, open HERON, load the job, then click **Refresh** in the pane.

### Using the pane

At the top, pick a dataset:

| Dataset                 | What it lists                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Quantity Takeoff</b> | Your billable takeoff: description, quantity, unit and level.                                       |
| <b>Material Takeoff</b> | The full material and labour breakdown from your <b>Budget</b> , with unit rate and amount.         |
| <b>Rates</b>            | Each bill item's <b>Description</b> , <b>Unit</b> , <b>Rate</b> and <b>Amount</b> from your budget. |

Type in **Search** to filter the list. Item names keep their original takeoff name in front, so you can still search for DPM or Topsoil after HERON has expanded them into full descriptions.

To link figures into your sheet:

1. Click the cell in your workbook where the figure should go.
2. Select one or more rows in the pane.
3. Click **Link Cell**. The cell now holds the linked value (several rows are added together).
4. To add more rows to the same cell as a running sum, select them and click **Add to Cell**.
5. To take rows back out, select them and click **Remove from Cell**. The cell re-totals.
6. To break the link on a cell, click it and click **Unlink Cell**.

The top of the pane shows how many cells and items are linked in this workbook.

## Keeping figures up to date

Change a quantity or rate in HERON, switch back to Excel, and the linked cells update on their own. Cells that did not change are left alone, so your Undo history is kept. You can also click **Refresh** in the pane, or **Refresh** on the **ADLM** ribbon tab, at any time.

### TIP

If the **ADLM** tab ever disappears from Excel (for example after an Office repair), close Excel and click **Connect Excel** in HERON. You can also double-click `Register-ExcelAddin.cmd` in `C:\ProgramData\Planswift Plugin`, then reopen Excel.

## Your HERON projects on the ADLM website

Everything you save to the cloud from HERON appears on the ADLM website under your HERON projects ( `/projects/planswift` , or **HERON** in the side rail once you are signed in). Anyone on your account can read the bill there without PlanSwift. Opening a project on the website gives you the bill grouped by section, the budget beside it, Excel exports and sharing.

## Learning samples

Your HERON projects page shows a **Learning samples** strip: worked duplex projects, one per foundation type, measured from PDF drawings. Open one to see every tab filled in. Samples are read-only. Click **Hide samples** to fold the strip away, and **Show N samples** to bring it back.

## Importing a bill from Excel

Sometimes the bill comes first: a client or another QS sends you their bill in Excel, and you need to price it or put real measurements behind it. Excel bill import is a HERON feature, done on the ADLM website, and then finished in HERON with **Build in PlanSwift**.

### NOTE

Excel bill import is switched on for your account by ADLM. It is not bought on its own, and it also needs a live HERON subscription. If you do not see **Import Excel BoQ · HERON** on your HERON projects page, contact ADLM support through [/support](#). There is no import button inside HERON itself.

### Step 1: Import the bill on the website

1. Sign in to the ADLM website and open your HERON projects ( `/projects/planswift` ).
2. Click **Import Excel BoQ · HERON**.
3. Optionally type a name in **Project name (optional)**. If you leave it blank, the file name is used.
4. Under **Excel workbook (.xlsx)**, choose your bill (.xlsx or .xlsm).
5. Click **Import project**.

The website reads the sections and lines of your bill and creates a HERON project from it. Where the workbook has no material and labour schedule, one is built for you (cement, sand, granite, blocks, formwork, rebar and labour), priced from your material constants and Rate Gen.

#### TIP

Click **Download the import template** if you want a workbook laid out the way the importer reads best.

#### IMPORTANT

Upload your own bill, not one exported from the ADLM website. An ADLM export is refused, because importing it would spend a project slot on a copy of a bill you already have.

### Step 2: Build the bill in PlanSwift

1. Open the matching job in PlanSwift.
2. In HERON, go to the **Dashboard** and find the imported project under **Recent Projects**.
3. Click **Build in PlanSwift** on the project card. (You can also click **Open** and answer **Yes** when HERON asks whether to build the bill now.)
4. The **Build Bill in PlanSwift** window lists every bill line. Each work section becomes a take-off folder and each bill line becomes a measurable item.
5. Check the **Template** chosen for each line and change it where the match is wrong. Lines with no match are built as plain measurable items. Where the template needs them, check **Measured as**, **Depth/Height**, **Width** and **Thickness**.
6. Choose which lines to build with **Select all**, **Select none** or **Only lines not in the job** (useful on a second run).
7. Click **Build in PlanSwift**.

HERON confirms what it built and opens **Quantity Take Off**. In PlanSwift, the new folders appear in the takeoff summary. If they do not show yet, refresh that panel in PlanSwift.

### What the build will and will not touch

- Lines that are missing are created.
- Lines already in the job are updated, never duplicated. Building again is safe.
- Lines that already have a measurement keep it. HERON never overwrites a measurement.
- Section headings become folders, not items.
- Costing rows (derived materials and labour) are skipped, because there is nothing on a drawing to measure for them.

### Step 3: Measure and save

1. Scale the drawings, exactly as for any job. The structure arrived ready-made, but scaling is still the step everything depends on.
2. Click **Measure** on a line in the **Build Bill in PlanSwift** window to send that item to PlanSwift ready for you to trace it on the drawing, or measure the items from the takeoff summary in PlanSwift.
3. Go to **Quantity Take Off**, click **Reload**, then **Save / Export** and **Save to Cloud**. Your measurements land on the same bill lines that came from the Excel file.

## What's new in 2.9.6

### HERON 2.9.6

- **The ADLM templates stay put when HERON starts.** Opening HERON with a saved sign-in could hide the ADLM template library from PlanSwift. It no longer does.

### Earlier 2.9 updates

- **Network signal in the header.** Bars show the quality of your connection to the ADLM Cloud, and tell you whether you are offline or the server is not answering.
- **Takeoff Time Log.** HERON records how long each takeoff takes (timings and counts only) and can show the time after an export. You can turn it off in **Settings**.
- **Rates load only when you ask.** The bill and the budget open with every rate at 0. Click **Load rates** to price them from your library, and **Reload rates** to refresh.
- **Below-cost items no longer block a save.** Items whose cost is higher than their rate are shaded amber and listed when you save, but the takeoff always saves.
- **Build in PlanSwift.** Turn a bill from your ADLM Cloud projects into take-off folders and measurable items in the open PlanSwift job.
- **Excel bill import moved to the website.** Import an Excel bill from your HERON projects page on the ADLM website, then build it in PlanSwift from HERON.
- **The Excel Takeoff Link is ready on install (2.9.2).** The add-in is registered as part of the install, so the **ADLM** tab is there the first time you open Excel.
- **Sign-in that follows the service (2.9.1).** HERON finds the ADLM server through a setting the Installer Hub writes, so a server move never needs a new download.

## Troubleshooting

### The ADLM templates are not in PlanSwift

Work through these in order:

1. Make sure you are signed in to HERON.
2. Close PlanSwift completely and open it again. PlanSwift only reads its templates when it starts.
3. In PlanSwift's templates panel, check the list at the top reads **Complete ADLM TakeOff Plugin**.
4. If it is still missing, sign out of HERON, sign in again, and restart PlanSwift. Signing in puts the templates back from the protected copy.
5. If you run two PlanSwift versions, check you have the one open that HERON installed into. If not, reinstall HERON from the Installer Hub with PlanSwift closed.

### The templates disappeared after I signed out

That is expected. Signing out hides them at the next PlanSwift restart. Sign in again and restart PlanSwift.

### "Could not retrieve the Job from PlanSwift."

No job is open in PlanSwift, or PlanSwift started after HERON. Open the job, then click **Reload** in **Quantity Take Off**.

### "PlanSwift connection lost. Please restart PlanSwift and click Reload."

PlanSwift closed or crashed while HERON was reading it. Restart PlanSwift, open the job, then click **Reload**. Nothing is lost: your takeoff lives in the PlanSwift job.

## HERON says the PlanSwift installation folder was not found

HERON could not find PlanSwift on this PC, so it cannot start. Install PlanSwift 10 or 11 (or repair it), then reinstall HERON from the Installer Hub.

## Quantities are far too big or too small

Almost always the scale. Check the page was scaled, check it with the Dimension tool, scale again if needed, then click **Reload** in HERON. If everything is out by about the same percentage, the page was scaled off a wrong dimension.

## HERON shows imperial units on a metric job

No page carried a usable Scale Units value, so HERON fell back. Scale your pages properly, or open **Settings** and pick **Metric** (m, m<sup>2</sup>, m<sup>3</sup>, mm, kg) under **Display Units**.

## An item is missing from Quantity Take Off

Click **Reload**. If it is still missing, check for a second item with the same name in the same folder: HERON merges duplicates.

## No rates loaded, or Load rates is missing

- **Load rates** appears only with an active Rate Gen subscription. The status line tells you if no Rate Gen subscription was found.
- You must be online and fully signed in (not in an offline session). Check the network signal in the header.
- If HERON reports that no rates matched, add or check the rates in Rate Gen, then click **Reload rates**.

## Save to Cloud refuses to save

- **Offline session:** sign out, then sign in again while connected.
- **Project name already exists:** choose another name, or open the existing project from the Dashboard and save from there.
- **Version conflict:** the project was changed on another device. Open it again from the Dashboard and save again.

## "This PlanSwift subscription is bound to a different device."

Your licence is tied to another PC. Contact ADLM support through [/support](#) to have the device lock reset.

## "Internet required for first sign-in (no valid offline license found)."

The first sign-in on a PC must be online. Connect to the internet and sign in again.

## "Your subscription has expired." or "No active PlanSwift subscription was found on this account."

Renew your HERON subscription from your account on the ADLM website ( [/dashboard](#) ), or check you signed in with the account that holds the licence.

## The ADLM tab or the Takeoff Link pane is missing from Excel

1. Close Excel completely.

2. In HERON, click **Connect Excel** and let it install or repair the add-in.
3. Open Excel and a workbook, then click **Takeoff Link** on the **ADLM** tab.

If that does not work, run `Register-ExcelAddin.cmd` from `C:\ProgramData\Planswift Plugin`, or reinstall HERON from the Installer Hub.

### The Takeoff Link pane shows no data

HERON must be open with your project loaded. Open HERON, open or start the project, then click **Refresh** in the pane.

### Build in PlanSwift says there is nothing to build

The cloud project has only section headings or costing rows, with no measurable bill lines. Check the bill on the website first. If HERON says no PlanSwift job is open, open the job in PlanSwift and try again.

### The Install button in the Hub fails with access denied

PlanSwift (or Excel) was open while the Hub was installing. Close both and click **Install** again.

## Frequently asked questions

### Does HERON replace PlanSwift?

No. You still need PlanSwift 10 or 11. You measure in PlanSwift with the ADLM templates, and HERON reads, prices and exports what you measured.

### Do I need Rate Gen?

Only for pricing from a rate library. Without Rate Gen you can measure, review, type rates by hand, export to Excel and save to the cloud.

### Can I work without the internet?

Yes, once you have signed in online at least once on that PC. Measuring, the takeoff, the review screen and Excel export all work offline. Loading rates and saving to the cloud need a connection and a full sign-in.

### Where is my work stored?

Your measurements live in the PlanSwift job file. Back that file up. HERON itself can always be reinstalled from the Hub. Projects you save to the cloud are also kept on your ADLM account, with a new version on every save.

### Can I use HERON on two computers?

Your licence is tied to one PC at a time. To move it, ask ADLM support to reset the device lock, then sign in on the new PC.

### Can I import an Excel bill from inside HERON?

No. Excel bill import happens on the ADLM website, on your HERON projects page, and only if ADLM has switched it on for your account. You then use **Build in PlanSwift** in HERON to turn that bill into measurable items.

### Does removing a line on the review screen delete my measurement?

No. The ✕ on the review screen only removes the line from that bill and hides it from the **Budget**. Hiding an item in the **Budget** also leaves PlanSwift untouched. To change the takeoff itself, edit it in PlanSwift and click **Reload**.

### Does HERON send my drawings to ADLM?

No. The Takeoff Time Log sends only timings and counts, and you can turn it off in **Settings**. A cloud save sends the bill and budget lines you chose to save.

### How do I get help?

Click **Feedback** at the bottom of the HERON sidebar to message ADLM support on WhatsApp, or raise a ticket at [/support](#). Tell us the job name, the folder and item affected, whether the page is scaled, the exact wording of any message and your PlanSwift version. A screenshot of PlanSwift's takeoff summary and HERON's **Quantity Take Off** answers most questions at once.



# ADLM Rate Gen

Build defensible rates from material, labour and plant prices for your location, keep your own library in step with the cloud, and price QUIV, HERON and ADLM Cloud projects from it.

ADLM Rate Gen is ADLM's rate build-up tool for quantity surveyors and estimators. It holds a library of material prices, labour day rates and plant hire rates, priced for the part of Nigeria you work in, and uses them to build up a rate for every item of work: ground works, concrete, blockwork, finishes, roofs, painting, steel, windows and doors, and more. Every rate shows exactly what it is made of, so you can check it, change it and defend it. You can also build rates of your own. Your library lives on your ADLM account, so QUIV, HERON and your ADLM Cloud projects all price from the same rates. Rate Gen comes as a Windows desktop app and as RateGen pages on the ADLM website. This guide covers Rate Gen 2.9.0, the version the Installer Hub installs today, and the website as it is now.

## Before you start

### What you need

| You need                                     | Notes                                                                                                                     |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| A Windows PC                                 | The Rate Gen desktop app runs on Windows.                                                                                 |
| An ADLM account with a Rate Gen subscription | The same email and password you use on the ADLM website. See <a href="#">Getting started</a> to buy one.                  |
| A pricing location on your profile           | Your state decides which regional prices you get. See <a href="#">Your location and price zones</a> .                     |
| An internet connection                       | Needed for your first sign-in and for every sync. Once you have signed in online, you can open Rate Gen and work offline. |
| A web browser (optional)                     | For the RateGen pages on adlmstudio.net.                                                                                  |

### Installing Rate Gen

Rate Gen is installed from the ADLM Installer Hub, never from a loose download. See [ADLM Installer Hub](#) for the Hub itself.

1. Open the Installer Hub and sign in.
2. Find **ADLM RateGen** and click **Install** (or **Update** if you already have an older version).
3. Accept the Windows permission prompt. The Hub needs it to install into the shared programs folder.
4. Wait for the Hub to report that the install has finished.

The Hub installs Rate Gen into `C:\ProgramData\ADLM RateGen` and puts an **ADLM RateGen** shortcut on the desktop.

#### NOTE

Updating keeps your data. Your library, your saved rates and your edits are stored in your own Windows profile, not in the program folder, and an update never touches them.

## The order to do things in

You only do steps 1 to 3 once.

1. Set your **Pricing location (State)** on your profile on the ADLM website.
2. Install Rate Gen from the Hub.
3. Open Rate Gen and sign in. Rate Gen loads the prices for your location.
4. Open a work section, check the rates, and change any build-up that does not match how you work.
5. Build custom rates for anything the library does not cover.
6. Price your projects in HERON or ADLM Cloud from the same library.

## Signing in

1. Open **ADLM RateGen** from the desktop shortcut.
2. Type your email or username in **Email or username**.
3. Type your password in **Password**.
4. Click **Log in**, or press  in either box.

Rate Gen checks your subscription, then loads the latest prices for your location. You will see **Sign in successful!** If your profile's location is different from the one Rate Gen last used, it asks **Update your RateGen prices to this zone now?** Click **Yes** to load that location's prices.

The sign-in screen has two links:

- **Forgot password?** opens the ADLM website, where you can reset your password. There is no password reset inside Rate Gen.
- **Create account** opens the sign-up page on the ADLM website.

## Working offline

After one successful online sign-in on a PC, Rate Gen can sign you in without the internet. You will see **Signed in (offline) via cached license**. Everything you have on the PC works offline. Syncing, loading new prices and sending your edits to the cloud wait until you are back online.

## Signing out

Click **Log out** at the bottom of the sidebar. Before it signs you out, Rate Gen sends any edits it has not yet synced to the cloud. You do not need to sign out at the end of each day. Just close the window.

## A tour of the Rate Gen window

### The sidebar

The sidebar on the left takes you around the app. Click the button at the top to fold it to icons and back, or drag its edge to resize it.

| Entry   | What it opens                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------|
| Library | Your material and labour library, with plant in the labour list. See <a href="#">The material and labour libraries</a> . |

| Entry                      | What it opens                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Item of works</i>       | A heading over the work sections below. Each section is a list of built-up rates.                                                                                       |
| <b>Ground</b>              | Ground works: excavation, filling, hardcore and similar.                                                                                                                |
| <b>Concrete</b>            | Concrete works.                                                                                                                                                         |
| <b>Block Works</b>         | Blockwork and walling.                                                                                                                                                  |
| <b>Finishes</b>            | Floor, wall and ceiling finishes.                                                                                                                                       |
| <b>Roofs</b>               | Roof coverings and roof carpentry.                                                                                                                                      |
| <b>Painting</b>            | Painting and decoration.                                                                                                                                                |
| <b>Steel</b>               | Structural steelwork.                                                                                                                                                   |
| <b>Window and Door</b>     | Windows, doors and ironmongery.                                                                                                                                         |
| <b>Saved Rates</b>         | The custom rates you have built. See <a href="#">Custom rates</a> .                                                                                                     |
| <b>Carbon &amp; Others</b> | Further rates, including mechanical, electrical and plumbing items, HVAC and fire protection. These reach you from ADLM Cloud on your next sync, without an app update. |
| <b>Export</b>              | Exports every section and your saved rates to Excel. See <a href="#">Exporting your rates</a> .                                                                         |
| <b>Help</b>                | Opens an email to ADLM support with your account email filled in.                                                                                                       |
| <b>Log out</b>             | Signs you out.                                                                                                                                                          |

## The header

| Control                     | What it does                                                                                                                                         |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Search here...</b>       | Searches across every section, including the mechanical, electrical and plumbing build-ups. Type part of a rate's name and pick a result to open it. |
| <b>Currency</b>             | Shows prices in NGN, USD, EUR, QAR, GHS or ZAR. This changes how prices are shown only. Your library is held in naira.                               |
| <b>Sync from Cloud</b>      | Fetches the latest rates and prices for your location and sends your edits up. See <a href="#">Syncing with the cloud</a> .                          |
| <b>Banner arrow</b>         | Folds the welcome banner away so the rate tables get more room. Rate Gen remembers your choice.                                                      |
| <b>Light or dark</b>        | Switches between the light and dark theme.                                                                                                           |
| <b>Reset all rate edits</b> | Puts every build-up you have edited, in every section, back to the shipped quantities.                                                               |
| <b>Undo library changes</b> | Restores an earlier copy of your material and labour libraries. See <a href="#">Undoing library changes</a> .                                        |
| <b>Bell</b>                 | Notifications: sync results, new published prices and updates from ADLM.                                                                             |
| <b>Profile</b>              | Opens your ADLM profile on the website, where you set your pricing location.                                                                         |

## Your location and price zones

Materials and labour cost different amounts in different parts of the country, so Rate Gen prices your library for your location.

- **You choose a state.** Your profile on the ADLM website asks for your **Pricing location (State)**: any of the 36 states or the FCT. See "Your profile and WhatsApp number" in [Getting started](#).
- **Prices come from the zone your state is in.** ADLM prices six geopolitical zones: South West, South East, South South, North Central, North East and North West. Kano and Katsina read the same today because both are North West. When ADLM has evidence for a single state, that state can be priced on its own without you doing anything.
- **There is one place to set it.** Rate Gen has no location picker of its own. You change your state on the website, and Rate Gen picks it up the next time you sign in or sync.

The library tells you where its prices came from, for example **Lagos: priced from south west rates. Set this on your ADLM profile.**

### TIP

If your rates look wrong for where you are, check the location line at the top of the library first. Then check the state on your profile.

## The material and labour libraries

Click **Library** in the sidebar. The library has two tabs, **Material** and **Labour**. Every rate in every section is priced from these two lists, so a price you change here flows into every rate that uses it.

### Material

The **Material** tab lists every material with its **Material Name**, **Material Unit**, **Price** and **Category**. Items with a size are sorted by size, so 12mm comes before 100mm.

### Labour and plant

The **Labour** tab lists trades and plant together, with **Labour Item**, **Labour Unit**, **Price** and **Category**. Labour is priced as a day rate. Plant (excavators, dozers, rollers, mixers, generators, cranes and so on) is in the same list, filed under its own categories such as earthmoving, compaction, lifting, haulage and power generation. There are 24 categories in all, so the **Category** filter narrows the list properly.

Open any labour or plant item to see a **Specification & expected output** panel. It says what the item is, what it typically produces in a day or an hour, what that output assumes, and what it burns in fuel or who staffs it. Outputs are given as ranges because soil, haul distance and gang size move them a long way.

### NOTE

The expected output is a sense check on your rate, not a guarantee. It does not feed any calculation.

## Finding an item

- Type in the search box above the table.
- Pick a **Category** to narrow the list.

- Click **Sort by** to change the order, or click a column heading.

## Changing a price

1. Find the item and click its **Edit** button.
2. Type your price.
3. Click **Update Library**.

Every rate that uses the item is priced again straight away. Your price stays yours: a sync or a sign-in never overwrites it.

## Adding an item

1. On the **Material** tab click **Add Material +**, or on the **Labour** tab click **Add Labour +**.
2. Fill in the name, unit, price and category.
3. For labour and plant you can also write a **Specification / notes (optional)**: what the item is and what it produces in a day. Your note shows beside the rate on the website too.
4. Click **Save to Database**.

To remove an item you added, click its **Delete** button. Only items you added can be deleted. Items that came with the library cannot be removed, but you can change their price.

## Fetching the latest prices

**Update Prices** fetches the latest published prices for your location. Prices you have set yourself are kept.

## When ADLM publishes a new price for something you changed

If ADLM publishes a new price for an item you have priced yourself, your price stays in use and a review card appears above the library. Each row shows the **Item**, **Unit**, **Your price**, the **Published** price and the **Change** between them.

1. Tick the rows you want to decide on, or use the box in the header to tick them all.
2. Click **Keep mine** to keep your figures and stop being asked about those rows, or **Use published** to take the published price.
3. Rows you leave unticked stay pending, and the card stays up with what is left.

Closing the card hides it without deciding. Reopen it from the bell: the notification has a **Review** button. If ADLM has not changed the price of an item you edited, you are never asked.

## Undoing library changes

Rate Gen copies your material and labour libraries before anything rewrites them: a sign-in, a sync, a change of location, or taking published prices. It keeps the last 20 copies.

1. Click the undo library changes button in the header.
2. Under **Restore an earlier library**, pick the point to go back to.
3. Click **Restore** and confirm.

Your current library is copied first, so a restore can itself be undone.

## Rates by section

Each work section (**Ground, Concrete, Block Works** and the rest) is a table of built-up rates:

| Column      | Meaning                                                                  |
|-------------|--------------------------------------------------------------------------|
| S/N         | The item number.                                                         |
| Description | The item of work. Click it to open the build-up.                         |
| Unit        | The unit the rate is for, such as m <sup>2</sup> , m <sup>3</sup> or nr. |
| Net Cost    | Materials, labour and plant for one unit.                                |
| Overhead    | The overhead added to the net cost.                                      |
| Profit      | The profit added to the net cost.                                        |
| Total Cost  | The rate: net cost plus overhead plus profit.                            |

Above the table:

- **Search description...** filters the rows.
- **Filter** orders the rows by net cost.
- **Sort by** cycles between the original order, overhead and total cost.
- **Overhead** and **Profit** boxes set the percentages for the whole section. See [Overheads and profit](#).
- **Add Custom Rate +** opens the custom rate form. See [Custom rates](#).

Every rate is priced from your library, so when you change a price in the library the rates follow.

### NOTE

The mechanical, electrical and plumbing rates say whether a library item is (supply) or (installed). A supply price is the material only, so allow for fixing. An installed price already includes fixing, so do not add a labour line on top.

## Reading and changing a build-up

Click a rate's **Description** to open its **Rate Composition**. Each line is a component: a material, a labour gang or an item of plant.

| Column     | Meaning                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------|
| Component  | What the line is. Click a component's name to open it in the library, ready to check or change its price. |
| Quantity   | How much of it one unit of work uses.                                                                     |
| Unit       | The component's unit.                                                                                     |
| Unit Price | Its price from your library.                                                                              |
| Total Cost | Quantity times unit price.                                                                                |

Totals, sub-totals and waste allowances are arithmetic, not library items, so they are not links.

## Changing quantities

When a rate is wrong for how you work, it is usually one quantity: a different mix, a different spacing, a different gang output.

1. Double-click a **Quantity** cell and type the new figure.
2. Click away from the cell. Sub-totals refresh straight away.
3. Click **Save**. You will see **Rate saved successfully. Your edits will sync to QUIV and HERON.**

The other buttons:

- **Cancel** throws away edits you have not saved and closes the build-up.
- **Reset This Rate** puts that item's quantities back to the shipped defaults, and sends the reset to the cloud.

To reset every edited rate in every section at once, click the reset all rate edits button in the header. It asks first, and it cannot be undone.

### IMPORTANT

Change prices in the library, and quantities in the build-up. A price lives in one place and is shared by every rate that uses it. A quantity belongs to one rate.

If you are offline when you save, you will see **Cloud Sync Pending**. Your edit is saved on the PC and goes to the cloud the next time you are connected.

## Overheads and profit

A rate is built in two steps:

**Net cost = materials + labour + plant**

**Rate = net cost + overhead + profit**

Overhead and profit are each a percentage of the net cost.

- **In a work section**, the **Overhead** and **Profit** boxes above the table set the percentages for every rate in that section. They start at 10% overhead and 25% profit. Type new figures and every row is worked out again.
- **In a custom rate**, you set the percentages for that rate alone. They start at 10% and 10%.
- **On the website**, a custom rate where you leave the boxes empty uses 10% and 10%.

### TIP

Set overhead and profit to match how your firm prices before you rely on the section totals.

## Custom rates

A custom rate is a rate you build yourself, for an item the library does not cover or one you price your own way. Custom rates are saved to your account, and HERON and ADLM Cloud price from them like any other rate.

## Building a custom rate in the desktop app

1. Click **Add Custom Rate +** in any section. The **Build New Custom Rate** form opens.
2. Type the **Name of Rate**.
3. Set **Overhead** and **Profit** as percentages of the material and labour total.
4. Type a **Description**: the measured description a client reads.
5. Under **Materials**, click **Add Material**. Type or pick a **Name of Material**. A name that is in your library fills in its **Rate** and **Unit**. Type the **Quantity** one unit of work uses.
6. Under **Labour**, click **Add Labour** and do the same. Plant is in the labour library, so add plant here too.
7. Check the totals at the foot of the form: **Material Total**, **Labour Total**, **Overall Total**, the overhead and profit, and the **Grand Total**.
8. Click **Save**.

You will see **New Rate Saved**. If you typed a priced material or labour line that is not in your library yet, Rate Gen adds it, and the message says how many items were added to your library. It never overwrites a price you already have.

## Saved Rates

Click **Saved Rates** in the sidebar to see every custom rate, with its **Title**, **Description**, **Overhead (%)**, **Profit (%)**, **Grand Total** and **Created Date**.

- Click a rate's title to open it and change it, then click **Save**.
- Select a rate and click **Delete Selected Rate** to remove it.
- Use **Search description...**, **Filter** and **Sort by** to find a rate.

Custom rates are priced from your library, so they follow when you change a library price.

## Building a custom rate on the website

You can also build a custom rate on the RateGen library page ( </work/library> ). Click **Build a custom rate**, then:

1. Type the **Name of rate** and its **Unit**, and pick a **Trade** if you want it filed under one.
2. Set overhead and profit, or leave them empty to use 10% and 10%.
3. Type the **Description**.
4. Add lines under **Materials** (quantity per unit), **Labour** (gang days per unit) and **Plant** (hours per unit). Materials and labour are picked from your library. A plant line takes the machine's name and your own price per hour.
5. Check the total strip, then click **Save rate**.

After saving, click **Open the build-up** to see the rate laid out line by line.

### IMPORTANT

Build each custom rate in one place. The desktop app keeps the cloud copy of your custom rates in step with its own **Saved Rates** list, and it does not download rates built elsewhere. So if you use the desktop app, build your custom rates there. A rate built on the website, or on another PC, can be removed from your account the next time the desktop app syncs.

## Plant lines on a rate

A plant line added to a custom rate on the website is kept when Rate Gen desktop syncs that rate, and the rate keeps its full value. The desktop app does not show a separate plant group, but it does not remove the line.

## Exporting your rates

Click **Export** in the sidebar to save your rates as an Excel workbook.

1. Click **Export**.
2. Choose where to save the file. It is named `ADLM_Rates_` followed by the date and time.
3. Click **Save**. You will see **Export completed**.

The workbook has one sheet per section (**Ground, Concrete, Block Works, Finishes, Roofs, Painting, Steel, Window & Door, Carbon & Others**) and a **Saved Rates** sheet.

### NOTE

The **Export** button is not always in the sidebar. It appears on the day you first sign in each calendar month, and again 20 days later.

## Syncing with the cloud

Your library is kept on your ADLM account as well as on your PC. That is what lets QUIV, HERON and ADLM Cloud price from the same rates as Rate Gen.

### What happens when you sync

Rate Gen syncs on its own when you sign in. Click **Sync from Cloud** in the header to sync at any time. A sync:

1. Fetches the rate library ADLM publishes for every section.
2. Fetches the rates that are calculated on ADLM Cloud, including **Carbon & Others**.
3. Fetches the master material and labour prices for your location, keeping every price you set yourself.
4. Sends your edited build-ups and your custom rates up to your account.

When you click **Sync from Cloud** yourself, a box lists each step and its result, for example **Rates: OK**, **Compute: OK** and **Master prices (south\_west): OK** with the number of materials and labour items. If a step fails, the box says why.

### What goes where

| You change it in                             | It reaches                                                |
|----------------------------------------------|-----------------------------------------------------------|
| The desktop library (a price)                | Every rate on that PC straight away.                      |
| A build-up in a desktop section (a quantity) | Your account on <b>Save</b> , then HERON and ADLM Cloud.  |
| A desktop custom rate                        | Your account on the next sync, then HERON and ADLM Cloud. |
| A price on the website ( /rategen )          | Rate Gen, QUIV and HERON from your next sign-in.          |

## You change it in

Your state on your profile

## It reaches

Rate Gen on your next sign-in or sync.

### NOTE

A project you have already priced keeps the rates it was priced with. A change in your library reaches a project only when that line is priced again, or when the project is set to follow Rate Gen changes. See [Using your rates in projects](#).

## The RateGen pages on the website

The ADLM website has two RateGen pages. Both need you to be signed in with an account that holds Rate Gen.

### The RateGen cloud dashboard ( /rategen )

This is where you set your own prices on the web, and see everything your account holds. **RateGen** in the left-hand menu brings you here.

The cards at the top show your **Pricing Location**, the size of **My Library**, **My Rates** (overrides and custom saved rates), the **Master Catalog**, and when the page was **Last Loaded**. Click **Refresh now** to reload, or tick **Auto refresh** to keep the page current.

The tabs:

| Tab                     | What it shows                                                                                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Master Materials</b> | The published material prices for your location.                                                                                                                                                    |
| <b>Master Labour</b>    | The published labour and plant rates for your location.                                                                                                                                             |
| <b>My Materials</b>     | Materials you added in Rate Gen desktop.                                                                                                                                                            |
| <b>My Labour</b>        | Labour and plant items you added in Rate Gen desktop.                                                                                                                                               |
| <b>My Custom Rates</b>  | Your custom rates, with their totals.                                                                                                                                                               |
| <b>Effective Rates</b>  | The rates your other ADLM software reads, with a <b>Source</b> of <code>master</code> (published), <code>user-override</code> (a published rate you edited) or <code>user-custom</code> (your own). |

Type in **Search this view...** to filter any tab.

### #### Setting your own price on the web

1. Open **Master Materials** or **Master Labour**.
2. Click a price.
3. Type what you actually pay and press `Enter`. Press `Esc` to cancel.

Your price is marked as yours. Click **reset** beside it to go back to the published price. Your price applies to the state on your profile only, so a correction you make for Kano does not follow you to a job in Lagos. If you have

no state set, it applies everywhere.

## The RateGen library page ( /work/library )

This page shows your whole library in one list: published rates, your edited copies and your own rates side by side. Open it from **Open RateGen** on your Overview, or go to </work/library>.

- The tabs are **Item of works**, **Materials**, **Labour** and **Plant**.
- Use the search box to find a rate, material, gang or description, then narrow with **Trade** (or **Category**) and **Sort by**.
- On **Item of works**, each rate shows its **Net cost**, **Overhead**, **Profit** and **Total**. Your own rates are marked **yours**, and published rates you have edited are marked **yours · edited**.
- On **Materials** and **Labour**, **Used in** tells you how many rates use each item. Prices you have set are marked **your price**.
- On **Plant**, each machine is read from the rates that use it, with its price and how many rates use it.

Click a rate to see its **Rate composition**: each line with its quantity and price, the net cost, the overhead and profit, and the rate per unit. Click **Open the build-up** to open the full rate page.

#### Changing many material prices at once

Market prices move. To move a whole category by a percentage:

1. Open the **Materials** tab and click **Update prices**.
2. Pick a **Category**, or **All materials**.
3. Type the percentage to **Change by**. Use a negative figure to reduce.
4. Click **Apply**.

A message says how many prices changed. Click **Undo** in that message to put them back. These are your own prices, not the published ones. Rates already built keep the cost they were built at until they are priced again.

**Edit the library** at the top takes you to the cloud dashboard ( /rategen ).

## A rate's build-up page

The build-up page lays one rate out in full: its materials, labour and plant as separate groups, each line with its quantity, unit price and amount, and the date each price was taken. Below the lines are the **Net cost**, **Overhead**, **Profit** and the **Rate**. The side panel, **What this rate is**, gives the item number, section, unit, the location it is priced for, and the total of any plant.

If a rate's stored net cost is more than its lines add up to, the difference is shown as **Not itemised**. It is part of the rate and is never dropped.

This page reads a rate. It does not change it. Quantities, overhead and profit are changed in the desktop app. The page has an **Open in Rate Gen** button, but Rate Gen 2.9.0 does not answer it yet. Open Rate Gen yourself and type the rate's name in **Search here....**

## RateGen Updates ( /rategen/updates )

Click **Updates** on the cloud dashboard to see rates ADLM has recently added. Search, filter by **Section** and sort by date or amount. Open an update to see its amounts, then click **Copy rate name**, open Rate Gen desktop, click **Sync from Cloud**, and search for the name. **Mark all as read** clears the count.

## Using your rates in projects

Your Rate Gen library is one library for your account. The other ADLM products read it directly, so there is nothing to export or import.

### In ADLM Cloud projects

In a project's **Bill of Quantity**, click a line's **Rate** cell and type part of a rate's name. Pick a rate from your library and it goes on the line.

Picking a Rate Gen rate also writes the line's build-up into the **Budget**: the materials at your library prices, one labour row and one plant row, with overhead and profit set so the budget reproduces the rate you picked. It replaces any budget rows ADLM had generated for that line, so nothing is counted twice. If the rate has no build-up, the rate still goes on the line and a message tells you the budget could not be priced.

A rate can only price a line measured in the same unit. A rate per m<sup>3</sup> is shown but cannot be picked for a line in m<sup>2</sup>.

The bill's **Rates** tab also has **Load RateGen rates**, **Sync rates**, **Auto-sync rates** and **Follow RateGen changes**. See [ADLM Cloud](#) for the full bill and budget.

#### TIP

Turn **Follow RateGen changes** off before a bill goes out, so later changes in your library do not move the prices you issued.

### In QUIV

QUIV measures your Revit model and saves the quantities to ADLM Cloud. You price the bill there, from your Rate Gen library, as above. See [QUIV for Revit](#).

### In HERON

HERON prices your PlanSwift takeoff straight from your Rate Gen library:

- In **Quantity Take Off**, click **Load rates** to price the bill, or use **Search Rate:** to apply a rate to one row.
- In the **Budget**, click **Load rates** to price materials and labour, or type a name in a **Price** cell to search Rate Gen.

**Load rates** and **Search Rate:** appear only when your account holds Rate Gen. See [ADLM HERON](#).

### Material and Services Constants

Rates say what work costs. The constants say what it takes: how many bags of cement go into a cubic metre of concrete, the waste allowances, formwork re-use, and the labour and plant outputs. They turn a bill into a material and labour schedule in HERON, QUIV and ADLM Cloud.

- **Material Constants** ( </rategen/material-constants> ): search for a constant, change its value, and click **Save changes**. The **ADLM default** column shows the starting figure, and **Reset** puts it back.
- **Services Constants** ( </rategen/services-constants> ): your standard lengths, joints and fitting uplift for pricing MEP services.

See "Material and Services Constants" in the [ADLM Cloud](#) guide.

## What's new in 2.9.0

### Rate Gen 2.9.0

- **What an item is, and what it produces.** Every labour and plant item has a specification and expected output panel, given as a range with the assumptions written beside it.
- **Your own notes on items you add.** Labour and plant you add can carry a specification of your own, and it shows on the website.
- **24 labour and plant categories** instead of 3, and the library sorts by size rather than by spelling.
- **Published prices one row at a time.** The review card shows your price, the published price and the change, and you decide row by row.
- **Sync moved to the header,** and tells you why a step failed.
- **More room for the rate tables.** Fold the welcome banner away from the header, and the last row of every table is no longer cut in half.
- **Screenshots carry the ADLM mark,** applied to the captured image, never to your screen.
- **Fixes:** sync failing on duplicated library rows, the library naming the wrong location, text hidden in dark mode, and misspelled plant names.

### On the website

- **Plant on a rate.** Custom rates built on the website can carry plant lines, and a desktop sync keeps them.
- **A Plant row in the Budget.** Plant from a Rate Gen rate is costed as plant, not counted as profit.
- **A picked rate prices the whole line.** Picking a Rate Gen rate on a bill writes the line's materials, labour and plant into the Budget, and replaces the budget rows it generated before instead of adding to them.

## Troubleshooting

### "Sign in failed. Please verify your credentials."

Check your email or username and password. If you have forgotten your password, click **Forgot password?** and reset it on the website, then sign in with the new one.

### "No active subscription for 'rategen'."

The account you signed in with does not hold an active Rate Gen subscription. Check you used the right account, or renew Rate Gen from your account on the ADLM website. See "Buying a product or subscription" in [Getting started](#).

### "Internet required for first sign-in (no valid offline license found)."

The first sign-in on a PC must be online. Connect to the internet and sign in again.

### "Your profile does not have a zone assigned."

You have not set a pricing location. Set your **Pricing location (State)** on your profile on the ADLM website, then sign out of Rate Gen and sign in again.

## "Signed in, but zone sync failed"

You are signed in, but the prices for your location could not be loaded. Check your connection, then click **Sync from Cloud**.

## Sync says "Please sign in again."

Your session has expired. Click **Log out**, then sign in again and sync.

## A sync step shows FAIL

The box names the step and the reason. Check your connection and try **Sync from Cloud** again. If the same step keeps failing, send ADLM support a screenshot of the box.

## My rates look wrong for my location

1. Look at the location line at the top of the **Library**.
2. If it names the wrong state or zone, change your **Pricing location (State)** on your profile on the website.
3. Click **Sync from Cloud**, or sign out and in again, and click **Yes** when Rate Gen offers to update your prices.

## A rate looks far too high or too low

It is nearly always one line. Open the rate, find the line that looks wrong, and click its component name to open it in the library. Check the price there, then check the quantity in the build-up.

## A material or labour line is priced at zero

The name on the line does not match any item in your library, so it has no price. In a custom rate, pick the name from the list rather than typing it, or add the item to your library with a price.

## A price I changed was replaced

Rate Gen keeps prices you set yourself through every sync and sign-in. If ADLM publishes a new price for that item, your price stays in use and the review card asks you to decide. If a price was replaced by mistake, use the undo library changes button in the header to restore an earlier copy.

## The Export button is missing

That is expected on most days. **Export** appears on the day you first sign in each calendar month, and again 20 days later.

## "Cloud Sync Pending" after saving a build-up

You were offline, or the server did not answer. Your edit is saved on the PC and will sync the next time you are connected. Click **Sync from Cloud** once you are back online.

## A custom rate I built on the website has gone

Rate Gen desktop keeps your account's custom rates in step with its own **Saved Rates** list, and does not download rates built elsewhere. Build the rate again in the desktop app, so it is in **Saved Rates** and stays on your account.

## Open in Rate Gen does nothing

Rate Gen 2.9.0 does not answer that button yet. Open Rate Gen yourself and type the rate's name in **Search** here....

## A rate cannot be picked on a bill line

The rate's unit is different from the line's unit. Pick a rate in the same unit, or build a custom rate in that unit.

## Frequently asked questions

### Do I need the desktop app, or is the website enough?

The desktop app is where rates are built and changed: build-up quantities, overhead and profit, custom rates and your library. The website lets you set your own prices, change material prices by category, build custom rates, and see every rate and its build-up. Most users work in the desktop app and use the website to look things up.

### Where is my data kept?

Your library, edits and saved rates are kept on your PC, in your Windows profile, and on your ADLM account. Reinstalling or updating Rate Gen does not touch them. Signing in on another PC brings back your edited build-ups and prices from your account.

### Does changing a price change my existing projects?

No. A project keeps the rates it was priced with. A change reaches a project only when a line is priced again, or when the project has **Follow RateGen changes** turned on.

### Does my price change affect other users?

No. Prices, build-ups and custom rates you change are yours. The published master prices are changed only by ADLM.

### Can I use rates for a different state?

Rate Gen prices for the state on your profile. To price a job elsewhere, change your **Pricing location (State)** on the website and sync. Prices you set on the website apply to the state they were set for.

### Why does my rate total differ from the line totals?

Overhead and profit are added to the net cost, and some published rates include an amount that is not itemised. On the website that amount is shown as **Not itemised**.

### Can I see what an item of plant should produce?

Yes. Open the item in the **Labour** tab of the library and read the **Specification & expected output** panel.

### Does Rate Gen send my drawings or projects anywhere?

No. Rate Gen sends your library edits, custom rates and prices to your own ADLM account, so your other ADLM products can use them.

## How do I get help?

Click **Help** in the Rate Gen sidebar to email ADLM support, or raise a ticket at </support>. Tell us the section and item number, what you expected the rate to be, and the exact wording of any message. A screenshot of the rate's build-up answers most questions at once.



# SERVIQ for Revit MEP

Measure ductwork, pipework, cable, fixtures and equipment straight from your Revit model, price them, and send the bill to ADLM Cloud.

The ADLM Revit MEP Suite is a Revit add-in for measuring the services in a building: ductwork, pipework, cable and containment, and every terminal, fixture and piece of equipment on them. It reads the quantities from the Revit model you have open, so you do not count symbols on a drawing. It is for services quantity surveyors, main-contract QSs pricing an M&E package, and services engineers who want a quick quantity check. You get a measured take-off, a priced bill with overhead and profit, a material schedule, an Excel workbook, and a project on your ADLM Cloud account.

This guide covers version 1.8.3, the version the Installer Hub installs today. In the Hub the product is listed as **ADLM Revit MEP Plugin**, and in Revit it adds a tab called **ADLM MEP & HVAC**. On the ADLM website the product is now shown as **SERVIQ**: it is the same subscription and the same add-in.

## NOTE

A new version, ServiQ 2.0, is being prepared and is not yet available to customers. Until the Installer Hub offers it, everything in this guide applies.

## Before you start

### What you need

| Item         | Requirement                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------|
| Computer     | A Windows PC that runs Autodesk Revit                                                                     |
| Revit        | Revit 2024, 2025, 2026 or 2027                                                                            |
| Model        | A Revit model with the services modelled as real elements (ducts, pipes, cable trays, fixtures and so on) |
| Internet     | Needed to sign in, to save to the cloud, and to search the Rate Gen library                               |
| Subscription | An active Revit MEP subscription on your ADLM account (shown on the website as <b>SERVIQ</b> )            |
| Optional     | An active <b>Rate Gen</b> subscription, if you want to pick rates from the Rate Gen library while pricing |

## NOTE

The MEP Suite is a separate subscription from QUIV for Revit. Having QUIV does not give you the MEP Suite, and having the MEP Suite does not give you QUIV. See [QUIV for Revit](#) for measuring the building itself.

### Get a subscription

1. Go to [adlmstudio.net/products](https://adlmstudio.net/products) and choose the SERVIQ (Revit MEP) plan.
2. Pay, then check that the subscription shows as active on your [dashboard](#).

## Install the add-in

The MEP Suite is installed through the ADLM Installer Hub, the same way as every other ADLM product. If you have not used the Hub before, read the [ADLM Installer Hub](#) guide first.

1. Close Revit completely. The Hub cannot replace an add-in that Revit is holding open.
2. Open the **ADLM Installer Hub** and sign in with your ADLM account.
3. Find **ADLM Revit MEP Plugin** and click **Install**.
4. When the Hub asks which Revit version you use, pick the year you actually open (2024, 2025, 2026 or 2027).
5. Click **Yes** when Windows asks for permission to make changes.
6. Wait for the Hub to say the install is complete.
7. Open Revit. If Revit asks whether to load the new add-in, choose **Always Load**.

### TIP

Use more than one Revit year? Install once for each year you use. Each year has its own copy of the add-in.

## Sign in

All the take-off buttons on the tab stay greyed out until you sign in.

1. Open the services model in Revit.
2. Click the **ADLM MEP & HVAC** tab.
3. On the **Tools** panel, click **Sign In**.
4. Type your **ADLM Email:** and **Password:**, then click **Sign In**.
5. Revit shows "User signed in successfully." Click **Close**.

The take-off buttons are now available, the **Sign In** button greys out, and a **Sign Out** button appears beside it. You sign in once each time you start Revit.

### IMPORTANT

Your subscription is tied to the computer you first sign in on. To move to a new computer, free the old one first. See [Moving to a new computer](#) in the Installer Hub guide.

## The ADLM MEP & HVAC tab

Everything the MEP Suite does starts from one ribbon tab with five panels. They follow the way a services bill is written: HVAC, plumbing, electrical, then your stored take-off.

| Panel         | Buttons                                                                              | What they do                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ADLM          | View Projects, User Guide                                                            | <b>View Projects</b> opens your Revit MEP projects on adlmstudio.net. <b>User Guide</b> opens the video walkthrough playlist. |
| HVAC Quantity | Air Terminal Quantity, Duct Fitting Quantity, Duct Quantity, HVAC Equipment Quantity | Measure the air-side services.                                                                                                |

| Panel               | Buttons                                                       | What they do                                                                |
|---------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------|
| Plumbing Quantity   | Plumbing Fixture Quantity, Pipe Works Quantity                | Measure pipework and sanitary fittings.                                     |
| Electrical Quantity | Lighting Quantity, Power Quantity, Cable Quantity             | Measure lighting, power points, distribution boards, cable and containment. |
| Tools               | Show Takeoff Database, Dock Take-Off Panel, Sign In, Sign Out | Open your stored take-off, dock it inside Revit, and sign in or out.        |

#### NOTE

Every button on the tab except **Sign In** is greyed out until you have signed in.

## The dockable workspace

The MEP Suite works inside a panel called **ADLM MEP Take-Off** that docks into Revit like the Properties or Project Browser panels. When you click a take-off button, the tool opens in this panel, so you can see the model and the quantities side by side.

### Dock or pop out

At the top of every MEP screen there is a header bar with two buttons:

| Button                | What it does                                                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Dock / Pop Out</b> | Moves the current screen between the Revit side panel and a large floating window. Your work stays as it is when you switch. |
| <b>Dark / Light</b>   | Switches the MEP screens between light and dark colours (Revit 2025 and later).                                              |

1. Click **Dock / Pop Out** to open the current screen in a full floating window, for example when you want a wide view of a long take-off.
2. Click **Dock / Pop Out** again to put it back in the side panel.

You can also drag, float or tab the docked panel the same way you do with Revit's own panels.

### Open the take-off panel directly

Click **Dock Take-Off Panel** on the **Tools** panel to open your stored take-off (the Take-Off Database, described below) straight into the docked panel.

### Dark mode

On Revit 2025, 2026 and 2027 you can work in dark colours, which suit Revit's own dark theme.

1. Open any MEP screen.
2. Click **Dark / Light** in the header bar.

The MEP Suite remembers your choice for next time. On Revit 2024 the MEP screens are light only and the **Dark / Light** button does not appear.

## Taking off a service

All nine take-off tools work the same way. Learn one and you know them all. Each tool has a narrow sidebar on the left with two pages: a model page (labelled **Model Quantities**, or with the tool's own name, such as **Pipe Works** or **Cables**) and **Result Info**.

### Step by step: measure the ductwork

1. On the **HVAC Quantity** panel, click **Duct Quantity**. The **Duct Work** tool opens in the docked panel on its **Model Quantities** page.
2. Under **Select a Floor Level**, open **Select Level** and choose a level, or **All Floors** for the whole building.
3. Open **Select Duct Type** and choose a duct type found on that level.
4. Read the **Duct Take Off Schedule**. It lists each duct name and size with its total length in metres. The matching ducts are selected in the Revit model so you can see what was measured.
5. Click a line in the schedule. Revit selects and zooms to just those ducts.
6. Click **Result Info** in the sidebar. This page shows every duct on the level you chose, grouped by name, type and size, with the **Total Length** at the top.
7. Click **Save to Take-Off**. A message tells you how many rows were saved to the Take-Off database for that level.
8. Go back to step 2, pick the next level, and repeat until every level is saved.

#### IMPORTANT

**Result Info** only opens after you have picked a level on the model page. If clicking it does nothing, go back and choose a level first.

#### TIP

Saving the same level twice does not double your quantities. Rows already saved for that level, with the same name, type and size, are skipped, and the message tells you how many were skipped.

### Export one tool's results to Excel

On the **Result Info** page of any tool, click **Export to Excel**, choose where to save the file, and click **Save**. The file is named after the tool and the level, for example `Ducts_All`. Use this when you need one service on its own. For the whole job, export from the Take-Off Database instead (see [Export the full bill to Excel](#)).

### Work in a sensible order

1. Start with the service you know best. You are checking how much you can trust the model as much as measuring it.
2. Look at the model page first. Does the count look like the job? A model with 40 air terminals when the drawings show 200 is telling you something before you measure anything.
3. Move to **Result Info** and check that sizes and types read the way your bill needs them.
4. Save each level, then move to the next tool. Work through HVAC, then plumbing, then electrical, so the bill comes out in a sensible order.

#### NOTE

The MEP Suite measures what is modelled. A service that exists only as a single-line diagram, a note or a specification clause is not an element in the model, so it cannot be measured.

## What each tool measures

Anything that runs is measured in metres. Anything that sits on a run is counted. That is how a services bill is written, and it is why the tools are split this way.

| Tool                      | Ribbon panel        | Pages in the sidebar                             | What it gives you                                                                                                                                  |
|---------------------------|---------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Duct Quantity             | HVAC Quantity       | Model Quantities, Result Info                    | Duct name, type and size, with length in metres (m).                                                                                               |
| Duct Fitting Quantity     | HVAC Quantity       | Model Quantities, Result Info                    | Fitting name, type and size, with the number of fittings (EA).                                                                                     |
| Air Terminal Quantity     | HVAC Quantity       | Model Quantities, Result Info                    | Diffusers, grilles and other terminals by name, type and size, with the number of terminals (EA).                                                  |
| HVAC Equipment Quantity   | HVAC Quantity       | Model Quantities, Result Info                    | Air handling units, fan coil units, VAV boxes and other equipment by name, type and size, with the number of units (EA).                           |
| Pipe Works Quantity       | Plumbing Quantity   | Pipe Works, Pipe Fittings, Result Info           | Pipe category, name and size with length in metres, plus pipe fittings counted by name, type and size.                                             |
| Plumbing Fixture Quantity | Plumbing Quantity   | Model Quantities, Result Info                    | WCs, basins, sinks and other fixtures by name and size, with the number of fixtures (EA).                                                          |
| Lighting Quantity         | Electrical Quantity | Lighting Fixtures, Light Switches, Result Info   | Lights by category, name and type with the number of lights, plus light switches counted separately.                                               |
| Power Quantity            | Electrical Quantity | Power Fixtures, Distribution Boards, Result Info | Sockets, outlets and other power points by category, name and type, plus distribution boards counted separately.                                   |
| Cable Quantity            | Electrical Quantity | Cables, Cable Trays, Conduits, Result Info       | Cable by category, name and size with length in metres; cable tray length in metres; conduit length in metres with the number of conduit fittings. |

#### TIP

Lines are grouped by the names, types and sizes the model uses. A model where half the supply ductwork is called "Duct 1" gives you a bill that says "Duct 1". Fix the names in Revit and run the tool again, rather than renaming lines every time the design changes.

#### NOTE

Duct and pipe fittings come out as a number, not a length, because that is how they are usually priced. If your bill needs fittings as a percentage on the duct run, convert the number once in the bill.

## The Take-Off Database

Everything you save with **Save to Take-Off** collects in one place, the Take-Off Database. This is where you price the job, check the totals and send the bill out of Revit.

### Open it

You can open it two ways:

- Click **Show Takeoff Database** on the **Tools** panel to open it in a floating window.
- Click **Dock Take-Off Panel** to open it in the docked panel.

### Find your way around

The left side of the Take-Off Database has three groups:

| Group                     | What is in it                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>PROJECT</b>            | The project name box, <b>Save to Cloud</b> , and <b>Load Latest Local</b> .                                                     |
| <b>VIEW</b>               | <b>Take-Off Data</b> , <b>Take-Off Summary</b> , <b>Budget</b> and <b>Material Schedule</b> : the four ways to look at the job. |
| <b>RECENT CLOUD SAVES</b> | Your projects already saved to ADLM Cloud. Click one to reopen it. The refresh button reloads the list.                         |

Along the top right are **Export to Excel**, which exports the whole job, and **Clear All**, which empties the take-off.

### Take-Off Data

This is the working bill. Each saved line shows its description, quantity and unit, with three columns on the right: **Unit Rate**, **Amount** and a **f** button.

- Tick the box at the start of a line to select it. Tick several to work on them together.
- Click **Delete Selected** to remove the ticked lines. You are asked to confirm first.

### Take-Off Summary

**Take-Off Summary** shows the **Totals** for the whole job and a **Details** list of every line, grouped by service (Air Terminals, Cable Works, Duct Work and so on). Use it for a quick read-through before you price.

### Start a new job

Click **Clear All** to empty the take-off and start again. It does not ask for confirmation, so export or save to the cloud first if you need the old take-off. If you clear by mistake, click **Load Latest Local** straight away to bring the rows back from the local backup.

## Pricing the bill

The MEP Suite turns your quantities into a priced bill and a budget, inside Revit. Prices are shown in naira (₦).

## Type a rate

1. Open the Take-Off Database and click **Take-Off Data**.
2. Click in the **Unit Rate** box on a line and type the rate.
3. Press **Tab** or click elsewhere. The **Amount** (quantity x rate) fills in.

A line with an empty **Unit Rate** is treated as unpriced. As you price, the **Bill Total** at the bottom of the list updates.

## Add overhead and profit

At the bottom of **Take-Off Data**:

1. Type your **Overhead %**.
2. Type your **Profit %**.
3. Read the **Budget Total**. It is the **Bill Total** plus overhead and profit.

## Build a rate from material and labour

Instead of typing one figure, you can build a rate up from the materials and labour that go into one unit of the item.

1. On a line in **Take-Off Data**, click the **f** button. The **Rate Build-Up** window opens.
2. Click **+ Material** to add a material row, or **+ Labour** to add a labour row.
3. For each row, fill in the **Description**, **Qty/unit** (how much goes into one unit of the line), **Unit** and **Unit Price**. The **Amount** works itself out.
4. Remove a row you do not want with X.
5. Check the totals at the bottom: **Material:**, **Labour:** and **Unit cost:**. The unit cost becomes the line's rate.
6. Click **Done**.

### TIP

Put waste and packing into **Qty/unit**. For example, 1.05 m of cable per metre run allows 5% waste. That figure carries through to the Material Schedule.

### #### Pick items from the Rate Gen library

If you also have an active Rate Gen subscription, the **Rate Build-Up** window shows a **SEARCH RATEGEN LIBRARY** box.

1. Type a material or labour name in the box.
2. Click a result. It is added to the build-up as a new row with its price.
3. Adjust **Qty/unit** for waste or for the way the item is sold.

Without a Rate Gen subscription the search box does not appear, and you type every row yourself. See [ADLM Rate Gen](#) for how the library works.

### #### Reuse a build-up on similar lines

1. In the **Rate Build-Up** window, click **Apply to similar lines**.
2. The same build-up is copied onto every other line in the same service that has no build-up yet. A message tells you how many lines it was applied to.

#### NOTE

Lines that already have a build-up are left alone.

## The Budget page

Click **Budget** under **VIEW** for the priced job broken down by trade.

| Part                                                 | What it shows                                                                                                                                                                                                                |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OVERHEAD % and PROFIT %</b>                       | The same margins as on Take-Off Data. Change them here or there.                                                                                                                                                             |
| <b>DISCIPLINE, LINES, SUBTOTAL</b>                   | One row per service, with the number of lines and the priced subtotal.                                                                                                                                                       |
| <b>COST COMPOSITION</b>                              | How much of the bill is <b>Material</b> , how much is <b>Labour</b> , and how much is <b>Unclassified (rate only)</b> , meaning lines priced with a typed rate and no build-up. Shown once at least one line has a build-up. |
| <b>Bill Subtotal, Overhead, Profit, Budget Total</b> | The roll-up from bill to budget.                                                                                                                                                                                             |

## The Material Schedule

Click **Material Schedule** under **VIEW** for a procurement list. It takes each material in each line's build-up, multiplies it by that line's quantity, and adds the same material together across the whole project. You get **MATERIAL, QTY, UNIT, UNIT PRICE** and **TOTAL** for every item, with the **Total Material Cost** at the bottom.

If the page says there are no material build-ups yet, use the **f** button on some lines first. Only lines with a build-up appear here.

## Saving your work

### The local backup

The MEP Suite keeps a backup of your take-off on your computer and updates it every time you add, delete or price a line. You do not have to do anything to make this happen.

Every Revit session starts with an empty Take-Off Database. To get your last take-off back:

1. Open the Take-Off Database.
2. Click **Load Latest Local**.
3. Check the project name, the save time and the number of rows in the message, then click **Yes** to replace what is on screen.

#### TIP

Revit crashed in the middle of a take-off? Open the Take-Off Database and click **Load Latest Local**. Your rows come back as they were at the last change.

#### NOTE

The backup holds one take-off: the last one you worked on on this computer. To keep several jobs, save each one to the cloud with its own name.

## Save to ADLM Cloud

1. Open the Take-Off Database.
2. Under **PROJECT**, replace "Untitled" with a clear project name, for example the job name and "Services".
3. Click **Save to Cloud**.
4. If a project with that name is already on your account, you are asked whether to overwrite it. Click **Yes** to overwrite, or **No** to cancel and rename.
5. Wait for the message that the project was saved to the cloud.

The project then appears under **RECENT CLOUD SAVES** and on the website at [adlmstudio.net/projects/mep](https://adlmstudio.net/projects/mep).

#### IMPORTANT

You must be signed in to save to the cloud. If the save fails, your work is still in the local backup, so nothing is lost. Try again when your connection is back.

## Reopen a cloud project

1. Open the Take-Off Database.
2. Under **RECENT CLOUD SAVES**, click the refresh button if the list looks out of date.
3. Click the project you want.
4. Click **Yes** to replace the current take-off with that project.

## On the website

Projects you save from Revit appear on [adlmstudio.net](https://adlmstudio.net) under **Revit MEP projects**, alongside your other ADLM projects on the same account. Click **View Projects** on the **ADLM** panel to go straight there, or open [/projects/mep](https://adlmstudio.net/projects/mep) in your browser.

#### TIP

Keep the services and the building as separate projects. They come from different models, change at different times, and are often valued as different packages.

## Learning samples

Your Revit MEP projects page can also show a **Learning samples** strip: worked services projects for sample duplex buildings, covering electrical, plumbing and drainage, and air conditioning. Open one to see a fully filled-in project. Samples are read-only, so you cannot change or break them. Click **Hide samples** to fold the strip away, and the **Show** button (for example "Show 3 samples") to bring it back.

## Export the full bill to Excel

1. Open the Take-Off Database.

2. Click **Export to Excel** at the top right.
3. Choose where to save. The file name starts as "Complete MEP BoQ".
4. Click **Save**. A message confirms the export.

The workbook has:

| Sheet                 | What is on it                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One sheet per service | Air Terminals, Cable Works, Lighting, Pipe Works, Plumbing Fixtures, Power Fixtures and the rest. Columns: S/N, Description, Quantity, Unit, Rate (₦) and Amount (₦), with a TOTAL row. |
| <b>Summary</b>        | Each service with its number of rows, total quantity and total amount, and a GRAND TOTAL.                                                                                               |
| <b>Budget Summary</b> | The bill subtotal, Overhead %, Profit % and the Budget Total. The Budget Total is a live Excel formula, so if you change the percentages in Excel, the total updates.                   |

Only the services you have saved lines for get a sheet.

## What's new in 1.8.3

### 1.8.3 (31 July 2026)

- **Sign-in that survives server moves.** The add-in now finds the ADLM server through a setting the Installer Hub puts on your computer, so a future server change does not need a new download.
- **Fixed:** signing in, licence checks and cloud saves failing after ADLM moved its server. Cloud saves and reopening cloud projects work again with this update.

### 1.8.2 (July 2026)

- **Fixed:** "already bound to another device" sign-in lockouts. Your licence used to be linked to whichever network connection was active, so docking a laptop, starting a VPN or switching between Wi-Fi and a cable could make the same computer look new. It now uses fixed hardware details, and your existing licence moves over automatically the first time you sign in.

### 1.2 (June 2026)

- The docked **ADLM MEP Take-Off** panel, with **Dock / Pop Out** to switch to a floating window.
- **Dark / Light** colours on Revit 2025 and later.
- One install covers Revit 2024, 2025, 2026 and 2027.
- Pricing in the Take-Off Database: unit rates, Bill Total, overhead and profit, the **Budget** page, and the Budget Summary sheet in the Excel export.
- Duct take-off now respects the level you choose, so quantities no longer spill across floors.
- Mouse-wheel scrolling works again in the take-off list.

# Troubleshooting

## The ADLM MEP & HVAC tab is missing

| Cause                                                     | Fix                                                                     |
|-----------------------------------------------------------|-------------------------------------------------------------------------|
| Installed for a different Revit year                      | Reinstall from the Hub and choose the year you are opening.             |
| Revit was open during the install                         | Close Revit completely and install again.                               |
| Revit's add-in security prompt was answered "Do Not Load" | Install again from the Hub, open Revit, and choose <b>Always Load</b> . |

## All the buttons on the tab are greyed out

You are not signed in. Click **Sign In** on the **Tools** panel. This is needed each time you start Revit.

### "Sign-in failed: Invalid email or password."

Check the email and password by signing in at [adlmstudio.net](https://adlmstudio.net). If you have forgotten the password, reset it on the website, then try again in Revit.

### "Sign-in failed: No active subscription for 'mep'."

Your account does not have an active Revit MEP subscription. It may have expired, or it may be on a different account. Check your [dashboard](#). Remember that QUIV and the MEP Suite are separate subscriptions.

### "No active MEP subscription or device mismatch."

Either the subscription is not active, or your licence is in use on another computer. Check your [dashboard](#) first. If the subscription is active, free the other computer as described in [Moving to a new computer](#), then sign in again.

### "Internet required for first sign-in" or "Network error"

The add-in could not reach the ADLM server. Check your internet connection, then click **Sign In** again. If you use a VPN or a company proxy, try without it.

## A tool finds nothing

The elements are probably in a linked model, not the one you have open. Open the linked services file directly and run the tool there. Also check that you picked a level and a type on the model page.

## Clicking Result Info does nothing

Pick a level on the model page first. **Result Info** needs a level to work from.

## Counts are far below the drawings

The service is probably drawn but not modelled, for example as a single-line diagram or annotation. There is nothing to measure. Price that part another way and say so in your bill.

## Lines are named "Duct 1", "Pipe 2" and so on

The model uses default names. Rename the types in Revit and run the tool again.

## Lengths look too long

Check whether risers and drops are modelled as part of the run. They usually should be. Click the line on the model page to see exactly which elements were measured.

## My take-off is empty when I open Revit

That is normal. Each Revit session starts empty. Click **Load Latest Local** in the Take-Off Database to bring back your last take-off, or click a project under **RECENT CLOUD SAVES**.

## "Enter a project name before saving to cloud."

The project name box under **PROJECT** is empty. Type a name and click **Save to Cloud** again.

## "You must be signed in to save to the cloud."

Sign in from the **Tools** panel, then save again. Your rows are kept while you sign in.

## "Cloud save failed"

The message ends with "Your data is still saved locally." Nothing is lost. Check your connection and try again. If it keeps failing, contact support and mention the time of the failure.

## "Not Confirmed" after Save to Cloud

The server accepted the save but the project did not appear in your project list afterwards. This usually means the Revit MEP subscription on your account is not active. Check your [dashboard](#), then save again. Your data is safe in the local backup.

## "No projects found on your cloud account yet."

You have not saved a project to the cloud from this account yet, or the server is still processing a save you just made. Wait a few seconds and refresh the list.

## The Rate Gen search box is missing from Rate Build-Up

The search box only appears if your account has an active Rate Gen subscription. You can still build rates by typing rows with + **Material** and + **Labour**.

## The Dark / Light button is missing

Dark mode is available on Revit 2025, 2026 and 2027 only. On Revit 2024 the MEP screens are light only.

## I cleared the take-off by mistake

Click **Load Latest Local** straight away. **Clear All** does not wipe the local backup, so your rows come back.

## Something else went wrong

Contact ADLM through [support](#) with your account email, your Revit year, and what you were doing. If asked, the add-in keeps a log file at %AppData%\ADLM\RevitMEP\cloud.log that helps support see what happened.

## Frequently asked questions

### Do I need QUIV to use the MEP Suite?

No. The MEP Suite works on its own with its own subscription. On a full bill you will usually want both: QUIV for the building and the MEP Suite for the services. They save as separate projects under the same account.

### Which Revit versions are supported?

Revit 2024, 2025, 2026 and 2027. When you install, the Hub asks which year you use and installs the right copy.

### Can I measure one floor at a time?

Yes. Pick a level under **Select Level** on the model page, save it with **Save to Take-Off**, then pick the next level. Choose **All Floors** to measure the whole building in one go.

### Will saving the same level twice double my quantities?

No. Rows already saved for that level with the same name, type and size are skipped, and the message tells you how many.

### Can I change a description before it goes into the bill?

Descriptions come from the names, types and sizes in the model. The best fix is to rename them in Revit and run the tool again, so the change sticks through every revision.

### Do I need a Rate Gen subscription to price?

No. You can type any rate, and you can build rates from material and labour by hand. Rate Gen only adds the library search inside the **Rate Build-Up** window.

### What currency are prices in?

Naira (₦).

### Does the MEP Suite work offline?

You need the internet to sign in and to save to the cloud. Once you are signed in, measuring and pricing happen on your computer, and your work is kept in the local backup.

### Where is my local backup kept?

On your computer, in your Windows user folder under `%AppData%\ADLM\RevitMEP`. You do not need to open it; use **Load Latest Local** instead.

### How do I move my licence to a new computer?

Free the old computer on the website, then install and sign in on the new one. The steps are in [Moving to a new computer](#).

### Can I import a services bill from Excel?

Not in the MEP Suite. It measures from the Revit model.

## Where are the video tutorials?

Click **User Guide** on the **ADLM** panel. It opens the ADLM video playlist in your browser.



# ADLM Time Pro

Record what your gangs really produce on site, then turn a bill quantity into a duration, a crew size and a Microsoft Project programme.

ADLM Time Pro is a Windows app for quantity surveyors, site engineers and project managers who need durations they can defend. You log what each gang actually produced on site, day by day. Time Pro averages those records for every item of work. Then you type in a quantity from your bill, and it tells you how many days the work will take and how many skilled and unskilled hands you need to finish in the time you want. When you are happy, you send the result to Excel or to Microsoft Project.

## Before you start

### What you need

| Item              | Requirement                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Computer          | A Windows 10 or Windows 11 PC                                                                                                 |
| Other software    | None. Time Pro runs on its own and does not need Revit, ArchiCAD or PlanSwift                                                 |
| Microsoft Project | Only if you want to open the programme file Time Pro exports. Any recent desktop version that opens Project XML files will do |
| Internet          | Needed for your first sign-in, for fetching weather, and for checking your licence. After that you can record tasks offline   |
| Subscription      | An active <b>ADLM Time Pro</b> subscription on your ADLM account                                                              |

### Get a subscription

Time Pro is a paid subscription. An ADLM account on its own is not enough.

1. Go to the Time Pro product page on [adlmstudio.net](https://adlmstudio.net) and choose a plan.
2. Pay and wait for the subscription to show as active on your [dashboard](#).

If you do not yet have an ADLM account, create one first with **Sign up** on the website.

### Install Time Pro

Time Pro is installed through the ADLM Installer Hub, the same way as the other ADLM products.

1. Open the **ADLM Installer Hub** and sign in with your ADLM account.
2. Find **ADLM Time Pro** in your subscriptions. It only appears once your subscription is active.
3. Click install and approve the Windows administrator prompt.
4. When it finishes, open Time Pro from the **ADLM Time Pro** shortcut on your desktop.

#### NOTE

Updates also come through the Installer Hub. Updating Time Pro does not delete your Task Log.

## Sign in

The first window you see is **Sign in to ADLM Time Management**.

1. Type your ADLM username or email address into **Username or Email**.
2. Type your **Password**.
3. Click **Sign in**.

Time Pro checks your account and your subscription, then opens on the **Task Log**.

- If you do not have an account yet, click **Sign up** under "Don't have an ADLM account?". It opens the sign-up page on the website.
- If your account is fine but has no active Time Pro subscription, you will see a message saying so and a **Subscribe to ADLM Time Pro** button that opens the product page.

## One computer per licence

Your Time Pro licence is tied to the computer you sign in on. Once you have signed in online on that PC, Time Pro remembers your licence, so:

- it can reopen without asking you to sign in again for up to about a day and a half (after that you will see **Session expired. Please sign in again.**);
- if the PC has no internet when you sign in, Time Pro can still let you in, as long as you have signed in online on this same PC before and your licence is still valid.

If you move to a new computer, contact [ADLM support](#) so the licence can be released from the old one.

## How Time Pro works

Time Pro has one simple idea, and it helps to understand it before you start typing.

1. **Record.** Every day, for every gang, you log the item of work, the crew, the hours and the output. This is the **Task Log**.
2. **Average.** Time Pro groups your records by item of work and works out the average output per day and the average crew that produced it. This is the **Duration Summary**.
3. **Forecast.** You type the quantity from your bill. Time Pro divides it by your average output to give the expected number of days, and tells you the crew needed for the number of days you actually want.
4. **Export.** You send the result to Excel, or to Microsoft Project as a ready-made programme.

### IMPORTANT

A brand-new Time Pro has no history, so it cannot forecast anything yet. Every expected duration will show as 0 until you have logged some tasks for that item of work. Start recording on site before you need the forecast. A week of honest records is enough to begin.

## What Time Pro does not do

Time Pro builds the durations and crews. It does not draw a Gantt chart, track progress against a baseline, or let you set your own task links. Those jobs happen in Microsoft Project after you export: see "Working with the programme in Microsoft Project" below. Time Pro also does not open or import Microsoft Project files. It only exports to them.

# Finding your way around

## The side menu

The side menu on the left takes you between the three screens. Use them in this order:

| Button           | What it is for                                                                      |
|------------------|-------------------------------------------------------------------------------------|
| Current Weather  | Look up a town and fetch today's weather, so it can be stamped on your task records |
| Task Log         | Your daily site records. Everything Time Pro knows comes from here                  |
| Duration Summary | One row per item of work, with averages and your forecast. Exports start here       |

At the bottom of the side menu are two more buttons:

- **Dark Mode** (or **Light Mode** when dark mode is on) switches the colour theme.
- **Sign Out** signs you out of Time Pro on this computer.

The screen you are on is highlighted in ADLM orange.

### TIP

If you make the Time Pro window narrow, the side menu shrinks to a strip of icons to give the tables more room. Hover over an icon to see its name. Widen the window again and the labels come back.

## The top bar

Across the top of the window you will find:

- **Search here...:** a search box. Start typing an item of work, a trade, a piece of equipment or a weather condition and Time Pro filters the table on the current screen and offers suggestions underneath. Click a suggestion to use it.
- A help button (the question mark). It shows a short note and a link to [adlmstudio.net](https://adlmstudio.net) for more ADLM software and training.
- Your name and email. Click them to open a small menu with **Logout**.

## Dark mode

Time Pro has a full dark theme that covers every screen, dialog and export window.

- The first time you open Time Pro, it follows your Windows light or dark setting.
- To switch, click **Dark Mode** or **Light Mode** at the bottom of the side menu.
- Time Pro remembers your choice the next time you open it.

## Fetching the weather

Weather changes what a gang produces. Time Pro lets you attach real conditions to each record so that later you can tell a wet-season output from a dry-season one. The weather comes from the free Open-Meteo service, so you need an internet connection at the moment you fetch it.

## Fetch today's weather

1. Click **Current Weather** in the side menu.
2. Type your town into the box under **SEARCH CITY** and click **Search**.
3. Choose the right place under **SELECT CITY**. Many town names exist in more than one country, so check the result.
4. Click **Fetch Weather**.

Time Pro shows the temperature and condition, with **Wind Speed**, **Humidity**, **Rainfall** and **Cloud Cover**. Your last few readings are listed under **Weather Snapshots**.

## Attach weather to your tasks

There are two ways:

- **Automatically.** Once you have fetched the weather, every new task you add picks up the current reading when you save it.
- **By hand.** Select a task in the **Task Log**, then click **Apply Current Weather to Selected Task** on the Task Log, or **Apply to Selected Task** on the **Current Weather** screen.

### TIP

Fetch the weather once at the start of the day. The reading stays in place while you move between screens, so every task you add that day gets it.

### NOTE

Weather does not change Time Pro's arithmetic. It applies no wet-weather factor. It simply records the conditions so you can judge whether an average fits the job you are planning.

## Recording site work in the Task Log

The **Task Log** is where you record daily site work: items of work, labour, hours and outputs. One row is one gang, on one item of work, for one day.

### Add a task

1. Click **Task Log** in the side menu.
2. Click **Add Task**. The **Add Task** window opens with today's date already filled in.
3. Type the **Item of Work**, for example "Blockwork 225mm".
4. Pick the **Trade** from the list, for example "Blockwork / Masonry / Bricklaying".
5. Enter the number of **Skilled Labour** and **Unskilled Labour** on this item for the day.
6. Enter the **Hours Worked** and the **Break Hours**. **Net Hours (auto)** fills itself in: hours worked minus breaks.
7. Type the **Equipment Used**, for example "Mixer", if any.
8. Enter the **Output** the gang produced and its **Output Unit**, for example 18 and m2.
9. Check the **Start Date** and **End Date**.
10. Click **Save Task**.

The new row appears in the table straight away and is saved on this computer.

### IMPORTANT

The Item of Work must be typed exactly the same way every time. Time Pro groups your records by this text. "Blockwork 225mm" and "225 blockwork" become two separate items, each with half the history. If you work in a team, agree the item names before anyone starts recording.

### TIP

Keep the output unit the same as the unit in your bill. If your bill measures blockwork in m2, record the output in m2, or the forecast will be wrong by the conversion.

## What each field is for

| Field                             | What to enter                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| Item of Work                      | The work being done. This is the field that drives all the averages                          |
| Trade                             | The trade, chosen from the list. It is for your reference and does not change the arithmetic |
| Skilled Labour / Unskilled Labour | How many of each worked on this item that day. Count the gang, not the whole site            |
| Hours Worked                      | Hours on the item for the day                                                                |
| Break Hours                       | Hours not worked out of those. Cannot be more than the hours worked                          |
| Net Hours (auto)                  | Worked out for you. You cannot type into it                                                  |
| Equipment Used                    | The plant used, such as a mixer, vibrator or hoist                                           |
| Output / Output Unit              | What the gang actually produced, and in what unit                                            |
| Start Date / End Date             | The days this record covers. The end date cannot be before the start date                    |

### IMPORTANT

Time Pro treats each row as one day's output when it works out **Avg Output / Day**. Record one row per gang per day. If you put a whole week's output into one row, that item's daily output will look far higher than it really is.

## The Task Log table

The table shows your records with these columns: **ID**, **Item of Work**, **Trade**, **Skilled**, **Unskilled**, **Hours**, **Break**, **Net**, **Equipment**, **Output**, **Unit**, **Start** and **End**. Time Pro numbers the **ID** for you. The table is for reading only: to change a record, use **Edit Selected**.

## Edit a task

1. Click the row you want to change.
2. Click **Edit Selected**. The **Edit Task** window opens with the record filled in.
3. Make your changes.
4. Click **Save Changes**.

The Duration Summary updates on its own. There is nothing to recalculate by hand.

## Delete a task

1. Click the row you want to remove.
2. Click **Delete Selected**.

### IMPORTANT

**Delete Selected** removes the row at once. Time Pro does not ask you to confirm and there is no undo. Check you have the right row selected first.

## If Save will not work

If the save button stays greyed out, or you see **Please fill Item of Work, check dates, and confirm hours/break values.**, check that:

- the **Item of Work** is not empty;
- the **End Date** is not before the **Start Date**;
- **Break Hours** is not more than **Hours Worked**;
- no number is negative.

## Turning records into durations: the Duration Summary

The **Duration Summary** turns your site records into durations from your BOQ quantities. It shows one row per item of work. The averages come from your Task Log. Three columns are yours to type in: **BOQ Qty (Requested)**, **Planned Duration (days)** and **Weather Scenario**. Everything else is worked out for you.

### Get a duration from a BOQ quantity

1. Click **Duration Summary** in the side menu.
2. Find the item of work.
3. Click the cell that says **Click to enter BOQ Qty...** in the **BOQ Qty (Requested)** column.
4. Type the quantity from your bill, for example the measured quantity from [QUIV for Revit](#), and press .

**Expected Duration (days)** fills in at once, and **Planned Duration (days)** starts at the same figure. **Required Skilled / Day** and **Required Unskilled / Day** show the crew needed.

### Squeeze or stretch the programme

If the client needs the work finished sooner, or you can only put a small gang on it:

1. Click the **Planned Duration (days)** cell for that item.
2. Type the number of days you want and press .

**Required Skilled / Day** and **Required Unskilled / Day** change to the crew needed to finish in that time.

### Weather Scenario

**Weather Scenario** starts with the **Predominant Weather** for that item. Overwrite it with a short note of the conditions you are planning for, such as "Rainy season". It is a note only and does not change any figure.

## What each column means

| Column                                  | How it is worked out                                                                                    |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| SN                                      | Row number                                                                                              |
| Item of Work                            | One row for each distinct item in your Task Log                                                         |
| BOQ Qty (Requested)                     | You type it: the quantity from your bill                                                                |
| Expected Duration (days)                | BOQ Qty divided by Avg Output / Day. How long the work takes at the rate you have really been achieving |
| Planned Duration (days)                 | You can change it. Starts equal to the expected duration                                                |
| Required Skilled / Day                  | Skilled workers per day needed to finish in the planned duration, rounded up to whole people            |
| Required Unskilled / Day                | The same for unskilled workers                                                                          |
| Weather Scenario                        | You can type here. A note of the conditions you are assuming                                            |
| Avg Output / Day                        | The average output across your Task Log rows for this item                                              |
| Avg Net Hrs / Day                       | Average hours actually worked, after breaks                                                             |
| Avg Skilled / Day / Avg Unskilled / Day | The average crew that produced that output                                                              |
| Predominant Weather                     | The weather recorded most often against this item                                                       |

## The arithmetic, with an example

Time Pro keeps the total effort the same and spreads it over the number of days you ask for:

- Expected duration = BOQ quantity ÷ average output per day
- Skilled man-days = average skilled per day × expected duration
- Required skilled per day = skilled man-days ÷ planned duration, rounded up

Say your records for "Blockwork 225mm" average 18 m<sup>2</sup> a day from 3 skilled and 2 unskilled workers, and the bill says 540 m<sup>2</sup>.

- Expected duration:  $540 \div 18 = 30$  days.
- Effort:  $3 \times 30 = 90$  skilled man-days, and  $2 \times 30 = 60$  unskilled man-days.
- Set **Planned Duration (days)** to 20, and the crew becomes 5 skilled ( $90 \div 20 = 4.5$ , rounded up) and 3 unskilled ( $60 \div 20 = 3$ ).

### NOTE

Halving the duration roughly doubles the crew. Time Pro does not know the point where a gang starts getting in its own way on a small workface. That judgement stays with you. A planned duration far below the expected one is a risk worth flagging.

## Enter quantities and export in one sitting

### IMPORTANT

In version 1.1.1, Time Pro does not keep the figures you type on the Duration Summary. **BOQ Qty (Requested), Planned Duration (days) and Weather Scenario** clear when you leave the Duration Summary for another screen, when you add, edit or delete a task, and when you close Time Pro. Enter your quantities, then export to Excel or Microsoft Project before you move away. Keep your bill quantities handy (for example in your BOQ spreadsheet) so you can type them in again.

## Exporting your work

Two buttons sit at the top right of the **Duration Summary: Excel** and **MS Project**.

### Export to Excel

The Excel export is your record and your evidence. Use it when someone asks where a duration came from.

1. Open the **Duration Summary** and enter any BOQ quantities you want included.
2. Click **Excel**.
3. Choose where to save the file. Time Pro suggests a name such as `ADLM_Time_Export_20261001_0930.xlsx`.
4. Click save. You will see **Export complete**.

The workbook has two sheets:

| Sheet            | What is on it                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Summary</b>   | One row per item of work: the averages, predominant weather, requested quantity, weather scenario, expected and planned durations and the required crews |
| <b>DataInput</b> | Every Task Log row exactly as recorded, including equipment, output, unit, dates and the weather attached to it                                          |

The sheets are set up to print neatly, with "Generated by ADLM Studios" and the print date in the page footer.

### TIP

The Excel export is also the easiest way to keep a backup of your Task Log, or to send your records to a colleague.

### Export to Microsoft Project

The **MS Project** button builds a programme file that Microsoft Project opens directly, with tasks, durations, links, resources and costs already in place.

The button only becomes active once at least one item has a BOQ quantity.

1. On the **Duration Summary**, enter a **BOQ Qty (Requested)** for each item you want in the programme.
2. Click **MS Project**. The **Export to MS Project** window opens.
3. Set the **Project Start Date**. It starts on today's date.
4. Choose the **Currency** your rates are in.
5. Under **Resource Rates (enter 0 to fill in MS Project later)**, type:

- **Skilled Labour – Rate per person/hour**
- **Unskilled Labour – Rate per person/hour**
- **Equipment – Rate per use**

6. Click **Export**.

7. Choose where to save the file. Time Pro suggests a name such as `ADLM_Project_20261001_0930.xml`.

8. Click save. Time Pro tells you how many work items it exported.

#### NOTE

The file is a Microsoft Project XML file ( `.xml` ), not an `.mpp` file. Microsoft Project opens it from **File**, then **Open**. Once it is open, save it as an `.mpp` from inside Project if you want.

#### #### Currencies you can choose

Nigerian Naira, US Dollar, Pound Sterling, Euro, Ghanaian Cedi, Kenyan Shilling, South African Rand, West African CFA Franc, UAE Dirham, Saudi Riyal, Indian Rupee, Canadian Dollar and Australian Dollar. Time Pro starts on the currency of your Windows region and remembers the one you pick. The currency goes into the file, so Microsoft Project shows costs in the same money you typed.

#### #### Typing rates

You can type rates the way you normally write money, for example `1,500.00` or `₦1500`. Enter 0 for any rate you would rather fill in later inside Microsoft Project.

#### #### Which items are exported

An item goes into the programme only if it has both a BOQ quantity and a planned duration above zero.

- If an item has a quantity but no logged output history, its expected duration is 0. Time Pro names these items and either skips them (telling you how many it skipped) or, if nothing can be exported, shows **Nothing to Export**.
- To include such an item, either type a number of days into its **Planned Duration (days)**, or log some tasks for it in the Task Log first.
- Items with no BOQ quantity are left out.

## Working with the programme in Microsoft Project

When you open the exported file in Microsoft Project, you get:

| In Project                                                                                                                                     | Where it comes from                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| A project summary task, "ADLM Project Summary"                                                                                                 | Added by Time Pro                                                         |
| One task per exported item of work                                                                                                             | The item's <b>Planned Duration (days)</b>                                 |
| Tasks linked finish-to-start, in the order of the Duration Summary                                                                             | Added by Time Pro, starting on your <b>Project Start Date</b>             |
| Up to three resources per item, such as "Blockwork 225mm – Skilled Labour", "Blockwork 225mm – Unskilled Labour" and "Blockwork 225mm – Mixer" | Labour as hourly work resources, equipment as a per-use material resource |

| In Project           | Where it comes from                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Resource assignments | The crew from <b>Required Skilled / Day</b> and <b>Required Unskilled / Day</b>                                         |
| Costs                | Your rates, in your chosen currency                                                                                     |
| Task notes           | The BOQ quantity, expected and planned durations, average output, crew, equipment and predominant weather for each item |
| Calendar             | A standard Monday to Friday calendar, 08:00 to 12:00 and 13:00 to 17:00 (8 hours a day)                                 |

From here, use Microsoft Project for everything Time Pro does not do:

- **Gantt chart:** the file opens in Project's Gantt view.
- **Dependencies:** the finish-to-start chain is a starting point. Change or add links in Project where trades can overlap.
- **Rescheduling:** because the tasks are linked, moving the first task moves the whole programme.
- **Baselines and progress:** set a baseline and record progress in Project as usual.

#### TIP

If your quantities change, export again and rebuild the programme rather than patching the old one by hand. It keeps your Time Pro figures and your programme in step.

## Your data

- Your Task Log is saved on this computer as you work, so closing Time Pro never loses a day's records. It is still there the next time you sign in on this PC.
- Signing out does not delete your Task Log.
- The Duration Summary figures you type (quantities, planned durations, scenarios) are not saved. See "Enter quantities and export in one sitting" above.
- Excel and Microsoft Project files are snapshots. Changing them does not change Time Pro.

## What's new in 1.1.1

Version 1.1.1 was released in August 2026.

- **Dark mode on every screen.** Follows your Windows setting at first, then remembers your choice. Switch from the bottom of the side menu.
- **Sign out from the side menu.** Sign Out now sits under the theme switch.
- **Choose your currency for Microsoft Project.** Thirteen currencies, starting from your Windows region. Costs show in Project in the same currency you typed.
- **Microsoft Project files now open.** The exported file previously could be rejected by Project. It now opens straight from **File**, then **Open**.
- **Tasks chain in Microsoft Project.** Each item is linked finish-to-start to the one before it.

- **Labour and equipment cost correctly.** Labour is now an hourly resource and equipment a per-use resource, so rates land in the right place.
- **Rates accept normal money formats** such as 1,500.00 and ¥1500.
- **Clearer messages when an item cannot be exported.** Time Pro now names the items with no duration and tells you what they need.
- **A readable side menu.** Every button shows its icon and name, with the current screen in ADLM orange.
- **The Project Start Date** now shows clearly in the export window.
- **The ADLM logo** now shows in dark mode.

## Version 1.0.1

- Time Pro became a paid subscription. You now sign in with your ADLM username or email and password, and Time Pro checks for an active subscription.
- Your licence is tied to your computer, and Time Pro can sign you in without internet on a PC where you have signed in before.

## Troubleshooting

### "Your ADLM account does not have an active ADLM Time Pro subscription"

Your username and password are correct, but the account has no active Time Pro subscription. Click **Subscribe to ADLM Time Pro**, complete the purchase, then sign in again. If you have just paid, check your [dashboard](#) shows the subscription as active. If you bought it under a different email address, sign in with that one.

### "Sign in failed. Please check your username/email and password."

Check the spelling of your email or username and your password. If you have forgotten your password, reset it on the website, then try again.

### "Internet is required for your first sign-in."

Your PC is offline and you have never signed in to Time Pro on it before. Connect to the internet and sign in once. After that, Time Pro can work offline on this PC.

### "No valid offline licence was found. Connect to the internet and sign in again."

You are offline and the licence saved on this PC has run out, or it belongs to another computer. Connect to the internet and sign in.

### "Session expired. Please sign in again."

Time Pro keeps you signed in for about a day and a half. Sign in again. Your Task Log is still there.

### Time Pro is not in my Installer Hub

Time Pro only appears once your subscription is active. Sign out of the Installer Hub and sign back in so it reads your subscriptions again. If it still does not appear, contact [support](#).

## Every Expected Duration shows 0

The item has no output recorded in the Task Log, or its rows have an output of 0. Log some tasks for that exact item of work, spelled exactly the same way. If you need to export it before you have records, type a number of days into **Planned Duration (days)**.

## The same item appears twice in the Duration Summary

The item has been typed two different ways in the Task Log, for example with an extra space or different wording. Edit the rows so they all use the same text, and the two lines will merge into one.

## My BOQ quantities disappeared

Time Pro clears the Duration Summary entries when you leave that screen, change the Task Log, or close the app. Type them in again and export before moving away.

## The MS Project button is greyed out

No item has a BOQ quantity yet. Enter at least one quantity in **BOQ Qty (Requested)** and press .

## "Nothing to Export"

Either no item has a BOQ quantity, or the items with a quantity have no duration because there are no logged tasks for them. Follow the message: enter quantities, or type a **Planned Duration (days)** for the items it lists.

## Microsoft Project will not open the file

Make sure you are on version 1.1.1 (update through the Installer Hub). Earlier versions produced files that Project could reject. Open the file from **File**, then **Open** in Project, and choose XML files in the file type list if the file does not show.

## Costs in Microsoft Project are zero

You left a rate at 0 in the export window. Either export again with your rates, or type the rates for each resource in Project's Resource Sheet.

## "Search failed" or "No cities found" on the weather screen

Check your internet connection and the spelling of the town. Try a larger nearby town if a small one is not found.

## The side menu only shows icons

The window is narrow, so the menu has shrunk to give the tables more room. Widen or maximise the window to bring the labels back.

## I deleted a task by mistake

There is no undo. Add the task again with **Add Task**. If you exported to Excel recently, the **DataInput** sheet has the old values to copy from.

## Frequently asked questions

### Does Time Pro draw a Gantt chart?

No. Time Pro works out durations and crews from your site records. Export to Microsoft Project to get the Gantt chart, change links, set baselines and track progress.

### Can I import a Microsoft Project file into Time Pro?

No. Time Pro exports to Microsoft Project but does not open or import Project files.

### Can I export to PDF?

Time Pro does not export PDF directly. Export to Excel and use Excel's print or Save as PDF, or print from Microsoft Project after exporting there. The Excel sheets are already set up to print.

### Do I need internet all the time?

No. You need it for your first sign-in and to fetch weather. After that you can record tasks on site without a connection, on the PC where you signed in.

### Can I use Time Pro on two computers?

Your licence is tied to one computer. To move it, contact [support](#) to release it from the old PC.

### Where does my Task Log live?

On the computer you use Time Pro on, in your Windows user profile. If two people share one Windows login on the same PC, they share one Task Log. Export to Excel regularly to keep a copy.

### How many days of records do I need?

There is no minimum, but more is better. A week of daily records on a live site is enough to start forecasting that trade. Records from different weather and different gangs make the average more reliable.

### Does the weather change my durations?

No. Time Pro records the weather so you can see it, but applies no adjustment. Use the **Weather Scenario** column to note what conditions your forecast assumes.

### Where do my BOQ quantities come from?

From your bill. You can measure them with [QUIV for Revit](#) or another ADLM takeoff tool, or take them from any bill of quantities, and type them into **BOQ Qty (Requested)**. Keep the unit the same as the unit you recorded output in.

### How much does Time Pro cost?

See the Time Pro page on [adlmstudio.net](https://adlmstudio.net) for current monthly and yearly prices.

# ADLM Cloud (overview)

Price, value, programme and share the projects you measure in QUIV, HERON, Revit MEP, CIVIQ and ArchiCAD, from any browser.

ADLM Cloud is the web side of your ADLM products. When you save a take-off to the cloud from QUIV, HERON, Revit MEP or CIVIQ, the project appears on adlmstudio.net. There you price the bill, plan the purchases, value the work done, issue interim certificates, run the programme and share progress with your client. You can do all of this from any computer or phone, without Revit or PlanSwift installed.

This guide covers the project workspace as it is on the live site at the start of October 2026.

## The ADLM Cloud guides

The Cloud guide comes in five parts. Read them in order the first time, or jump to the one you need.

| Guide                                                  | What it covers                                                                                                  |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <a href="#">Projects and your workspace</a>            | Find your way around the ADLM Cloud, open your projects, and use the project dashboard, 3D model and work area. |
| <a href="#">The bill and the budget</a>                | Price the Bill of Quantity, pick Rate Gen rates, plan materials and labour in the Budget, and export reports.   |
| <a href="#">Valuations and contract administration</a> | Value the work done, issue interim payment certificates, and run variations and the contract lock.              |
| <a href="#">Programme and project management</a>       | Track progress with the PM Dashboard and PM Tracker, and see every project at once on the Portfolio dashboard.  |
| <a href="#">Sharing, team and Ada</a>                  | Share a dashboard with your client, invite collaborators, control what they see, and get answers from Ada.      |

To learn on real data first, open a learning sample: see [Sample projects](#).

## What's new in 2026.09

### September 2026

- **Picking a Rate Gen rate prices the whole line.** A rate picked on the bill now prices the line's material, labour and plant in the Budget at the same time, and the build-up reproduces the rate to the kobo.
- **Learning samples.** Read-only worked projects for QUIV and HERON, then Revit MEP and CIVIQ, with every tab filled in.
- **New project screens.** A views sidebar you can hide, and a new **Work area** that joins the model, the bill, the schedule and Ada.
- **Continue where you left off.** Projects reopen on the tab you were last using.
- **Contract administration in the Valuation view.** **Certificates**, **Variations** and **Final account** sit together, and only approved variations change the contract.
- **A redesigned buy schedule.** Purchases can be ticked off, and the lead time is remembered.

- **A Summary box on the bill.** It totals the bill and adds **Mark as tendered**.
- **Your work is safer.** Re-saving a project from the plugin no longer wipes the rates, purchase marks and supplier notes you added on the website.
- **Money on shared projects is protected.** Collaborators without Rate Gen now see dashes instead of totals on project lists and cannot download priced exports. Certificates on merged contracts count in **Certified to date** again.

## July 2026

- PDF project, PM and management reports.
- Ada replaces the old help bot.
- Extra project storage slots in blocks of ten.
- The portfolio dashboard includes every project type.

## Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# ADLM Cloud: projects and your workspace

Find your way around the ADLM Cloud, open your projects, and use the project dashboard, 3D model and work area.

This is one part of the [ADLM Cloud guide](#). It covers the workspace as it is on the live site at the start of October 2026.

## Before you start

### What you need

- **An ADLM account.** Use the same email and password you use in the desktop products. Sign in at [adlmstudio.net](#).
- **A licence for the product the project came from.** To open a QUIV project you need an active QUIV licence; a HERON project needs HERON, and so on. Projects a colleague shares with you also need the matching product licence.
- **A Rate Gen licence (optional but recommended).** Without one you can still see and value a project, but you cannot pull rates from the Rate Gen library, and on projects shared with you the rates stay hidden. See [ADLM Rate Gen](#).
- **A browser.** Any up-to-date browser works. The 3D model viewer needs a browser with WebGL, which every modern desktop browser has.

### How a project gets here

You do not create projects on the website. A project starts in a desktop product:

| Product                                    | Where you measure | How it reaches the cloud                        |
|--------------------------------------------|-------------------|-------------------------------------------------|
| QUIV                                       | Revit             | Run QUIV and extract, then Save to ADLM Cloud   |
| HERON                                      | PlanSwift         | Take off with HERON, then Save to ADLM Cloud    |
| Revit MEP (shown on the website as SERVIQ) | Revit             | Run SERVIQ and extract, then Save to ADLM Cloud |
| CIVIQ                                      | Civil 3D          | Run CIVIQ and extract, then Save to ADLM Cloud  |

Every save from the plugin creates a new version of the project. Re-saving does not throw away the rates, purchase marks or supplier notes you added on the website.

#### NOTE

The steps for each plugin are in its own guide: [QUIV for Revit](#), [HERON](#) and [Revit MEP](#).

## Finding your way around

The signed-in pages have a menu down the left side. On a phone, open it with the menu button at the top.

- **Work**
  - **Overview** ( /work ): what needs you today, across all your projects.
  - **My tools**: one page each for **QUIV**, **HERON**, **RateGen**, **Constants**, **SERVIQ** (the website's name for Revit MEP) and **CIVIQ**. A tool you have no licence for is greyed out and marked **Add**.
  - **Projects** ( /work/projects ): every project on your account, whichever tool it came from.
- **Learn**: your courses, assignments, certificates and guides.
- **Manage: Overview, Products & seats, Team, Billing & invoices, Downloads and Support**.
- **Account settings** and **Sign out** sit at the bottom.

The search box at the top (**Search projects, rates and programmes**) jumps to a page by name. The bell next to it shows assignment reminders.

Signing in takes you to **Overview** under **Manage** ( /manage ). The old /dashboard address now opens the same page.

## Your work overview

**Overview** ( /work ) is one page, read from top to bottom, that answers "what am I in the middle of?".

The headline figures across the top:

| Figure                             | What it shows                                                          |
|------------------------------------|------------------------------------------------------------------------|
| <b>Measured work, all projects</b> | The value of measured work across every project (quantity times rate). |
| <b>Certified to date</b>           | The total of your approved and paid interim certificates.              |
| <b>Items waiting for a rate</b>    | Bill lines that have no rate yet. Click to go to the rate library.     |
| <b>Needs a decision</b>            | How many things are waiting on you, and how many are urgent.           |

Below the figures are these panels:

- **Needs a decision**: one table of things to act on, such as lines to price, variations to decide or overdue tasks. Each row has a button that opens the exact place where you deal with it (for example **Price**, **Decide** or **See task**). When the table is clear it says **Nothing is waiting on you**.
- **Continue where you left off**: the projects you last had open in this browser, reopening on the tab you were using.
- **Projects**: every project with its source, stage, how much is priced and complete, the measured value and the certified value. **All projects** opens the full gallery.
- **Valuations and variations**: your latest certificates and variations, most recent first, with their status (**Draft**, **Approved**, **Paid**, **Pending** or **Rejected**).
- **Programme**: tasks that are overdue, due in the next two weeks, or under way.
- **RateGen**: rates you changed recently and where they are used.
- **Learning**: assignments due.

If your account has no projects yet, the page says so and offers **Install the plugins** and **Open the rate library**.

# Projects

## The projects gallery

**Projects** ( /work/projects ) shows every project on your account as cards.

1. Type in **Search projects** to find a project by name.
2. Use **Source** (**All tools** or one product) to show one product's projects.
3. Use **Stage** to show projects at one stage: **Takeoff**, **Priced**, **Tendered**, **Contract locked**, **Valuations** or **Final account**.
4. Sort by **Recently updated**, **Estimated value**, **Name** or **Stage**.
5. Switch between **Grid** and **List** with the layout buttons. Your choice is remembered in this browser.
6. Click a card to open the project.

Each card shows the product, the stage, whether the project is **Yours** or **Shared with you**, its version, how much is valued, the estimated value and when it was last updated. Flags tell you when a **Share link on** is active, when a project carries a **Material schedule**, or when **Money hidden** applies (see [Sharing and collaborators](#)).

## A tool's page

Click a tool under **My tools**, for example **QUIV**, to see only that tool's projects. The page shows **Projects**, **Estimated** and **Last saved**, and a short **How a project starts** section with the three steps for that plugin. If you do not have that product, the page says so and offers a link to add it. Projects a consultant shares with you still open here.

## A product's project list

Each product also has a full project list:

| Product   | Page title         | Address             |
|-----------|--------------------|---------------------|
| QUIV      | QUIV projects      | /projects/revit     |
| HERON     | HERON projects     | /projects/planswift |
| Revit MEP | Revit MEP projects | /projects/mep       |
| CIVIQ     | CIVIQ projects     | /projects/civil3d   |

At the top of the list are summary tiles: the combined value of the visible projects, **Completed to date** with the share of lines marked, and **Outstanding balance**.

Under the tiles, **Cloud Storage** shows how many project slots you have used, for example "12 of 30 projects". At 80% you see **Upgrade storage**. When you reach the limit you see **Buy more storage**, and you must delete a project or buy more slots before you can save another. Extra slots are sold in blocks of ten on the purchase page.

The header buttons on this page:

- **Refresh projects** reloads the list, for example straight after a save from the plugin.
- **Portfolio Dashboard** opens the roll-up of all your projects (see [Portfolio dashboard](#)).
- **PM Tracker · QUIV** (QUIV list only) opens the standalone [PM Tracker](#).

- **Add shared project** adds a project a colleague shared with you by code.

## Selecting, merging and deleting

1. Click **Select** on each project card you want, or **Select all** for every project shown. **Clear** empties the selection.
2. Click **Delete selected** to delete them, or **Delete all** to delete every project in the list. You are asked to confirm, and this cannot be undone.
3. To delete one project, click **Delete** on its card. Only the owner sees this button; shared projects cannot be deleted by collaborators.

**Merge selected** combines two or more projects into one, for example the architectural and structural models of one building, or several buildings on one job.

1. Select two or more projects.
2. Click **Merge selected**.
3. Answer the first question: click **OK** if they are separate buildings on one job (each gets its own sheet in the exported bill), or **Cancel** if they are disciplines of one structure.
4. Type a name for the merged project and confirm.

The originals are kept and still open on their own in the plugin. Inside the merged project you can change the order of the buildings or disciplines with the up and down arrows.

## Sample projects

Above your own projects you may see **Learning samples**: worked projects with every tab filled in, so you can see what a priced, valued and certified job looks like before your own gets there. QUIV and HERON samples are worked duplex jobs, one per foundation type. Revit MEP samples cover the services for those duplexes, and CIVIQ samples are road jobs (asphalt, concrete, interlock and laterite).

- Click a sample card to open it. A banner reads **Sample project · Read-only learning material**, and **What to look at in this sample** lists the things worth looking at.
- You can open every tab, filter and export, but nothing can be changed.
- Click **Hide samples** to fold the strip away, or **Show N samples** to bring it back. This only hides them in this browser.

Samples appear for products you have an active licence for.

## Inside a project

### The project screen

When you open a project, the bar at the top holds:

- **Projects**: back to the list.
- The contract status: **Draft (editable)** or **Contract locked**.
- **Copy project ID**: copies the ID used by **Open from Cloud** in the Windows plugins.
- **Collaborators**: invite colleagues (owner only).
- **Delete**: delete the project (owner only).

- **Project report:** preview and download a PDF progress report. On the PM Dashboard this button reads **PM report**.
- **Export (Workbooks):** Excel exports, described in [Exports and reports](#).
- **Save changes:** saves your rates, progress and settings. It reads **Saved** when there is nothing new to save.

#### IMPORTANT

Changes to rates, progress and valuation settings are not kept until you click **Save changes**. Save before you leave the page.

Down the side is **Views**, the list of the project's screens in three groups. Click **Hide** to give the bill the full width, and **Views** to bring the list back.

| Group      | View                    | What it is for                         |
|------------|-------------------------|----------------------------------------|
| Overview   | <b>Dashboard</b>        | Overview and progress                  |
| Commercial | <b>Bill of Quantity</b> | Rates and line items                   |
| Commercial | <b>Budget</b>           | Cost plan & procurement                |
| Commercial | <b>Valuation</b>        | Certificates and settings              |
| Delivery   | <b>Work area</b>        | Model, bill, schedule and Ada together |
| Delivery   | <b>3D Model</b>         | View & verify the BIM model            |
| Delivery   | <b>PM Dashboard</b>     | Schedule, EVM, risks, issues           |

**3D Model** does not appear for HERON projects or for projects imported from an Excel bill, because they are measured from drawings rather than a model.

## Dashboard

The Dashboard answers three questions: what is the job worth, how much is done, and what is left.

- **Estimated total:** measured work, sums, preliminaries, contingency, VAT and approved variations.
- **Completed to date:** completed items plus executed PC sums, preliminaries and variations.
- **Outstanding balance:** the estimated total less what has been earned.
- **Progress:** the share of the work done.

**Progress overview** counts completed, remaining and total work items. If you record actual quantities or rates, **Actual vs planned performance** compares them with the plan. Switch the chart between **Donut**, **Bars** and **Trend**.

On QUIV and HERON projects, **Linked Services & Works** lets you link another project, such as the Revit MEP services for the same building, so the **Linked total** shows the whole job. Click **+ Link a project**, choose it from **Select a project...**, then click **Link**. **Unlink** removes it.

On Revit MEP projects, **Price services from RateGen** prices the services bill from your Rate Gen library. Click **Price services**. The lengths, joints and fittings it allows for come from your Services Constants (see [Material and Services Constants](#)).

The **Share dashboard** button on this view creates a public link for your client. See [Sharing a dashboard with your client](#).

## 3D Model

The 3D Model view opens the attached model in your browser.

1. Pick the discipline (**Architectural** or **Structural**).
2. Click a line in the **Bill of Quantity** panel to light up its elements in the model, or click an element to find the lines it was measured for.
3. **Material breakdown** shows the material lines behind the selection.
4. Click **Clear highlight** to start again.

## Work area

The Work area puts the model, the priced bill and the construction schedule on one screen. Pick a bill line and its elements light up. Pick a task and the lines it builds light up. Click an element and the lines and tasks behind it are found.

Ada sits beside them. Type a question in **Ask about this project**, or click **Ask Ada to explain** on the duration, time, cost and material summary. Ada only uses items that exist in your account; she proposes and you decide.

## Troubleshooting

### My project is not on the website

Check that the plugin's save finished. The plugin confirms when a project is saved to the cloud. Then click **Refresh projects** on the product's list. Make sure you are signed in to the website with the same account as the plugin. If the plugin started offline, sign out and back in with a connection, then save again.

### I cannot save another project: storage is full

The **Cloud Storage** bar shows you are at your limit. Delete projects you no longer need, or click **Buy more storage** to add slots in blocks of ten.

### My changes disappeared

Rates, progress and settings are only kept when you click **Save changes**. Make your edits again and save before leaving the project.

## Frequently asked questions

### Do I need Revit or PlanSwift to use ADLM Cloud?

No. You need the plugin to measure and save a project, but once it is on the cloud you can price, value, programme and share it from any browser, on any computer or phone.

### Can I delete a sample project?

No. Samples are read-only and cannot be deleted. Click **Hide samples** to fold them away.

### Why is the 3D Model view missing?

HERON projects are measured from 2D drawings, so they have no model to show.

### Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# ADLM Cloud: the bill and the budget

Price the Bill of Quantity, pick Rate Gen rates, plan materials and labour in the Budget, and export reports.

This is one part of the [ADLM Cloud guide](#). It covers the workspace as it is on the live site at the start of October 2026.

## The bill and the budget

### Bill of Quantity

The bill lists every measured line with its quantity, unit and rate. Above it are the totals (**Measured**, **PC**, **Variations**, **Project total**) and a ribbon of bill tools. Click **Hide tools** or **Show tools** to fold the ribbon.

The ribbon has six tabs:

| Tab         | What is on it                                                                              |
|-------------|--------------------------------------------------------------------------------------------|
| Home        | Only fill empty rates, Show actual qty / rate, and grouping By element or By trade         |
| Rates       | Rate Gen tools: Auto-sync rates, Sync rates, Follow RateGen changes and Load RateGen rates |
| Navigate    | Jump to a category or trade, and Top / Bottom of the bill                                  |
| Contract    | Lock contract or Unlock, Preliminaries % and the contract sum                              |
| Variations  | Add variation, Go to list and Approve / reject                                             |
| Provisional | Add PC sum, Add provisional sum and Go to Summary                                          |

To find a line, type in **Search items (description / group / S/N)...**

**Grouping.** Group the bill by building element (Substructure, Superstructure and so on) or by trade (Concrete Works, Formwork, Masonry and so on). To move a line to another group, drag it onto that category, or change its **Category:** or **Trade:**. ADLM remembers the change for next time. Use **+ New category** to add your own.

**Rows.** You can move a row up or down, or drag it into a new place. **Delete row (you'll be able to undo)** removes a line; the undo bar lets you bring it back.

**Summary.** The **Summary** box at the foot of the bill adds up measured work, **Preliminaries**, **PC sums**, **Provisional sums**, **Contingency**, **Approved variations** and **VAT** to an **Estimated total**. Contingency and VAT are percentages you can change there. When the priced bill goes out, click **Mark as tendered** so the project shows at the Tendered stage. **Not tendered after all** reverses it.

**Preliminaries.** **Preliminaries %** on the **Contract** tab sets the preliminaries as a percentage of measured work plus the sums (typically 5 to 10%). Under the bill, the preliminary items checklist shares that pool between items: set each **Alloc %**, record **Actual N** and tick **Done** as each item is executed. **Even split** shares the pool equally.

**PC and provisional sums.** On the **Provisional** tab, **Add PC sum** adds a prime-cost sum for a nominated supplier or subcontractor, and **Add provisional sum** adds an allowance for work not yet defined. Tick each one once it has been executed, so it counts towards earned value.

## Pricing lines and picking Rate Gen rates

Click a line's **Rate** cell. You can do three things there:

- **Type a number** for the rate.
- **Type a formula** starting with `=`, for example `=1.2*1.5*95000`. Only numbers and `+` `-` `*` `/` `(` `)` `%` are allowed. Press  or click away to apply it.
- **Type a name** to search your Rate Gen library (needs a Rate Gen licence). Pick a rate from the list.

When you pick a Rate Gen rate on the bill, ADLM prices that line's material, labour and plant in the Budget at the same time. The budget build-up reproduces the rate you picked, so the bill and the budget agree. If the rate has no build-up, the rate still goes on the line and a message tells you the budget could not be priced.

Rates you type or pick stay fixed on the line. To let the budget set the rate again, clear the cell. The line then says the rate has been released and will be priced from the budget build-up when you save.

On the **Rates** tab:

- **Load RateGen rates** loads your library so searches are fast.
- **Sync rates** fills rates from the library. With **Only fill empty rates** ticked (on the **Home** tab), lines you have already priced are left alone.
- **Follow RateGen changes**: while this is on, a rate you change in your Rate Gen library is applied to this bill too. Turn it off before the bill goes out, so the prices stay as issued.

### TIP

To price many similar lines at once, link them. A line marked "Linked: rate changes propagate to similar items" passes its rate on to the others.

## Budget

The Budget is the **Material & Labour breakdown** of every bill item, laid out in the same order and sections as the bill. For each bill item:

**Bill Rate = Material + Labour + O&P**

Each row is a resource: **Material**, **Labour**, **Plant**, **Equipment** or **Consumable**.

1. Price a row by typing a rate, entering a `=` formula, or searching Rate Gen in the rate cell.
2. Set Overhead and Profit for the item, or use **Global Overhead & Profit** and click **Apply to all** to write the same figures onto every item.
3. Click **Save changes**. The new bill rate flows up to the Bill of Quantity.

**Marking purchases.** A bill item only counts as complete when every line under it is marked procured or done. Buying the materials is not enough until the labour is done too.

- Tick a line to mark it procured.
- Use **Mark all** for a whole bill item, or **Unmark all** to undo.

**Buy schedule.** Switch from **Breakdown** to **Buy schedule** to see what to buy and when:

- **Should already be bought**: late purchases.
- **Buy this week**: what is due now.
- **Not yet scheduled**: materials for work with no dates yet.

- **Lead time:** how many days before work starts each material should be bought (14 days unless you change it).

The buy schedule takes its dates from the tasks on the PM Dashboard, so plan the programme first.

#### NOTE

Once the contract is locked, procurement marking on the Budget is frozen. Bill items you mark complete on the Bill of Quantity still show as done here.

## Material and Services Constants

**Material Constants** (Constants under **My tools**, </work/constants>) are the factors behind every material and labour schedule, such as how many bags of cement go into a cubic metre of concrete. They are the same constants QUIV and HERON use on the desktop.

1. Find a constant with **Search constants....**
2. Change its value. The **ADLM default** column shows the starting figure.
3. Click **Save changes**. **Reset** on a row, or **Reset all to defaults**, puts the ADLM figures back.

New schedules use your constants straight away. An existing project keeps its figures until its schedule is rebuilt.

**Services Constants** (</rategen/services-constants>) are your house standards for pricing MEP services: standard lengths, how connectors are counted and the fitting uplift. They feed **Price services** on Revit MEP projects. Choose the **Unit system**, edit the rows and click **Save constants**.

## Exports and reports

**Excel workbooks.** Click **Export** in a project's top bar and choose:

| Group         | Options                                                                                                                                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bill & Budget | <b>Export bill &amp; budget workbook:</b> your bill with its own sections and totals, plus separate Material, Labour and Plant schedules, a Schedule of Current Prices and a Material Summary. <b>Export bill &amp; budget (by trade):</b> the same, sectioned by trade. |
| Generic BoQ   | <b>Export generic BoQ (by category)</b> and <b>Export generic BoQ (by trade)</b>                                                                                                                                                                                         |
| Elemental BoQ | <b>Bungalow</b> or <b>Multi-storey</b> , grouped by building element                                                                                                                                                                                                     |
| Trade BoQ     | <b>Bungalow (Trade format)</b> or <b>Multi-storey (Trade format)</b> , one bill per work section                                                                                                                                                                         |
| Milestone BoQ | <b>Bungalow (Milestone format)</b> or <b>Multi-storey (Milestone format)</b> , one priceable bill per construction stage, as the basis for a payment schedule                                                                                                            |

**PDF reports.**

- **Project report:** the Project Progress Report for the open project.
- **PM report:** the Project Management Report (schedule and earned value), from the PM Dashboard.
- **Management report:** across all your projects, from the Portfolio Dashboard.

Each opens a preview first. Click **Download PDF** to save it.

**Valuations and certificates.** **Print valuation** and **Export Excel** in the daily valuation log, the certificate downloads in **Certificates**, and the final account **Download** are described above.

## Troubleshooting

### I cannot change rates or delete lines

The contract is probably locked: the top bar says **Contract locked**. Raise a variation instead, or unlock the contract from the **Contract** tab on the bill ribbon. If the project is shared with you at **View only**, you cannot edit it at all; ask the owner for **Full access**.

### The budget says "Re-save this project from the plugin"

The project was saved before the material and labour breakdown was sent. Open it in the plugin and save it to the cloud again.

### Picking a rate did not price the Budget

The rate has no build-up in Rate Gen, it is no longer in your library, or you only have view access. The rate still goes on the line; price the Budget rows yourself, or choose a rate that has a build-up.

## Frequently asked questions

### Does re-saving from the plugin overwrite my prices?

No. Each save is a new version. Rates, purchase marks, supplier details and notes you added on the website are kept.

### Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# ADLM Cloud: valuations and contract administration

Value the work done, issue interim payment certificates, and run variations and the contract lock.

This is one part of the [ADLM Cloud guide](#). It covers the workspace as it is on the live site at the start of October 2026.

## Valuations and the contract

### Valuation

The Valuation view is where you value work done and run the contract.

**Recording progress.** Progress is recorded on the Bill of Quantity:

1. Tick a line's box to mark it fully done (**Mark as completed**), or type a percentage in its **Done** box. Partial values are paid pro rata.
2. Click **Save changes**. Each save is logged against that day in the daily valuation log.

**Valuation workspace.** At the top of the Valuation view:

- **Valuation basis:** choose **By Bill of Quantity** (each line valued by its own % complete) or **By Budget (Material & Labour)** (each line valued from its material and labour lines marked on the Budget).
- **Show daily valuation log** and **Show valuation settings** turn those sections on and off.

**Valuation settings.** These are saved per project:

| Setting           | Default | What it does                                         |
|-------------------|---------|------------------------------------------------------|
| Client / Employer | blank   | The client's name on the valuation and exported bill |
| Retention %       | 5       | Held back from the gross value                       |
| VAT %             | 7.5     | Added to the subtotal                                |
| Withholding tax % | 2.5     | Deducted from the subtotal                           |

Set these once, when the contract is agreed, so every later valuation is worked out the same way.

### The daily valuation log.

1. Choose a day in **Valuation date**.
2. Read down the figures: **Gross value of works to date**, **Net valuation to date** (after retention), **Less previous payments** ("Not applicable for first valuation" on the first one), **Subtotal before taxes**, and **Total amount due for payment**.
3. Click **Print valuation** for the formal Interim Payment Application, or **Export Excel** for an editable workbook.

## Contract administration

Below the valuation is **Contract administration**, with four views: **Certificates**, **Variations**, **Final account** and **BIM models**.

**Locking the contract.** Locking freezes the priced bill as the contract sum. After that, new items become variations and re-measured quantities are recorded as actuals, so the contract figure never moves.

1. Open the **Bill of Quantity** and click the **Contract** tab on the ribbon.
2. Click **Lock contract**.
3. Choose a 4-digit PIN and type it twice. If your account uses email verification for sensitive actions, you enter the code sent to your email instead.

### IMPORTANT

Keep the PIN safe. You need the same 4 digits to unlock the contract, and a lost PIN cannot be recovered without asking ADLM support.

To unlock, click **Unlock** on the same tab and enter the PIN (or the emailed code).

### Interim certificates.

1. Open **Certificates**.
2. Click **Issue certificate**. It certifies the current value to date as the next IPC.
3. Set each certificate's status as it moves along: **Draft**, **Approved**, **Paid**.
4. Download any certificate as an Excel workbook.

Each certificate row shows **Cumulative**, **Less prev.**, **This cert**, **Retention**, VAT, withholding tax and **Net payable**. Only the latest certificate can be deleted, which keeps the sequence honest.

### TIP

Lock the contract before the first certificate, so every certificate is measured against the same baseline.

### Variations.

1. Open **Variations** and click **Add variation**.
2. Fill in **What changed**, the **Instruction reference**, the **Type** (**Addition** or **Omission**) and the **Value (N)**.
3. A new variation waits for approval and is not counted yet. Open it and click **Approve** or **Reject**.

Only approved variations move the project total. The tiles show **Approved, net**, **Additions**, **Omissions** and **Waiting for approval**. Lines added to a locked contract from the plugin are raised as variations for you.

**Final account.** This shows the closing settlement: **Measured work**, **Preliminaries**, **Provisional and PC sums**, contingency and **Approved variations**, set against the **Contract sum**, with **Certified so far**. The movement reads as an over-run, **Under-run (savings)** or **On budget**.

1. When the job is finished, click **Close the final account**. The contract must be locked first.
2. Confirm. Items, variations and certificates are frozen.
3. Click **Download** for the final account workbook.

To make an adjustment later, click **Reopen**. No new certificate can be issued while the final account is closed.

**BIM models** (QUIV, Revit MEP and CIVIQ projects). When you save from QUIV, the model is exported to IFC and attached to the project automatically, one slot each for **Architectural**, **Structural** and MEP. Each slot

shows whether it is **Attached** and whether its element IDs match the bill. If the model does not cover the bill, the missing element IDs are listed.

If the plugin was not used, click **Advanced: manual upload** and then **Upload manually** (or **Replace manually**) to attach an IFC file yourself. Files can be up to 100 MB.

## Troubleshooting

### I forgot the contract PIN

The PIN cannot be recovered on the website. Contact ADLM support from **Support** ( </manage/support> ) to have it reset.

## Frequently asked questions

### What happens when I lock the contract?

The priced bill becomes the contract sum and stops moving. New items become variations, re-measured quantities are recorded as actuals, and certificates are measured against the locked figure. HERON shows the lock too and saves your edits as actuals.

### Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# ADLM Cloud: programme and project management

Track progress with the PM Dashboard and PM Tracker, and see every project at once on the Portfolio dashboard.

This is one part of the [ADLM Cloud guide](#). It covers the workspace as it is on the live site at the start of October 2026.

## Project management

### PM Dashboard

The PM Dashboard is the programme for the project, tied to the bill.

To build the task list, use one of these buttons:

- **Generate from BoQ** creates one task per bill item.
- **Import MS Project** brings in a programme from a Microsoft Project file (.xml or .mpp). Once tasks exist, the button reads **Update MS Project**, which refreshes the schedule and keeps your progress and links. **Delete imported tasks** removes the imported tasks again.
- **Add Task**, **Add Risk** and **Add Issue** add items by hand.

The dashboard shows tiles for **Progress**, **Budget Used**, **Overdue**, **CPI**, **SPI** and **Tasks Done**. Below them are charts for **Tasks**, **Budget**, **Tasks by priority** and **Burndown**, and an **Earned Value Summary**.

Earned value in plain terms:

| Figure    | Meaning                                                              |
|-----------|----------------------------------------------------------------------|
| BAC       | Budget at completion: what the job is meant to cost in total.        |
| PV        | Planned value: what should have been done by now.                    |
| EV        | Earned value: the budget value of the work actually done.            |
| AC        | Actual cost of that work.                                            |
| EAC       | Estimate at completion: what the whole job now looks likely to cost. |
| VAC       | Variance at completion: EAC against BAC.                             |
| CPI / SPI | Cost and schedule performance. Below 1 means over budget or behind.  |

Click **View Details** for **WBS / Tasks**, **Risk Register** and **Issue Log**.

- In the task table, set each task's %, its status (**Not started**, **In progress**, **Completed** or **Blocked**) and its actual days. Setting 100% marks the task Completed.
- **Export calendar** downloads the tasks as a calendar file.
- **Reschedule** moves the tasks after a date changes.
- Click **Save changes** when you are done.

## PM Tracker

The PM Tracker ( /pm-tracker ) is for jobs you want to programme without a bill: tasks, risks and issues only. It is a QUIV feature, so you need an active QUIV licence.

1. Open your **QUIV projects** list and click **PM Tracker · QUIV**.
2. Click **New Project**, type a **Project name** and click **Create project**.
3. Open the project. Use **Dashboard** for the PM dashboard described above and **Details** for the task, risk and issue tables.

You can keep up to 10 PM Tracker projects. Delete one to add another.

**Sharing a PM Tracker project.** Click **Share**.

- **View Only:** tick **Enable public link** and copy the link. Your client sees a read-only dashboard without signing in. Untick it to switch the link off.
- **Invite Editor:** type a colleague's email and click **Send Invite Email**. They receive a single-use join link, tied to that email address, that lets them edit tasks, risks and issues.

## Portfolio dashboard

**Portfolio Dashboard** ( /portfolio-dashboard ) rolls up every project you can see, across all products. Open it from the button on any project list.

- **Overview** shows the overall progress ring and tiles for **Total projects**, **Completed items**, **Remaining items**, **Total work items**, **Planned total**, **Completed to date**, **Outstanding balance** and **Overall progress**. The **Project breakdown** table lists each project with its product, status, items, BoQ total, completed value and progress.
- **Charts** compares products: **BoQ Total vs Completed**, **by Product**, **Projects by Product**, **Overall Delivery** and **Progress by Product**.
- **Report** gives an executive summary and a breakdown by product.

Buttons in the header:

- **Refresh** reloads the figures.
- **Export Excel** downloads the portfolio as a workbook.
- **Management report** opens a PDF report across your whole organisation, which you can preview and download.

## Troubleshooting

### My model was refused, or a slot says the model does not cover the bill

The model is missing element IDs that priced lines were measured from. It is usually an older revision or the wrong file. Re-export the IFC from the same Revit model the take-off came from, then save again from QUIV or use **Upload manually**.

## Frequently asked questions

### Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# ADLM Cloud: sharing, team and Ada

Share a dashboard with your client, invite collaborators, control what they see, and get answers from Ada.

This is one part of the [ADLM Cloud guide](#). It covers the workspace as it is on the live site at the start of October 2026.

## Sharing and collaborators

There are two ways to let someone else see a project. They are not the same.

|                | Public dashboard link         | Collaborator                                                 |
|----------------|-------------------------------|--------------------------------------------------------------|
| Who it is for  | A client with no ADLM account | A colleague or consultant who will work on the project       |
| Sign-in needed | No                            | Yes, with their own account and the matching product licence |
| What they see  | A summary dashboard only      | The whole project, at <b>View only</b> or <b>Full access</b> |
| Can edit       | No                            | Only with <b>Full access</b>                                 |

### Sharing a dashboard with your client

1. Open the project's **Dashboard**.
2. Click **Share dashboard**.
3. Tick **Enable public link**.
4. Click **Copy link** and send it to your client.

The button then reads **Shared · link on**. Your client sees a read-only Project Dashboard: overall status, progress, the contract sum and its make-up, cost to date and forecast, interim certificate totals, and the planned and actual spend. They cannot edit anything.

#### IMPORTANT

Anyone who has the link can open the dashboard. When you no longer want it shared, untick **Enable public link**. Ticking it again brings back the same link.

Only the project owner can turn the public link on or off.

### Inviting collaborators

1. Open the project and click **Collaborators**.
2. Under **Access level**, choose **View only** or **Full access**.
3. Optionally fill in **Label (optional)** (for example the firm's name), **Restrict to emails (optional)** and **Max uses (0 = unlimited)**.
4. Click **Generate code**.
5. Under **Active codes**, click **Copy** to copy the code, or **Link & QR** for a join link and QR code. **Copy link** copies the link and **Download QR** saves the QR image.

Send the code or link to your colleague. **People with access** lists everyone who has joined. You can change each person between **View only** and **Full access**, or click **Remove**. **Revoke** stops a code being used again.

## Joining a project someone shared with you

You can use either a link or a code.

- **With a link or QR code:** open it. Sign in if asked. The page shows **Joining project...** and then opens the project.
- **With a code:** open the project list for that product, click **Add shared project**, type the **Share code** and click **Add project**.

If you do not have the matching product, you see **Subscription required** and a button to get it. A code that has been revoked, has reached its use limit or is restricted to other emails is refused.

## What collaborators can see and do

|                                    | Owner | Full access                  | View only                    |
|------------------------------------|-------|------------------------------|------------------------------|
| Edit rates and progress            | Yes   | Yes                          | No                           |
| Download exports and reports       | Yes   | Yes                          | No                           |
| Invite people, delete, public link | Yes   | No                           | No                           |
| See rates and money                | Yes   | Only with a Rate Gen licence | Only with a Rate Gen licence |

A collaborator without an active Rate Gen licence sees the message **Rates hidden. A RateGen subscription is required to view rates.** For them, rates and money totals show as a dash (on project cards this appears as **Money hidden**), bill and budget exports are refused, and they cannot raise variations, rebuild the schedule or price services.

## Team

**Team** under Manage ( </manage/team> ) shows your account's seats and the machines they are activated on.

- **Members** lists the account holder. Adding colleagues to one account is not available yet, so each person signs in with their own ADLM account. To work together on a project, use [collaborators](#).
- **Machines** lists the computers holding a seat. To move a seat to a new computer, click **Free the seat** on the old one and then install on the new one. Nothing is deleted.
- **Seats per product** and **Free activations** show how many seats you own and how many are free right now. **Buy more seats** opens the purchase page.

## Ada, your assistant

Ada is ADLM's assistant. Click **Ask Ada** at the bottom of any page, or in the left-hand menu.

- Anyone can ask Ada about products, prices, trainings, downloads and material quantities. The suggestion buttons **Products**, **Trainings**, **Software downloads** and **Material quantities** are a quick start.
- When you are signed in, Ada can also answer questions about your own projects and account, for example which projects have a 3D model, or what a project's bill or budget comes to. She only reads your own data.

- Ada answers from what ADLM publishes. If she does not know, she says so.
- To talk to a person, choose **Chat on WhatsApp** in one of her answers.

Ada is not a quote. Use the quote page for prices you can rely on.

#### NOTE

ADLM can set a usage allowance for Ada. If you reach it, she tells you and offers WhatsApp instead until the allowance resets.

## Troubleshooting

### Rates show as dashes on a shared project

You are a collaborator without an active Rate Gen licence. Rates and money stay hidden until you have one. The project owner always sees them.

### "Subscription required" when I open a share link

You need an active licence for the product the project was made in (for example HERON for a HERON project). Click the button on the page to get it, then open the link again.

### The share code does not work

The code may have been revoked, reached its use limit, or been limited to other email addresses. Ask the owner for a new code.

## Frequently asked questions

### Can my client see the project without an ADLM account?

Yes. Turn on **Share dashboard** and send the link. Your client sees a summary dashboard and cannot change anything.

### What is the difference between View only and Full access?

**View only** collaborators can look but cannot edit or download. **Full access** collaborators can edit rates and progress and download exports. Neither can invite others, delete the project or share it publicly; only the owner can.

### Can I add colleagues to my account?

Not yet. Each person uses their own ADLM account. Share individual projects with them as collaborators.

### Where do I get help?

Ask Ada, chat with us on WhatsApp, or raise a ticket from **Support** ( </manage/support> ).

# Sample projects

Twenty finished, read-only jobs you can open on ADLM Cloud to see a bill, a budget, valuations and a programme filled in from start to finish.

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Sample projects are complete, worked jobs that ADLM Studio has put on ADLM Cloud for you to learn from. Each one is a real-looking Nigerian job: a priced bill of quantities, a budget with suppliers, a locked contract, interim certificates, variations, a programme with risks and issues, and, for the model-based products, a 3D model. When your own first project has only a bill in it, a sample shows you what every other tab will look like once the job is running.

There are four samples for each product that has a cloud project screen: QUIV for Revit, HERON for PlanSwift, SERVIQ for Revit MEP, CIVIQ for Civil 3D and QUIV for ArchiCAD. Every sample is read-only. You can open it, read it, filter it and export it, but nobody can change it, save over it or delete it.

This guide lists every sample, says what is in it and what it teaches, and ends with five short exercises.

## Before you start

### What you need

- **An ADLM account.** Sign in at [adlmstudio.net](https://adlmstudio.net).
- **An active licence for the product.** Samples appear only for products you hold an active licence for. A QUIV licence shows you the QUIV samples, a HERON licence the HERON samples, and so on. If your licence has expired, the samples for that product disappear with your own projects until you renew. See [Getting started](#) and **Products** at [/manage](#).
- **Nothing else.** You do not need Revit, PlanSwift, Civil 3D or ArchiCAD installed to open a sample. A Rate Gen licence is not needed either: samples show every rate to everyone.

### What a sample is, and what it is not

- A sample is **learning material**. The figures are worked properly (rates come from a material and labour build-up, the contract sum is calculated the same way as on your own jobs, and the certificates add up), but the clients, sites and suppliers are made up.
- A sample is **not a template**. You cannot copy it into your account or start your own project from it. Your own projects still start in the desktop plugin.
- A sample is **not yours**. It does not appear in your project list or gallery, it does not use any of your cloud storage slots, and it is not counted in your totals on the Overview page.

## Finding the samples

### On a product's project list

1. Sign in at [adlmstudio.net](https://adlmstudio.net).
2. Open the product's project list, for example **QUIV projects** ( [/projects/revit](#) ), **HERON projects** ( [/projects/planswift](#) ), **Revit MEP projects** ( [/projects/mep](#) ) or **CIVIQ projects** ( [/projects/civil3d](#) ).

3. Above your own project cards, look for the **Learning samples** strip. It reads, for example, "Worked duplex projects, one per foundation type, each measured from its own 3D model. Open one to see every tab filled in. Samples are read-only."
4. Click a sample card to open it.

## On a tool's page

1. Under **My tools**, click the tool, for example **QUIV** ( </work/tool/quiv> ).
2. Scroll past **How a project starts**. The **Learning samples** strip sits there too.
3. Click a sample card to open it.

## For QUIV for ArchiCAD

ArchiCAD samples are on the **QUIV for ArchiCAD** page ( </archicad> ), above **Your ArchiCAD projects**. See [Early-access products](#).

## Reading a sample card

Each card shows:

| On the card                        | What it means                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| The flag at the top left           | The foundation type, pavement type or building the sample is about, for example <b>Raft foundation</b> or <b>Interlocking paving</b> |
| <b>Read-only</b> (with an eye)     | You can look but not change                                                                                                          |
| The name                           | The job, for example "5-Bedroom Duplex - Raft Foundation"                                                                            |
| The two short lines under the name | The site, then the stage the job has reached                                                                                         |
| <b>Contract</b>                    | The locked contract sum, in millions of naira                                                                                        |
| <b>Lines</b>                       | How many priced lines are in the bill                                                                                                |
| <b>Certificates</b>                | How many interim certificates have been issued                                                                                       |
| <b>3D model</b>                    | Shown when the sample has a model you can open                                                                                       |

## Hiding and showing the strip

- Click **Hide samples** to fold the strip away once you no longer need it.
- Click **Show 4 samples** to bring it back. The number is how many samples that product has.

### NOTE

Hiding the strip only hides it in the browser you are using. On another computer or phone it starts open again.

## Read-only: what you can and cannot do

When a sample opens, an orange banner across the top reads **Sample project · Read-only learning material**, followed by the foundation or job type and the stage. Under it is a one-line summary of the job and the ground

conditions, and a fold-out list called **What to look at in this sample**. The banner ends: "You can open every tab, filter, and export, but nothing can be changed. Sync your own model from the plugin to start a project of your own."

| You can                                                                                                                                  | You cannot                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Open every view: <b>Dashboard, Bill of Quantity, Budget, Valuation, Work area, 3D Model</b> (where there is one) and <b>PM Dashboard</b> | Change a rate, a quantity, a percentage done or a setting                                |
| Search and filter the bill and the budget                                                                                                | Save changes. The server refuses with "Sample projects are read-only learning material." |
| See every rate and every money total, with or without Rate Gen                                                                           | Delete the sample, merge it with another project or select it for bulk actions           |
| Download the Excel workbooks and reports                                                                                                 | Add certificates, variations, tasks, risks or issues                                     |
| Open the 3D model and pick elements                                                                                                      | Invite collaborators or share it with a code                                             |

A few more points:

- **Nobody can edit a sample.** It is the same for every customer, and ADLM Studio does not edit it by hand either. If one is ever updated, it is replaced as a whole.
- **Samples never reach the desktop plugins.** The project list the plugins read never includes samples, so you cannot open a sample in Revit or PlanSwift and save your work over it.

#### TIP

Because you cannot break a sample, it is the safest place to try things. Click every tab, open every certificate, and download every export. Then do the same on your own project.

## How samples relate to your own projects

A sample is laid out exactly like one of your own projects. The same views, the same columns, the same tiles. The difference is that a sample has already been through the whole job.

| Stage of a job  | Your new project                                      | A sample                                                        |
|-----------------|-------------------------------------------------------|-----------------------------------------------------------------|
| Takeoff         | Lines arrive from the plugin                          | Done                                                            |
| Priced          | You price the bill, or let the budget build the rates | Done: every line priced from a material and labour build-up     |
| Contract locked | You lock the contract sum                             | Done: locked at the agreed sum                                  |
| Valuations      | You record progress and issue certificates            | One to four certificates already issued                         |
| Final account   | You agree the final account                           | Done on the pile duplex, its MEP services and the laterite road |

To start a project of your own:

1. Open your model or drawings in the desktop product.
2. Measure with the plugin.

3. Save to ADLM Cloud. The project appears on the same page as the samples, below the **Learning samples** strip.

The product guides explain each step: [QUIV](#), [HERON](#), [SERVIQ for Revit MEP](#), and the cloud workspace in [ADLM Cloud](#).

## The four duplexes

QUIV, HERON and QUIV for ArchiCAD all use the same four duplex designs, one for each common foundation type. SERVIQ uses the services for the same four buildings. Learning one duplex once helps you read it in every product.

| Duplex                                                       | Client                    | Site                     | Ground                                          | Stage reached                                                                      |
|--------------------------------------------------------------|---------------------------|--------------------------|-------------------------------------------------|------------------------------------------------------------------------------------|
| 4-Bedroom Duplex - Strip Foundation                          | Mr & Mrs Adebayo Ogunleye | Oluyole Estate, Ibadan   | Firm lateritic clay, 150 kN/m2                  | Interim Certificate 1 issued: substructure complete, ground floor frame under way  |
| 4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation | Engr. Chidi Nwosu         | Lokogoma District, Abuja | Stiff sandy clay over weathered rock, 200 kN/m2 | Interim Certificate 2 issued: first floor slab cast, first floor walls rising      |
| 5-Bedroom Duplex - Raft Foundation                           | Mrs Funmilayo Bakare      | Lekki Phase 1, Lagos     | Soft silty clay, 60 kN/m2, water table 1.8 m    | Interim Certificate 3 issued: roofed, doors and windows going in, finishes started |
| 4-Bedroom Terrace Duplex - Bored Pile Foundation             | Harbourview Homes Ltd     | Ikate, Lekki, Lagos      | Loose reclaimed sand over peat to 9 m           | Practical completion reached and final account agreed                              |

Every duplex job is priced the same way: bill rates come from a material and labour build-up with 10% overhead and 15% profit. The contract adds 7.5% preliminaries, 5% contingency and 7.5% VAT. Valuations hold 5% retention, add 7.5% VAT and deduct 2.5% withholding tax. Each duplex also carries the same three PC and provisional sums: electrical installation (₦4.5m), plumbing and water supply (₦3.8m), and external works, soakaway and septic tank (₦2.6m).

The programme on the **PM Dashboard** follows seven phases: Substructure; Ground floor frame and blockwork; First floor slab, beams and staircase; First floor frame, blockwork and roof beam; Roof carpentry and covering; Doors and windows; and Finishes. Each phase is broken down into the trades a planner would draw, such as formwork, reinforcement and the concrete pour, rather than one long bar.

## QUIV samples (Revit)

QUIV samples are measured from their own 3D Revit models. Each has an **Architectural** and a **Structural** model attached, and every bill line remembers the elements it was measured from. The strip reads "Worked duplex projects, one per foundation type, each measured from its own 3D model."

The bill uses QUIV's own descriptions, for example "Foundations – Excavation (trench)", grouped into the work sections Earthworks, Concrete Works, Formwork, Reinforcement, Masonry, Damp-proofing, Carpentry & Roofing, Joinery, the three finishes sections (wall, floor and ceiling) and Decoration.

## Strip foundation

- **Name on the card:** 4-Bedroom Duplex - Strip Foundation. Flag: **Strip foundation**.
- **The job:** a two-storey, four-bedroom duplex in Ibadan on firm clay.
- **What is in it:** 65 bill lines (12 concrete, 11 reinforcement, 9 formwork, 7 earthworks and the rest across masonry, roofing, joinery and finishes), 222 budget lines, a contract sum of about ₦127.6m, one approved certificate (about 19% of the contract valued), and two variations: AI-01 deepens the strip footing at the rear wall after a soft spot was found, and AI-02 adds a concrete apron round the building.
- **What it teaches:** the start of a job. Substructure is complete, the ground floor frame is under way, and only one certificate has been issued, so you can see what a first valuation looks like and how a variation found during excavation is priced at the bill rate.

## Pad and strip foundation

- **Name on the card:** 4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation. Flag: **Pad and strip foundation**.
- **The job:** a larger four-bedroom duplex with a family lounge in Abuja, with pad bases under the columns and strips under the walls.
- **What is in it:** 67 bill lines (including separate excavation to trenches and to pits), 228 budget lines, a contract sum of about ₦157.4m, two certificates (the first paid, the second approved, about 42% valued), and two variations: a cantilever balcony to the master bedroom, and burglary-proof laminated glass to the ground floor windows.
- **What it teaches:** how one certificate follows another. Look at **Less previous payments** on certificate 2. The risk register also shows a risk that has been closed (weathered rock met at 1.2 m).

## Raft foundation

- **Name on the card:** 5-Bedroom Duplex - Raft Foundation. Flag: **Raft foundation**.
- **The job:** a five-bedroom duplex in Lekki Phase 1 on soft clay with a high water table, so it sits on a raft.
- **What is in it:** 64 bill lines (including excavation to reduce levels for the raft and disposal of surplus material), 221 budget lines, a contract sum of about ₦209.0m, three certificates (two paid, one approved, about 68% valued), and three variations: dewatering during the raft excavation, porcelain floor tiles to the living room, and extra rainwater goods to the rear. The electrical PC sum has been executed.
- **What it teaches:** a job well into finishes. The budget shows materials for later phases being bought ahead of time, and the issue log has three issues at different stages (resolved, in progress and open).

## Bored pile foundation

- **Name on the card:** 4-Bedroom Terrace Duplex - Bored Pile Foundation. Flag: **Pile foundation**.
- **The job:** a terrace duplex in Ikate, Lekki, on reclaimed sand over peat, carried on bored piles.
- **What is in it:** 72 bill lines (the most of the four, because piling adds concrete, reinforcement and formwork lines), 246 budget lines, a contract sum of about ₦146.6m, four certificates (all paid, about 90% valued), and three variations: pile integrity testing, extending the piles on grid A by 2 m after deeper peat was found, and a granite kitchen worktop. All three PC and provisional sums are executed.
- **What it teaches:** the end of a job. The last certificate releases half the retention at practical completion, and the **Final account** is agreed, with a final contract value of about ₦132.4m, roughly ₦14.2m under the contract sum.

## HERON samples (PlanSwift)

HERON samples are the same four duplexes, measured from PDF drawings in PlanSwift instead of from a model. The strip reads "Worked duplex projects, one per foundation type, measured from PDF drawings."

| Name on the card                                             | Flag                     | Bill lines | Budget lines | Contract      | Certificates | Final account |
|--------------------------------------------------------------|--------------------------|------------|--------------|---------------|--------------|---------------|
| 4-Bedroom Duplex - Strip Foundation                          | Strip foundation         | 65         | 222          | about ₦127.6m | 1            | No            |
| 4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation | Pad and strip foundation | 67         | 228          | about ₦157.4m | 2            | No            |
| 5-Bedroom Duplex - Raft Foundation                           | Raft foundation          | 64         | 221          | about ₦209.0m | 3            | No            |
| 4-Bedroom Terrace Duplex - Bored Pile Foundation             | Pile foundation          | 72         | 246          | about ₦146.6m | 4            | Yes           |

The quantities, rates, certificates, variations, risks and issues match the QUIV sample of the same duplex. What differs is how the bill reads:

- **Descriptions are written out in full**, the way a QS writes them from drawings. "Foundations – Excavation (trench)" in QUIV becomes "Excavate trenches for foundations, not exceeding 2.00m deep" in HERON.
- **Lines are grouped by takeoff folder**: Substructure, Frame, Blockwork, Roofing, Doors & Windows and Finishes.
- **There is no 3D model**, so the **3D Model** view does not appear.

**What they teach:** that a drawing-based takeoff ends up in exactly the same commercial workflow as a model-based one. Open the HERON and QUIV samples of the same duplex side by side and compare the bill descriptions against the identical totals.

## SERVIQ samples (Revit MEP)

SERVIQ samples are the building services for the same four duplexes: electrical, plumbing and drainage, and air conditioning. Each has an MEP model attached, so you can pick a bill line and see the fittings and pipe runs it was measured from. On the website the product appears as **Revit MEP**. The strip reads "Services for the worked duplexes: electrical, plumbing and drainage, and air conditioning."

Every MEP sample has 45 bill lines in three sections (17 Electrical Installations, 24 Plumbing & Drainage and 4 HVAC) and 110 budget lines. The PC and provisional sums are a utility meter and service connection (₦0.85m), a borehole and submersible pump (₦2.4m) and a solar inverter with battery backup (₦4.2m). The programme has six phases: electrical first fix, plumbing and drainage rough-in, electrical second fix, sanitary fittings and pumping, air conditioning and ventilation, and testing and commissioning.

### Strip duplex services

- **Name on the card:** 4-Bedroom Duplex - Strip Foundation - MEP Services. Flag: **Strip duplex**.
- **What is in it:** 36 light points, 36 sockets, 7 split units and 16 sanitary fittings. Contract about ₦29.7m, one approved certificate (about 6% valued), one variation (an outdoor socket and garden light circuit).

- **What it teaches:** services at their earliest stage, with conduits cast into the first floor slab and plumbing sleeves in. A first certificate on a services contract is small; this one shows why.

### Pad and strip duplex services

- **Name on the card:** 4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation - MEP Services. Flag: **Pad and strip duplex.**
- **What is in it:** 40 light points, 42 sockets, 9 split units and 22 sanitary fittings. Contract about ₦34.9m, two certificates (about 21% valued), two variations (a rain shower upgrade and CCTV conduit for six cameras).
- **What it teaches:** first fix complete and pipework pressure-tested. The issue log records a leak found during the pressure test and resolved.

### Raft duplex services

- **Name on the card:** 5-Bedroom Duplex - Raft Foundation - MEP Services. Flag: **Raft duplex.**
- **What is in it:** 48 light points, 52 sockets, 9 split units and 22 sanitary fittings. Contract about ₦35.9m, three certificates (about 55% valued), two variations (lifting the pump chamber above flood level and an extra split unit). The utility connection sum is executed.
- **What it teaches:** second fix and sanitary fittings going in, and how site conditions (a high water table) turn into a services variation.

### Pile duplex services

- **Name on the card:** 4-Bedroom Terrace Duplex - Bored Pile Foundation - MEP Services. Flag: **Pile duplex.**
- **What is in it:** 28 light points, 32 sockets, 7 split units and 10 sanitary fittings. Contract about ₦26.5m, four paid certificates (about 90% valued), two variations (relocated inspection chambers and surge protection). All three sums are executed.
- **What it teaches:** services complete, commissioned and the final account agreed, at about ₦23.8m, roughly ₦2.7m under the contract sum.

## Early-access products: CIVIQ and QUIV for ArchiCAD

CIVIQ and QUIV for ArchiCAD are not on general sale yet. Their samples appear only for accounts that have early access to those products. If you do not have early access, you will not see these samples, and nothing is wrong with your account.

### CIVIQ samples (Civil 3D)

CIVIQ samples are four road jobs, one for each common pavement type. Each carries a corridor model: pavement layers, kerbs, drains and culverts between 20 m stations, which you can open in **3D Model**. Lines are measured per chainage band, the way CIVIQ extracts them. They appear on **CIVIQ projects** (</projects/civil3d>). The strip reads "Worked road and drainage jobs, measured chainage by chainage from the corridor."

The bill has two sections, Earthworks and External Works. The PC and provisional sums are relocation of existing services (₦3.5m), street lighting (₦6.0m) and materials testing for CBR, compaction and cores (₦1.2m). The programme has six phases: site clearance and topsoil strip; earthworks; drainage and culverts; sub-base and base course; surfacing; and kerbs, markings and road furniture. Variations on road jobs are site instructions (SI-01, SI-02).

| Name on the card                                   | Flag                               | Site and client                                               | Road                                                                          | Bill lines | Budget lines | Contract      | Certificates                   | Stage                                      |
|----------------------------------------------------|------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------|------------|--------------|---------------|--------------------------------|--------------------------------------------|
| Estate Access Road - Asphalt Pavement              | <b>Flexible pavement (asphalt)</b> | Akobo, Ibadan, for Greenfield Court Residents Association     | 480 m long, 7.3 m carriageway, 322 m3 cut, 317 m3 fill                        | 39         | 116          | about ₦352.2m | 2 (about 45% valued)           | Drains and culvert in, sub-base being laid |
| Industrial Estate Road - Rigid Concrete Pavement   | <b>Rigid pavement (concrete)</b>   | Agbara Industrial Estate, Ogun, for Agbara Logistics Park Ltd | 360 m long, 7.3 m carriageway, 218 m3 cut, 327 m3 fill                        | 49         | 168          | about ₦315.2m | 1 (about 10% valued)           | Earthworks and drainage under way          |
| Housing Estate Street - Interlocking Paving Stones | <b>Interlocking paving</b>         | Gwarinpa, Abuja, for Cedar Grove Homes Ltd                    | 300 m long, 6 m carriageway, 178 m3 cut, 464 m3 fill                          | 30         | 100          | about ₦130.9m | 3 (about 78% valued)           | Paving laid, kerbs and markings going in   |
| Rural Feeder Road - Laterite with Culverts         | <b>Laterite feeder road</b>        | Ikire-Apomu, Osun, for Osun State Rural Access Agency         | 1,200 m long, 6 m carriageway, 90 m3 cut, 1,927 m3 fill, two stream crossings | 20         | 48           | about ₦170.8m | 4 (all paid, about 90% valued) | Road handed over and final account agreed  |

What each one teaches:

- **Asphalt road:** a big jump between certificates once drainage goes in. The variation SI-01 replaces soft material at Ch 0+300 with laterite, priced at the bill rate for filling.
- **Concrete road:** the most detailed bill of the four (49 lines), and an early job with one certificate. SI-01 thickens the slab at the loading bay, not yet executed.
- **Interlocking street:** a nearly finished job with both site instructions executed and the services relocation sum spent.
- **Laterite feeder road:** a long, mostly fill road with culverts. An extra culvert barrel was added after a flood survey, half the retention is released on the last certificate, and the final account closes at about ₦154.4m, roughly ₦16.4m under the contract sum.

## QUIV for ArchiCAD samples

ArchiCAD samples are the same four duplexes, measured from the ArchiCAD model and costed as an ArchiCAD bill. They appear on the **QUIV for ArchiCAD** page ( /archicad ). The strip reads "Worked duplex projects, one per foundation type, measured from the ArchiCAD model."

Clicking a card opens the sample's ArchiCAD bill page, with the same orange **Sample project · Read-only learning material** banner. The bill is grouped into the ArchiCAD work sections: Substructure, Frame, Upper Floors, External Walls, Internal Walls, Roof, Windows & External Doors and Internal Doors. Each line keeps the

elements it came from, and the costed bill is saved as a BoQ version. **Dashboard** opens the cost breakdown with **Cost by category** and the **Budget tracker**.

| Name on the card                                             | Flag                            | Bill lines | Bill total       | Contract         | Certificates |
|--------------------------------------------------------------|---------------------------------|------------|------------------|------------------|--------------|
| 4-Bedroom Duplex - Strip Foundation                          | <b>Strip foundation</b>         | 55         | about<br>₦70.3m  | about<br>₦98.5m  | 1            |
| 4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation | <b>Pad and strip foundation</b> | 57         | about<br>₦88.4m  | about<br>₦120.5m | 2            |
| 5-Bedroom Duplex - Raft Foundation                           | <b>Raft foundation</b>          | 54         | about<br>₦126.9m | about<br>₦167.2m | 3            |
| 4-Bedroom Terrace Duplex - Bored Pile Foundation             | <b>Pile foundation</b>          | 62         | about<br>₦90.3m  | about<br>₦122.8m | 4            |

ArchiCAD bills have fewer lines than QUIV for the same building, because they cover the model's structure, walls, roof and openings rather than every finish. The clients, sites, variations, risks and issues match the duplex of the same name.

**What they teach:** how an ArchiCAD model becomes a costed, sectioned bill, and how the cost of each work section compares. Substructure is the biggest section on the raft and pile duplexes, which is exactly what you would expect on poor ground.

## Exercises

These five exercises take ten to fifteen minutes each. Any product's samples work, but the steps below use the QUIV samples. On HERON the steps are the same without the 3D model.

### Exercise 1: Open a sample

1. Go to **QUIV projects** ( /projects/revit ).
2. In the **Learning samples** strip, click **5-Bedroom Duplex - Raft Foundation**.
3. Read the orange banner. Note the stage: "Interim Certificate 3 issued".
4. Click **What to look at in this sample** and read the list.
5. Under **Views**, click each view in turn, from **Dashboard** to **PM Dashboard**.
6. Click **Projects** to go back to the list.

### Exercise 2: Read the bill

1. Open **4-Bedroom Duplex - Strip Foundation**.
2. Click **Bill of Quantity**.
3. Find the first lines under Earthworks, for example "Foundations – Excavation (trench)". Note the quantity in m3 and the rate.
4. Scroll through the work sections and note which section carries the most money.
5. Click **3D Model**, then pick a bill line to see the elements it was measured from.
6. Now open the HERON sample of the same duplex and find the same excavation line. It reads "Excavate trenches for foundations, not exceeding 2.00m deep", with the same quantity and rate.

#### TIP

This is the best way to see that QUIV and HERON produce the same bill. Only the descriptions and grouping differ.

### Exercise 3: Compare the budget

1. Open **4-Bedroom Duplex with Family Lounge - Pad & Strip Foundation**.
2. Click **Budget**.
3. Pick one bill item and read the rows under it: the **Material**, **Labour** and **Plant** lines that make up its rate. Check that **Bill Rate = Material + Labour + O&P** (10% overhead and 15% profit).
4. Look at which rows are ticked as procured and which are only part bought. Materials for phases already started are bought; later phases are not.
5. Switch to **Buy schedule** and read **Should already be bought** and **Buy this week**.

### Exercise 4: Follow the valuation

1. Open **4-Bedroom Terrace Duplex - Bored Pile Foundation**.
2. Click **Valuation**.
3. Under **Contract administration**, open **Certificates**. There are four, all paid.
4. For each certificate, read the value this period, the retention held, VAT added and withholding tax deducted.
5. On certificate 4, find the retention released: half the retention comes back at practical completion.
6. Open **Variations** and read AI-01 to AI-03. Note that AI-02 (extending the piles) is priced at the bill rate for piling.
7. Open **Final account** and compare the final contract value with the contract sum.

### Exercise 5: See the PM dashboard

1. Open **5-Bedroom Duplex - Raft Foundation**.
2. Click **PM Dashboard**.
3. Read the tiles: **Progress**, **Budget Used**, **Overdue**, **CPI**, **SPI** and **Tasks Done**.
4. Read the **Earned Value Summary**. A CPI or SPI below 1 means over budget or behind programme.
5. Click **View Details** and open **WBS / Tasks**. See how the Substructure phase is split into trades.
6. Open **Risk Register** and **Issue Log**. Find the risk about the high water table (closed) and the issue about window frames out of square (in progress).

## What's new in 2026.10

- **September 2026:** sample projects arrived for QUIV and HERON, four duplexes each.
- **Late September 2026:** samples added for Revit MEP, CIVIQ and QUIV for ArchiCAD, and the **Learning samples** strip added to each tool's page under **My tools**.
- **End of September 2026:** the sample programmes now show each phase broken into trades (formwork, reinforcement, pours and so on), the way a real programme of works is drawn, instead of one bar per phase.

# Troubleshooting

## I cannot see the Learning samples strip

- **Check your licence.** Samples only appear for a product you hold an active licence for. Open **Products** at [/manage](#) and check the product is active and not expired.
- **Check you are on the right page.** The strip is on a product's project list (for example [/projects/revit](#) ), on the tool's page under **My tools**, and on [/archicad](#) for ArchiCAD. It is not on the **Projects** gallery ([/work/projects](#) ) or the **Overview**.
- **Check whether you hid it.** If you see a button reading **Show 4 samples**, click it.
- **CIVIQ and ArchiCAD:** these samples appear only with early access to those products.
- If none of these apply, click **Refresh projects** or reload the page.

## I changed a rate in a sample and it will not save

That is expected. Samples are read-only. The message "Sample projects are read-only learning material." means the server refused the change. Try the same thing on one of your own projects.

## I want to use a sample as a starting point for my job

You cannot copy a sample. Download its workbooks from **Export** if you want a reference, then measure your own job in the plugin and save it to ADLM Cloud.

## The sample has no 3D Model view

HERON samples are measured from PDF drawings and have no model. QUIV, Revit MEP and CIVIQ samples do. ArchiCAD samples open on the ArchiCAD bill page instead of the project screen.

## I cannot find a sample from inside the plugin

Samples are kept out of the desktop plugins on purpose, so nobody can save over them. Open them on the website.

# Frequently asked questions

**Do samples count towards my cloud storage?** No. Your project slots count only the projects you own.

**Do samples show up in my totals on the Overview page?** No. The **Overview** and the **Projects** gallery count only your own projects and projects shared with you.

**Can my colleagues see the same samples?** Yes. Every customer with the product sees the same samples. Because nobody can change them, what you see is what they see.

**Can I see the rates without Rate Gen?** Yes. Samples show every rate and money total to everyone. On your own projects and shared projects, the normal Rate Gen rules apply. See [ADLM Rate Gen](#).

**Can I download a sample?** Yes. The Excel workbooks and reports work on samples just as they do on your projects.

**Are the clients and sites real?** No. The names, sites and suppliers are made up. The quantities are measured from real models of the four duplexes and four roads, and the money is worked out the same way the website works out your own jobs.

**Why are the QUIV and HERON figures identical?** They are the same buildings measured two ways. That is the point: a drawing takeoff in HERON and a model takeoff in QUIV lead to the same bill, budget and valuations.

**Why do the ArchiCAD figures differ from QUIV?** The ArchiCAD bill covers what the ArchiCAD model holds (structure, walls, roof and openings) and is grouped into ArchiCAD's work sections, so it has fewer lines and a lower total.

**Will the samples change?** ADLM Studio may refresh a sample to show a new feature. It is replaced as a whole and keeps the same name and link.

**I need help.** Contact us through </support>.

# The ADLM QS handbook

Key QS formulas, a formula library, constants, grade and unit conversions, and real use cases for every ADLM product.

## About this handbook

This is a working reference for quantity surveyors and estimators in Nigeria. It sets out the formulas you use every day, the conversion factors behind a material schedule, and the conversions you need when a drawing, a supplier or a client speaks a different unit.

Two kinds of figure appear in it, and each one is labelled:

- **ADLM value.** A default held in ADLM software. The handbook names the product: Rate Gen **Material Constants** on the website, QUIV (the Revit plugin), or HERON. These are the figures your ADLM software will use unless you change them.
- **Typical.** A widely used industry figure, with its assumption stated. Use it as a starting point and check it against your specification, your supplier and your site.

### IMPORTANT

Nothing here replaces the contract documents. A specification, a structural engineer's bar schedule or a design mix always overrides a rule of thumb. Where a figure in this handbook and your drawings disagree, the drawings win.

### NOTE

ADLM's material defaults were recalibrated in September 2026 against the build-up formulas in several hundred real Nigerian bills. If your copy of QUIV, HERON or Rate Gen shows a different figure, you may be on an earlier version, or someone on your account may have changed it. You can see and edit every value in **Material Constants**.

Prices in the worked examples are illustrative only. Use current prices from your own Rate Gen library or your suppliers.

## Key QS formulas

Each formula below gives the units, a worked example and the mistake that most often goes with it.

### Areas and volumes of common shapes

| Shape     | Area or volume                                            | Units          |
|-----------|-----------------------------------------------------------|----------------|
| Rectangle | $A = L \times B$                                          | m <sup>2</sup> |
| Triangle  | $A = \frac{1}{2} \times \text{base} \times \text{height}$ | m <sup>2</sup> |
| Trapezium | $A = \frac{1}{2} \times (a + b) \times h$                 | m <sup>2</sup> |
| Circle    | $A = \pi \times d^2 \div 4$                               | m <sup>2</sup> |

| Shape                       | Area or volume                                          | Units |
|-----------------------------|---------------------------------------------------------|-------|
| Annulus (ring)              | $A = \pi \times (D^2 - d^2) \div 4$                     | $m^2$ |
| Sector                      | $A = (\theta \div 360) \times \pi \times r^2$           | $m^2$ |
| Prism or slab               | $V = \text{cross-section area} \times \text{length}$    | $m^3$ |
| Cylinder (pile, column)     | $V = \pi \times d^2 \div 4 \times h$                    | $m^3$ |
| Cone or pyramid             | $V = \frac{1}{3} \times \text{base area} \times h$      | $m^3$ |
| Frustum (tapered pad, cone) | $V = h \div 3 \times (A1 + A2 + \sqrt{(A1 \times A2)})$ | $m^3$ |
| Sphere                      | $V = \pi \times d^3 \div 6$                             | $m^3$ |

**Worked example.** A bored pile 600 mm in diameter and 10 m deep:

$$V = \pi \times 0.6^2 \div 4 \times 10 = 0.2827 \times 10 = \mathbf{2.83 \text{ m}^3} \text{ of concrete per pile.}$$

HERON's piling template uses the same formula: concrete = number of piles  $\times$  depth  $\times \pi \times (D \div 2)^2$ .

#### TIP

Convert every dimension to metres before you multiply. A 600 mm diameter entered as 600 turns  $2.83 \text{ m}^3$  into  $2,827,433 \text{ m}^3$ .

**Common mistake:** using the radius where the formula wants the diameter (or the reverse).  $\pi \times d^2 \div 4$  and  $\pi \times r^2$  give the same answer only if you use the right one.

## Girth and centre-line method for trenches and walls

For a wall or a trench that runs all round a building, measure along the centre line. The centre line is shorter than the outside perimeter and longer than the inside perimeter.

| You know                         | Centre-line girth of a rectangular building |
|----------------------------------|---------------------------------------------|
| External dimensions $L \times B$ | $2 \times (L + B) - 4t$                     |
| Internal dimensions $L \times B$ | $2 \times (L + B) + 4t$                     |

Here  $t$  is the wall thickness (or the trench width, for a trench measured from its own edges). Each external corner takes  $2 \times \frac{1}{2}t$  off the outside girth, and a rectangle has four corners, so  $4t$  comes off in total. A building with set-backs still loses  $4t$  overall, because every extra outside corner is balanced by an inside corner.

**Worked example.** A bungalow measures  $12.00 \times 8.00 \text{ m}$  over 225 mm external walls. The strip footing trench is 675 mm wide and 900 mm deep, centred on the wall.

1. External perimeter =  $2 \times (12.00 + 8.00) = 40.00 \text{ m}$ .
2. Centre-line girth =  $40.00 - 4 \times 0.225 = \mathbf{39.10 \text{ m}}$ .
3. Excavation =  $39.10 \times 0.675 \times 0.90 = \mathbf{23.75 \text{ m}^3}$ .
4. Blockwork in foundation, 900 mm high =  $39.10 \times 0.90 = \mathbf{35.19 \text{ m}^2}$ .

**Internal walls.** An internal wall runs between the inner faces of the external walls. Its trench runs only between the inner edges of the external trenches, which are further apart.

For one internal wall across the 8.00 m width of the example:

- Wall length =  $8.00 - 2 \times 0.225 = 7.55 \text{ m}$ .
- The external trench's inner edge sits  $0.1125 + 0.3375 = 0.45 \text{ m}$  in from the outside face, so the trench length =  $8.00 - 2 \times 0.45 = 7.10 \text{ m}$ .

**Common mistake:** measuring the internal trench at the wall length (7.55 m). The 0.45 m at each end has already been dug as part of the external trench, so it would be measured twice.

## Excavation, bulking and compaction

| Quantity                   | Formula                                                                        | Units        |
|----------------------------|--------------------------------------------------------------------------------|--------------|
| Trench excavation          | $(w + 2 \times \text{working space}) \times \text{depth} \times \text{length}$ | $\text{m}^3$ |
| Pad excavation             | $(L + 2ws) \times (B + 2ws) \times \text{depth}$                               | $\text{m}^3$ |
| Earthwork support (trench) | $2 \times \text{depth} \times \text{length}$                                   | $\text{m}^2$ |
| Loose volume to cart away  | $\text{in-situ volume} \times (1 + \text{bulking \%})$                         | $\text{m}^3$ |
| Loose fill to order        | $\text{compacted volume} \times \text{compaction factor}$                      | $\text{m}^3$ |
| Disposal                   | $\text{excavated volume} - \text{volume backfilled}$                           | $\text{m}^3$ |

A working space of **0.30 m** each side of strip and pad excavation is a typical allowance, and earthwork support is measured as  $2 \times \text{depth} \times \text{length}$ .

### NOTE

Methods of measurement treat working space differently. Some measure the net size of the foundation and leave working space in the rate; others allow it as a separate item. Follow the method named in your contract (BESMM4, NRM2 or SMM7) and say which one you used in your preambles.

**Typical bulking** (in-situ to loose, after digging):

| Material                      | Typical bulking |
|-------------------------------|-----------------|
| Sand and gravel               | 10 to 15%       |
| Ordinary soil, loam, laterite | 20 to 30%       |
| Clay                          | 30 to 40%       |
| Broken rock                   | 50 to 70%       |

**Typical compaction.** Laterite and sand fill lose roughly 15 to 30% of their loose volume when compacted in layers, so a compaction factor of about **1.2 to 1.3** loose  $\text{m}^3$  per compacted  $\text{m}^3$  is common. Hardcore varies with its grading.

**Worked example.** From the trench above ( $23.75 \text{ m}^3$  in-situ, ordinary soil):

1. Loose volume =  $23.75 \times 1.25 = 29.69 \text{ m}^3$ .
2. With a tipper carrying about  $6 \text{ m}^3$  loose (ask your haulier for the real figure), that is  $29.69 \div 6 = 4.9$ , so **5 trips**.

**Filling in tonnes.** QUIV, HERON and Rate Gen **Material Constants** all convert measured fill to tonnes at **1.6 t/ $\text{m}^3$**  (hardcore, laterite and general fill).

Hardcore 150 mm thick inside the example building (7.55 × 11.55 m):

- Area =  $7.55 \times 11.55 = 87.20 \text{ m}^2$ .
- Volume =  $87.20 \times 0.15 = 13.08 \text{ m}^3$ .
- Weight =  $13.08 \times 1.6 = \mathbf{20.93 \text{ t}}$ .

#### IMPORTANT

Never treat trips per  $\text{m}^3$  as tonnes per  $\text{m}^3$ . A common bill template converts fill with " $\text{m}^3 \div 3.51$ ", which gives tipper TRIPS. Reading that 0.285 as tonnes per  $\text{m}^3$  makes the filling about six times too light. ADLM's software had this error and it was corrected in September 2026; check any spreadsheet you inherit for the same thing.

## Concrete volume, dry volume and mix materials

**Volume** is length × width × depth, in  $\text{m}^3$ . Measure net. Do not deduct the space taken by reinforcement. SMM7 and NRM2 do not deduct voids of  $0.05 \text{ m}^3$  or less; check the rule in your contract's method.

**Dry volume.** Cement, sand and granite shrink when they are mixed with water, because the fine material fills the voids between the coarse. One cubic metre of placed concrete needs about **1.54  $\text{m}^3$**  of dry materials. QUIV, HERON and Rate Gen **Material Constants** all use 1.54. That factor already carries the normal allowance for waste, so ADLM's concrete waste factor is **1.00**.

**ADLM's formula** (QUIV, HERON and Rate Gen **Material Constants**), for a mix c : s : g and V  $\text{m}^3$  of concrete:

- Cement (bags) =  $V \times (c \div (c + s + g)) \times 1.54 \times 1,440 \div 50$
- Sharp sand (t) =  $V \times (s \div (c + s + g)) \times 1.54 \times 1,440 \div 1,000$
- Granite (t) = sharp sand tonnes × 2

The constants are: cement 1,440 kg/ $\text{m}^3$ , sharp sand 1,440 kg/ $\text{m}^3$ , a 50 kg bag, and a granite-to-sand multiplier of 2.

**Materials per  $\text{m}^3$  of concrete (ADLM values):**

| Mix     | Cement (bags) | Sharp sand (t) | Sharp sand ( $\text{m}^3$ loose) | Granite (t) |
|---------|---------------|----------------|----------------------------------|-------------|
| 1:1:2   | 11.09         | 0.55           | 0.39                             | 1.11        |
| 1:1.5:3 | 8.06          | 0.60           | 0.42                             | 1.21        |
| 1:2:4   | 6.34          | 0.63           | 0.44                             | 1.27        |
| 1:3:6   | 4.44          | 0.67           | 0.46                             | 1.33        |
| 1:4:8   | 3.41          | 0.68           | 0.47                             | 1.37        |

**Worked example.** The strip footing from the girth example is 675 mm wide and 225 mm thick, in 1:2:4.

1. Volume =  $39.10 \times 0.675 \times 0.225 = \mathbf{5.94 \text{ m}^3}$ .
2. Cement =  $5.94 \times 6.34 = 37.6$ , so order **38 bags**.
3. Sharp sand =  $5.94 \times 0.63 = \mathbf{3.76 \text{ t}}$ .
4. Granite =  $5.94 \times 1.27 = \mathbf{7.53 \text{ t}}$ .
5. At an illustrative N11,500 a bag, the cement costs  $38 \times 11,500 = \mathbf{N437,000}$ .

#### NOTE

The granite rule "sand  $\times$  2" is right for every mix in the table, because each has twice as much stone as sand. For a mix such as 1:2:3, work the granite out from its own share:  $V \times (3 \div 6) \times 1.54 \times \text{density}$ .

#### Common mistakes:

- **Leaving out the dry-volume factor.** Applying 1:2:4 to the wet volume gives 4.3 bags/m<sup>3</sup> instead of about 6.3, which under-orders cement by a third.
- **Confusing cement density with bags per m<sup>3</sup> of concrete.** 1 m<sup>3</sup> of cement weighs 1,440 kg, which is 28.8 bags. That is the bulk density of the cement itself, not the cement in a cubic metre of concrete. A bill that prints 28.8 bags/m<sup>3</sup> for 1:2:4 has made exactly this mistake.

### Blinding

Blinding is a thin, lean concrete layer, usually 50 to 75 mm, under footings and slabs. Its volume is area  $\times$  thickness, in m<sup>3</sup>.

QUIV, HERON and Rate Gen **Material Constants** treat blinding as a lean **1:4:8** mix:

| Per m <sup>3</sup> of blinding (ADLM value) | Before waste | With 5% waste |
|---------------------------------------------|--------------|---------------|
| Cement                                      | 3.25 bags    | 3.41 bags     |
| Sharp sand                                  | 0.65 t       | 0.68 t        |
| Granite                                     | 1.30 t       | 1.37 t        |

**Worked example.** Blinding 50 mm thick under the strip footing:

1. Volume =  $39.10 \times 0.675 \times 0.05 = \mathbf{1.32 \text{ m}^3}$ .
2. Cement =  $1.32 \times 3.41 = 4.5$ , so **5 bags**.
3. Sharp sand =  $1.32 \times 0.68 = \mathbf{0.90 \text{ t}}$ .
4. Granite =  $1.32 \times 1.37 = \mathbf{1.80 \text{ t}}$ .

#### Common mistakes:

- **Costing blinding as 1:2:4.** That gives 6.3 bags/m<sup>3</sup> for a layer that needs about 3.4.
- **Leaving out the granite.** Blinding at 1:4:8 or 1:10 still contains stone. QUIV, HERON and Rate Gen once carried about 0.9 bag/m<sup>3</sup> and no granite; this was corrected in September 2026.

### Blockwork

Nigerian sandcrete blocks have a face of 450  $\times$  225 mm. With 10 to 15 mm joints, a square metre of wall takes about 9.3 blocks, and practice orders **10 per m<sup>2</sup>** to cover breakage on delivery. The count is the same for 225 mm and 150 mm blocks, because the face size is the same.

**ADLM values (QUIV, HERON and Rate Gen Material Constants):**

| Per m <sup>2</sup> of wall | Value                               |
|----------------------------|-------------------------------------|
| Blocks                     | 10 nr                               |
| Blockwork waste factor     | 1.03 (QUIV, Rate Gen); 1.05 (HERON) |

| Per m <sup>2</sup> of wall | Value    |
|----------------------------|----------|
| Mortar cement              | 0.20 bag |
| Mortar sharp sand          | 0.055 t  |

ADLM applies the mortar figure to every block thickness. A 150 mm wall uses less mortar than a 225 mm wall. A typical allowance is about **two-thirds** of the 225 mm figure, roughly 0.13 bag of cement and 0.036 t of sand per m<sup>2</sup>, assuming the same 1:6 mortar and the same joint thickness. If you price a lot of 150 mm walling, set your own figure in **Material Constants**.

**Area to measure.** Measure the net face area of the wall, on its centre line, and deduct openings. SMM7 and NRM2 do not deduct voids of 0.5 m<sup>2</sup> or less. Check the rule in the method named in your contract, including BESMM4.

**Worked example.** Ground-floor 225 mm walls 3.00 m high on the 39.10 m centre line, with four 1.20 × 1.20 m windows and two 0.90 × 2.10 m doors.

1. Gross area =  $39.10 \times 3.00 = 117.30 \text{ m}^2$ .
2. Openings =  $4 \times 1.44 + 2 \times 1.89 = 5.76 + 3.78 = 9.54 \text{ m}^2$ .
3. Net area =  $117.30 - 9.54 = 107.76 \text{ m}^2$ .
4. Blocks =  $107.76 \times 10 \times 1.03 = 1,110 \text{ blocks}$ .
5. Cement =  $107.76 \times 0.20 = 21.6$ , so **22 bags**.
6. Sharp sand =  $107.76 \times 0.055 = 5.93 \text{ t}$ .
7. At an illustrative ₦1,300 a block, the blocks cost  $1,110 \times 1,300 = \text{₦1,443,000}$ .

**Common mistake:** using the outside perimeter instead of the centre line. On this building that adds  $0.90 \times 3.00 = 2.7 \text{ m}^2$  of walling that does not exist, and on a large building the error grows with every corner.

## Plaster, render and screed

**ADLM per m<sup>2</sup> values (QUIV, HERON and Rate Gen Material Constants):**

| Work                        | Cement per m <sup>2</sup> | Sand per m <sup>2</sup> |
|-----------------------------|---------------------------|-------------------------|
| Render or plaster, one face | 0.13 bag                  | 0.05 t plaster sand     |
| Floor screed (40 mm, 1:3)   | 0.40 bag                  | 0.057 t sharp sand      |
| Tile bed (floor)            | 0.40 bag                  | 0.057 t sharp sand      |
| POP wall screeding          | 0.04 bag                  | none                    |

These are per m<sup>2</sup> of the face measured, the way Nigerian bills state them. The render figure includes the usual allowance for dubbing out uneven blockwork.

**From first principles**, using HERON's method:

Cement (bags) = area × thickness × dry factor × waste ×  $(c \div (c + s)) \times 28.8$

The 28.8 is the number of 50 kg bags in a cubic metre of cement ( $1,440 \div 50$ ). HERON's defaults are:

- **Screed:** 40 mm, 1:3, dry factor 1.30, waste 1.05.
- **Render:** 12 mm, 1:6, dry factor 1.30, waste 1.05.

For screed:  $1 \times 0.040 \times 1.30 \times 1.05 \times 0.25 \times 28.8 = 0.39 \text{ bag/m}^2$ , which agrees with the 0.40 in the table.

For a 12 mm 1:6 render, the same sum gives about  $0.07 \text{ bag/m}^2$  before any dubbing out. That is why the bill figure of 0.13 is higher.

**Worked example.** Screed 40 mm thick to the  $87.20 \text{ m}^2$  floor:

1. Cement =  $87.20 \times 0.40 = 34.9$ , so **35 bags**.
2. Sharp sand =  $87.20 \times 0.057 = 4.97 \text{ t}$ .

For a different screed thickness, scale it: about **0.01 bag of cement per mm per  $\text{m}^2$  at 1:3**.

**Common mistake:** pricing render on both faces of a wall from one face area, or the reverse. Measure each face you will plaster.

## Reinforcement weight, laps, cutting lengths and bends

**Unit weight.** Steel weighs  $7,850 \text{ kg/m}^3$ , so a bar of diameter  $d \text{ mm}$  weighs:

$$\text{kg/m} = d^2 \div 162 \text{ (exactly } 0.006165 \times d^2\text{)}$$

QUIV, HERON and Rate Gen **Material Constants** all use a coefficient of  **$0.00617 \text{ kg/m per mm}^2$** , which is the same rule. Many ADLM templates write it as  $(d^2 \div 36) \times 0.2222$ , which is  $d^2 \div 162$  in another form.

**Bar schedule table** (12 m bars):

| Bar   | kg/m ( $d^2 \div 162$ ) | kg/m (BS exact) | kg per 12 m bar | Bars per tonne |
|-------|-------------------------|-----------------|-----------------|----------------|
| 6 mm  | 0.222                   | 0.222           | 2.67            | 375            |
| 8 mm  | 0.395                   | 0.395           | 4.74            | 211            |
| 10 mm | 0.617                   | 0.617           | 7.41            | 135            |
| 12 mm | 0.889                   | 0.888           | 10.67           | 94             |
| 16 mm | 1.580                   | 1.578           | 18.96           | 53             |
| 20 mm | 2.469                   | 2.466           | 29.63           | 34             |
| 25 mm | 3.858                   | 3.853           | 46.30           | 22             |
| 32 mm | 6.321                   | 6.313           | 75.85           | 13             |

Bars per tonne are rounded. When you order, round up to whole bars for each diameter.

**Stock length.** ADLM's standard stock length is **12 m** (QUIV, HERON and Rate Gen **Material Constants**). Some mills deliver bars a little short of 12 m. That is why some ADLM bar counts use a usable length of 11 to 11.7 m.

**Laps.** Where a run is longer than a stock bar, bars are lapped. A typical tension lap is **40 to 50 bar diameters** (40d for a Y16 is 640 mm). The real figure depends on concrete grade, bar type and position, and the structural drawings state it.

**Waste and lap allowances held by ADLM:**

| Allowance                   | Value | Product                           |
|-----------------------------|-------|-----------------------------------|
| General reinforcement waste | 1.05  | QUIV, Rate Gen Material Constants |

| Allowance                              | Value                             | Product                                  |
|----------------------------------------|-----------------------------------|------------------------------------------|
| General reinforcement waste            | 1.10                              | HERON                                    |
| Beam main bars, laps and waste         | 1.15                              | QUIV                                     |
| Staircase bars, laps and waste         | 1.10                              | QUIV                                     |
| Lap/bend extra on column and pile bars | 0.30 m per bar                    | HERON templates                          |
| Lap/bend extra on beam bars            | 1.20 m per bar                    | HERON templates                          |
| Binding wire                           | 0.01 kg per kg of steel (10 kg/t) | QUIV, HERON, Rate Gen Material Constants |

**Rates per m<sup>3</sup> of concrete.** When there is no bar schedule yet, an estimate can price steel as kilograms per m<sup>3</sup> of concrete. Typical starting rates, used by ADLM as defaults, are:

| Element       | Typical starting rate |
|---------------|-----------------------|
| Strip footing | 40 kg/m <sup>3</sup>  |
| Slab          | 80 kg/m <sup>3</sup>  |
| Beam          | 160 kg/m <sup>3</sup> |
| Column        | 180 kg/m <sup>3</sup> |

Replace them with the engineer's bar schedule as soon as you have it. Real rates vary widely with spans, loads and the designer's detailing.

**Link cutting length.** HERON's beam and column link templates use:

$$\text{link length} = 2 \times ((b - 2c) + (h - 2c)) + 0.15 \text{ m}$$

where b and h are the member's width and depth and c is the cover. The 0.15 m allows for the two hooks. The number of links is (length ÷ spacing) + 1.

**Bends.** Under BS 8666 (shape code 11, an L-bar), total length = A + B – 0.5r – d, where r is the bending radius. With the standard radius of 2d for bars up to 16 mm, this works out as a deduction of **2d for each 90° bend**. Larger bars use a larger radius, so take the length from the bar schedule.

**Worked example.** A beam 225 × 450 mm, 6.00 m long. It has 3 Y16 bottom bars and 2 Y16 top bars, with R10 links at 200 mm centres and 50 mm cover (HERON's default).

1. Main bars = 5 × 6.00 × 1.580 × 1.15 (QUIV's beam laps and waste) = **54.5 kg**.
2. Link length = 2 × ((0.225 – 0.10) + (0.45 – 0.10)) + 0.15 = 2 × 0.475 + 0.15 = **1.10 m**.
3. Number of links = 6.00 ÷ 0.20 + 1 = **31**.
4. Links = 31 × 1.10 × 0.617 = **21.0 kg**.
5. Total = 54.5 + 21.0 = **75.5 kg**, plus 0.76 kg of binding wire.

An L-bar Y12 with legs of 2.00 m and 0.30 m has a cutting length of 2.30 – 2 × 0.012 = **2.276 m**.

**Common mistakes:**

- **Mixing kilograms and tonnes.** 75.5 kg is 0.0755 t. A rate per tonne applied to a quantity in kg is a thousand times too high, and a tonne quantity read as kg is a thousand times too low. An older HERON build made this error when an item was measured in tonnes; it was corrected in September 2026.
- **Applying a lap allowance twice.** That happens when you add laps bar by bar and then also apply a percentage on top.

## Formwork areas

Formwork is measured as the area of concrete face in contact with it, in m<sup>2</sup>.

| Member                      | Formwork area                      |
|-----------------------------|------------------------------------|
| Beam (soffit and two sides) | $(2h + w) \times L$                |
| Column                      | $2 \times (a + b) \times h$        |
| Pad footing sides           | $2 \times (L + B) \times h$        |
| Suspended slab soffit       | net soffit area                    |
| Slab edges                  | perimeter $\times$ slab thickness  |
| Pile casing (if used)       | $\pi \times d \times \text{depth}$ |

QUIV uses  $(2h + w) \times L$  for beams and  $2 \times (w + d) \times h$  for columns.

### Board and sundries (ADLM values, QUIV, HERON and Rate Gen Material Constants):

| Item           | Value                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Board coverage | 2.88 m <sup>2</sup> per sheet (2.4 $\times$ 1.2 m)                                                                                                                                                            |
| Nails          | 0.0576 bag per sheet, about 0.02 bag per m <sup>2</sup> (QUIV, HERON, Rate Gen). HERON stores 0.0524 per sheet because it already adds 10% board waste, which comes to the same 0.02 bag per m <sup>2</sup> . |
| Bracing timber | 6.3 m per sheet (QUIV, Rate Gen); 0.60 piece per m <sup>2</sup> (HERON)                                                                                                                                       |
| Formwork waste | 1.10 (HERON)                                                                                                                                                                                                  |

**Worked example.** Formwork to the 225  $\times$  450 mm beam, 6.00 m long:

1. Area =  $(2 \times 0.45 + 0.225) \times 6.00 = 1.125 \times 6.00 = \mathbf{6.75 \text{ m}^2}$ .
2. Sheets =  $6.75 \div 2.88 = \mathbf{2.34 \text{ sheets}}$ .
3. Nails =  $2.34 \times 0.0576 = \mathbf{0.13 \text{ bag}}$ .
4. Bracing =  $2.34 \times 6.3 = \mathbf{14.8 \text{ m}}$ .

#### TIP

If your beam depth is the overall depth including a slab cast with it, deduct the slab thickness from h for the sides. The slab soffit formwork already covers that strip.

**Common mistake:** measuring formwork to the top face of a slab or footing. Only the faces the concrete is cast against are measured.

## Roofing: pitch factor, rafter length and sheets

**Pitch factor.** The area on the slope = plan area × pitch factor, where pitch factor =  $1 \div \cos(\text{pitch})$ . This works for gable and hipped roofs, as long as every slope has the same pitch. Use the plan area measured over the eaves.

| Pitch | Pitch factor | Slope (%) |
|-------|--------------|-----------|
| 10°   | 1.015        | 17.6      |
| 15°   | 1.035        | 26.8      |
| 20°   | 1.064        | 36.4      |
| 22.5° | 1.082        | 41.4      |
| 25°   | 1.103        | 46.6      |
| 30°   | 1.155        | 57.7      |
| 35°   | 1.221        | 70.0      |
| 40°   | 1.305        | 83.9      |
| 45°   | 1.414        | 100.0     |

**Rafter length.** QUIV's roof module uses:

- rafter length = (half span + overhang) ÷ cos(pitch)
- rise = half span × tan(pitch)

The span is measured between wall plates.

**ADLM roof defaults:**

| Item                                          | Value                              | Product                     |
|-----------------------------------------------|------------------------------------|-----------------------------|
| Rafter spacing                                | 1.2 m                              | HERON                       |
| Purlin spacing                                | 0.9 m                              | HERON                       |
| Eaves overhang                                | 0.6 m                              | HERON                       |
| Timber stock length                           | 3.6 m                              | QUIV, HERON                 |
| Covering per sheet                            | 2.0 m <sup>2</sup>                 | QUIV, HERON                 |
| Nails per sheet or bundle                     | 30                                 | QUIV, HERON                 |
| Ridge cap cover per piece                     | 0.8 m                              | HERON                       |
| Roof sheet per m <sup>2</sup> of covered area | 1.0 m <sup>2</sup> /m <sup>2</sup> | Rate Gen Material Constants |

**Worked example.** The 12.00 × 8.00 m bungalow (wall plates 8.00 m apart), a 0.60 m overhang all round, and a 30° pitch.

1. Plan over eaves =  $13.20 \times 9.20 = 121.44 \text{ m}^2$ .
2. Area on slope =  $121.44 \times 1.155 = 140.2 \text{ m}^2$ .
3. Rafter length =  $(4.00 + 0.60) \div \cos 30^\circ = 4.60 \div 0.866 = 5.31 \text{ m}$ .

4. Rise =  $4.00 \times \tan 30^\circ = 2.31$  m.

5. At  $2.0 \text{ m}^2$  of cover per sheet,  $140.2 \div 2.0 = 70.1$ , so **71 sheets**.

**Laps on long-span sheets.** Rate Gen's  $1.0 \text{ m}^2$  per  $\text{m}^2$  assumes your measured area already covers the laps, or that the sheet is priced per  $\text{m}^2$  of cover. If you buy by sheet width, allow for the side lap: a sheet about 1.0 m wide typically covers 0.85 to 0.9 m. Allow an end lap of about 150 to 200 mm where sheets join along the slope. Together these usually add **10 to 15%** to the net area on the slope. Check the manufacturer's effective cover width.

**Common mistake:** pricing roof covering on the plan area. At  $30^\circ$ , that leaves out 15.5% of the roof.

## Painting coverage

**ADLM values:**

| Product                           | Value                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------|
| QUIV, Rate Gen Material Constants | 0.026 drum (20 L) per $\text{m}^2$ , all coats, which is $0.52 \text{ L/m}^2$ |
| HERON                             | $2.0 \text{ m}^2$ per litre, all coats ( $0.5 \text{ L/m}^2$ ), plus 5% waste |
| HERON primer                      | $3.57 \text{ m}^2$ per litre, plus 5% waste                                   |

**Typical.** A good emulsion covers about 10 to  $12 \text{ m}^2$  per litre per coat on smooth, sealed plaster. Expect less on rough render. A mist coat plus two full coats on new render comes close to ADLM's  $0.5 \text{ L}$  per  $\text{m}^2$ .

**Worked example.**  $250 \text{ m}^2$  of emulsion:

- QUIV and Rate Gen:  $250 \times 0.026 = 6.5$  drums (130 L).
- HERON:  $250 \times 1.05 \div 2.0 = 131.25$ , so **132 L**.

**Common mistake:** applying a per-coat coverage figure ( $10 \text{ m}^2/\text{L}$ ) to a total that needs three coats. That under-orders paint by about two-thirds.

## Tiling with wastage

**ADLM values:**

| Item                                    | Value                                  | Product                                  |
|-----------------------------------------|----------------------------------------|------------------------------------------|
| Tiles per $\text{m}^2$ of floor or wall | $1.10 \text{ m}^2$ (10% cutting waste) | QUIV, HERON, Rate Gen Material Constants |
| Tile pack coverage                      | $1.44 \text{ m}^2$ per pack            | QUIV, HERON                              |
| Bed (cement)                            | 0.40 bag per $\text{m}^2$              | QUIV, Rate Gen Material Constants        |
| Bed (sharp sand)                        | $0.057 \text{ t}$ per $\text{m}^2$     | QUIV, Rate Gen Material Constants        |
| White cement grout                      | 0.02 bag per $\text{m}^2$ (about 1 kg) | QUIV, HERON, Rate Gen Material Constants |
| Floor tile bed                          | 40 mm, 1:3                             | HERON                                    |
| Wall tile bed                           | 25 mm, 1:3                             | HERON                                    |

A pack of  $1.44 \text{ m}^2$  is four  $600 \times 600 \text{ mm}$  tiles. To count individual tiles: number = area  $\times$  waste factor  $\div$  area of one tile.

**Worked example.** 600 × 600 mm floor tiles to 87.20 m<sup>2</sup>:

1. Tiles = 87.20 × 1.10 = **95.92 m<sup>2</sup>**.
2. Packs = 95.92 ÷ 1.44 = 66.6, so **67 packs**.
3. Bed = 87.20 × 0.40 = **35 bags** of cement and 87.20 × 0.057 = **4.97 t** of sand.
4. Grout = 87.20 × 0.02 = **1.74 bags** of white cement.

Allow more waste (12 to 15%) for small rooms, diagonal laying or large-format tiles.

**Common mistake:** ordering exactly the net area. Cuts at walls and around fittings always produce offcuts you cannot use.

## Earthworks volumes: average end area and prismoidal

For cuttings, embankments and road works, take cross-sections at regular intervals and calculate the volume between them.

| Method                                             | Formula                                                                   |
|----------------------------------------------------|---------------------------------------------------------------------------|
| Average end area (two sections)                    | $V = L \times (A1 + A2) \div 2$                                           |
| Average end area (series, spacing d)               | $V = d \times (A1 \div 2 + A2 + A3 + \dots + An \div 2)$                  |
| Prismoidal (end areas A1, A2, true mid-section Am) | $V = L \div 6 \times (A1 + 4Am + A2)$                                     |
| Simpson's rule (odd number of sections, spacing d) | $V = d \div 3 \times (A1 + 4(A2 + A4 + \dots) + 2(A3 + A5 + \dots) + An)$ |
| Spot levels on a grid (cell area a)                | $V = a \div 4 \times (\Sigma h1 + 2\Sigma h2 + 3\Sigma h3 + 4\Sigma h4)$  |

In the grid formula, h1 to h4 are depths at points shared by 1, 2, 3 and 4 grid cells.

**Worked example.** Three cross-sections 20 m apart: 12.5 m<sup>2</sup>, 15.0 m<sup>2</sup> and 18.2 m<sup>2</sup>.

1. Average end area:  $20 \times (12.5 \div 2 + 15.0 + 18.2 \div 2) = 20 \times 30.35 = \mathbf{607 \text{ m}^3}$ .
2. Prismoidal over 40 m, with 15.0 m<sup>2</sup> as the mid-section:  $40 \div 6 \times (12.5 + 4 \times 15.0 + 18.2) = 6.667 \times 90.7 = \mathbf{605 \text{ m}^3}$ .

Average end area usually gives slightly more than prismoidal. The difference grows where the sections change shape sharply.

**Common mistake:** using a mid-section in the prismoidal formula that was averaged from the two end sections. It must be a surveyed section at the true midpoint, or the formula gives the same answer as average end area.

## Pipework and cable lengths with fittings allowances

**Lengths.** Measure along the run, including the vertical drops to each outlet, switch or fitting. BESMM and SMM measure pipework and cable in metres, with fittings counted as extra over the pipe.

**Stock lengths and connectors.** These are ADLM values from the Rate Gen **Service Constants** on the website:

| Service | Standard length | Connectors                  |
|---------|-----------------|-----------------------------|
| Pipe    | 6 m             | one per joint (lengths – 1) |
| Conduit | 3 m             | one per joint               |

| Service                | Standard length | Connectors    |
|------------------------|-----------------|---------------|
| Cable tray or trunking | 3 m             | one per joint |
| Duct                   | 1.2 m           | one per joint |
| Cable                  | 100 m drum      | none          |

The fittings allowance in **Service Constants** is **0%** by default. Rate Gen adds nothing for fittings unless you set a percentage. ADLM's MEP plugin takes lengths from the Revit model and adds no fittings allowance of its own.

**Typical allowances** where fittings are not measured separately:

- Pipework fittings: 10 to 15% of the pipe cost.
- Cable: 5 to 10% on length, for terminations, loops and snaking.
- Conduit: about 10% for bends and boxes.

These are rules of thumb for early estimates. A priced bill should measure fittings.

**Worked example.**

1. 47 m of 20 mm conduit:  $47 \div 3 = 15.7$ , so **16 lengths** and **15 couplers**.
2. 64 m of pipe:  $64 \div 6 = 10.7$ , so **11 lengths** and **10 couplings**.
3. 230 m of 2.5 mm<sup>2</sup> cable with a 10% allowance:  $230 \times 1.10 = 253$  m, which is 3 drums of 100 m.

**Common mistake:** counting one coupler per length. A straight run of n lengths has n – 1 joints.

## Formula library

One row for each formula in this handbook.

| Item                              | Formula                                               | Units          | Notes                                |
|-----------------------------------|-------------------------------------------------------|----------------|--------------------------------------|
| Rectangle                         | $L \times B$                                          | m <sup>2</sup> |                                      |
| Triangle                          | $\frac{1}{2} \times \text{base} \times \text{height}$ | m <sup>2</sup> |                                      |
| Trapezium                         | $\frac{1}{2} \times (a + b) \times h$                 | m <sup>2</sup> |                                      |
| Circle                            | $\pi \times d^2 \div 4$                               | m <sup>2</sup> | d is the diameter                    |
| Cylinder                          | $\pi \times d^2 \div 4 \times h$                      | m <sup>3</sup> | Piles, round columns                 |
| Cone or pyramid                   | $\frac{1}{3} \times \text{base area} \times h$        | m <sup>3</sup> |                                      |
| Frustum                           | $h \div 3 \times (A1 + A2 + \sqrt{(A1 \times A2)})$   | m <sup>3</sup> | Tapered pads                         |
| Centre-line girth (external dims) | $2(L + B) - 4t$                                       | m              | Rectangle; any plan loses 4t net     |
| Centre-line girth (internal dims) | $2(L + B) + 4t$                                       | m              |                                      |
| Trench excavation                 | $(w + 2ws) \times d \times L$                         | m <sup>3</sup> | ws = working space, typically 0.30 m |

| Item                      | Formula                                                                              | Units          | Notes                                                |
|---------------------------|--------------------------------------------------------------------------------------|----------------|------------------------------------------------------|
| Earthwork support         | $2 \times d \times L$                                                                | m <sup>2</sup> | Both sides of a trench                               |
| Loose volume              | in-situ $\times$ (1 + bulking)                                                       | m <sup>3</sup> | Bulking 10 to 70% by soil type                       |
| Fill to order             | compacted $\times$ 1.2 to 1.3                                                        | m <sup>3</sup> | Typical compaction factor                            |
| Fill weight               | m <sup>3</sup> $\times$ 1.6                                                          | t              | ADLM value                                           |
| Concrete cement           | $V \times c \div \Sigma \times 1.54 \times 1,440 \div 50$                            | bags           | ADLM; 6.34 bags/m <sup>3</sup> for 1:2:4             |
| Concrete sand             | $V \times s \div \Sigma \times 1.54 \times 1.44$                                     | t              | ADLM                                                 |
| Concrete granite          | sand t $\times$ 2                                                                    | t              | ADLM; valid when g = 2s                              |
| Blinding (1:4:8)          | $V \times 1.05 \times (3.25 \text{ bags}, 0.65 \text{ t}, 1.30 \text{ t})$           | bags, t        | ADLM                                                 |
| Blocks                    | area $\times$ 10 $\times$ 1.03                                                       | nr             | ADLM; same for 225 and 150                           |
| Blockwork mortar          | area $\times$ 0.20 bag + area $\times$ 0.055 t                                       | bags, t        | ADLM, 225 mm                                         |
| Render                    | area $\times$ 0.13 bag + area $\times$ 0.05 t                                        | bags, t        | ADLM, per face                                       |
| Screed (40 mm 1:3)        | area $\times$ 0.40 bag + area $\times$ 0.057 t                                       | bags, t        | ADLM                                                 |
| Mortar (first principles) | $A \times t \times \text{dry} \times \text{waste} \times c \div (c + s) \times 28.8$ | bags           | HERON method                                         |
| Rebar weight              | $d^2 \div 162 \times \text{length}$                                                  | kg             | d in mm                                              |
| Bars per tonne            | $1,000 \div (d^2 \div 162 \times 12)$                                                | nr             | 12 m bars                                            |
| Lap length                | 40d to 50d                                                                           | mm             | Typical; drawings govern                             |
| Link length               | $2((b - 2c) + (h - 2c)) + 0.15$                                                      | m              | HERON template                                       |
| Number of links           | $L \div \text{spacing} + 1$                                                          | nr             |                                                      |
| L-bar cutting length      | $A + B - 0.5r - d$                                                                   | m              | BS 8666 shape 11; 2d per 90° bend for d $\leq$ 16 mm |
| Binding wire              | steel kg $\times$ 0.01                                                               | kg             | ADLM                                                 |
| Beam formwork             | $(2h + w) \times L$                                                                  | m <sup>2</sup> | QUIV                                                 |
| Column formwork           | $2(a + b) \times h$                                                                  | m <sup>2</sup> | QUIV                                                 |
| Formwork sheets           | area $\div$ 2.88                                                                     | sheets         | ADLM                                                 |
| Pitch factor              | $1 \div \cos(\text{pitch})$                                                          | factor         |                                                      |
| Roof slope area           | plan area over eaves $\times$ pitch factor                                           | m <sup>2</sup> | Uniform pitch                                        |
| Rafter length             | $(\frac{1}{2} \text{ span} + \text{overhang}) \div \cos(\text{pitch})$               | m              | QUIV                                                 |

| Item             | Formula                                              | Units          | Notes                                           |
|------------------|------------------------------------------------------|----------------|-------------------------------------------------|
| Roof rise        | $\frac{1}{2} \text{ span} \times \tan(\text{pitch})$ | m              | QUIV                                            |
| Roof sheets      | $\text{slope area} \div 2.0$                         | sheets         | QUIV, HERON                                     |
| Paint            | $\text{area} \times 0.026$                           | 20 L drums     | QUIV, Rate Gen; 0.52 L/m <sup>2</sup> all coats |
| Tiles            | $\text{area} \times 1.10$                            | m <sup>2</sup> | QUIV, Rate Gen                                  |
| Tile packs       | $\text{area} \times 1.10 \div 1.44$                  | packs          | QUIV, HERON                                     |
| Average end area | $L \times (A1 + A2) \div 2$                          | m <sup>3</sup> |                                                 |
| Prismoidal       | $L \div 6 \times (A1 + 4Am + A2)$                    | m <sup>3</sup> | Am measured, not averaged                       |
| Pipe lengths     | $\text{run} \div \text{stock length, rounded up}$    | nr             | 6 m pipe, 3 m conduit                           |
| Couplers         | $\text{lengths} - 1$                                 | nr             | Per straight run                                |

## Constants

### Material densities and conversions

| Material                                    | Value                            | Source                                          |
|---------------------------------------------|----------------------------------|-------------------------------------------------|
| Cement, bulk                                | 1,440 kg/m <sup>3</sup>          | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Sharp sand, for concrete conversion         | 1,440 kg/m <sup>3</sup>          | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Hardcore, laterite, general fill            | 1.6 t/m <sup>3</sup>             | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Sharp sand and granite, for rate conversion | 1.6 t/m <sup>3</sup>             | ADLM (QUIV rate conversion)                     |
| Dry volume factor, concrete                 | 1.54                             | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Dry volume factor, mortar, screed, render   | 1.30                             | ADLM (HERON)                                    |
| Dry volume factor, tile beds                | 1.20                             | ADLM (HERON)                                    |
| Plain concrete                              | 2,400 kg/m <sup>3</sup>          | Typical                                         |
| Reinforced concrete                         | 2,500 kg/m <sup>3</sup>          | Typical (about 2% steel)                        |
| Steel                                       | 7,850 kg/m <sup>3</sup>          | Typical                                         |
| Water                                       | 1,000 kg/m <sup>3</sup>          | Typical                                         |
| Loose sand, dry                             | 1,450 to 1,600 kg/m <sup>3</sup> | Typical; varies with moisture                   |

| Material                 | Value                            | Source  |
|--------------------------|----------------------------------|---------|
| Granite chippings, loose | 1,450 to 1,600 kg/m <sup>3</sup> | Typical |
| Compacted laterite       | 1,800 to 2,000 kg/m <sup>3</sup> | Typical |

#### NOTE

ADLM uses 1,440 kg/m<sup>3</sup> for sharp sand because that is the median of the trip-to-tonne conversions Nigerian firms use in their own bills (they range from 1.36 to 1.67 t/m<sup>3</sup>). Damp sand bulks and weighs differently, so check what your supplier means by a tonne.

## Bag, roll and pack sizes

| Item                    | Size                                                         | Source                             |
|-------------------------|--------------------------------------------------------------|------------------------------------|
| Cement bag              | 50 kg                                                        | ADLM                               |
| Emulsion paint drum     | 20 L                                                         | ADLM                               |
| POP glue (Top Bond)     | 10 kg                                                        | ADLM                               |
| Formwork board          | 2.4 × 1.2 m (2.88 m <sup>2</sup> )                           | ADLM                               |
| BRC mesh roll           | 48 m <sup>2</sup> gross, 45.7 m <sup>2</sup> net of 5% laps  | ADLM                               |
| DPM roll                | 50 m <sup>2</sup> gross, 45.5 m <sup>2</sup> net of 10% laps | ADLM                               |
| DPC roll                | 50 m                                                         | ADLM (HERON)                       |
| Waterproofing felt roll | 40 m <sup>2</sup>                                            | ADLM (Rate Gen Material Constants) |
| Tile pack               | 1.44 m <sup>2</sup>                                          | ADLM (QUIV, HERON)                 |
| Reinforcement bar       | 12 m                                                         | ADLM                               |
| Roof timber             | 3.6 m                                                        | ADLM                               |
| Conduit, trunking       | 3 m                                                          | ADLM (Rate Gen)                    |
| Pipe                    | 6 m                                                          | ADLM (Rate Gen)                    |
| Cable drum              | 100 m                                                        | ADLM (Rate Gen)                    |

## Wastage allowances

| Work                      | Factor | Source                                          |
|---------------------------|--------|-------------------------------------------------|
| Concrete (on top of 1.54) | 1.00   | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Blinding                  | 1.05   | ADLM                                            |
| Blockwork                 | 1.03   | ADLM (QUIV, HERON, Rate Gen Material Constants) |

| Work                                                                  | Factor       | Source                                          |
|-----------------------------------------------------------------------|--------------|-------------------------------------------------|
| Reinforcement                                                         | 1.05         | ADLM (QUIV, Rate Gen Material Constants)        |
| Reinforcement (on the number of 12 m bars, covering laps and offcuts) | 1.10         | ADLM (HERON)                                    |
| Beam main bars, laps and waste                                        | 1.15         | ADLM (QUIV)                                     |
| Staircase bars, laps and waste                                        | 1.10         | ADLM (QUIV)                                     |
| Finishes (render, screed)                                             | 1.05         | ADLM                                            |
| Formwork (whole plywood sheets, cutting and striking loss)            | 1.10         | ADLM (HERON)                                    |
| Tiles                                                                 | 1.10         | ADLM (QUIV, HERON, Rate Gen Material Constants) |
| Paint and primer                                                      | 1.05         | ADLM (HERON)                                    |
| Roofing sheets, laps                                                  | 1.10 to 1.15 | Typical                                         |
| Timber                                                                | 1.10         | Typical                                         |

#### TIP

Never stack waste factors. If a factor already carries waste (like the 1.54 dry-volume factor for concrete), adding 5% on top counts it twice. That is exactly why ADLM moved its concrete waste factor from 1.05 to 1.00.

## Labour outputs per day

HERON holds these outputs per gang for an 8-hour working day. It works out the labour rate per unit as the gang's cost per day divided by its output per day.

| Work          | Output per day    |
|---------------|-------------------|
| Concrete      | 10 m <sup>3</sup> |
| Filling       | 15 m <sup>3</sup> |
| Blockwork     | 12 m <sup>2</sup> |
| Reinforcement | 250 kg            |
| Formwork      | 12 m <sup>2</sup> |
| Rendering     | 20 m <sup>2</sup> |
| Plastering    | 20 m <sup>2</sup> |
| Screeding     | 25 m <sup>2</sup> |
| Painting      | 40 m <sup>2</sup> |
| Floor tiling  | 8 m <sup>2</sup>  |
| Wall tiling   | 7 m <sup>2</sup>  |

| Work              | Output per day    |
|-------------------|-------------------|
| Skirting          | 30 m              |
| Wall plate        | 30 m              |
| Roof              | 15 m <sup>2</sup> |
| Surface treatment | 50 m <sup>2</sup> |
| DPC               | 30 m              |
| DPM               | 50 m <sup>2</sup> |
| Wire mesh         | 50 m <sup>2</sup> |

**Worked example.** A blockwork gang of one mason and two labourers costs, say, ₦45,000 a day (an illustrative figure). At 12 m<sup>2</sup> a day, the labour rate is  $45,000 \div 12 = \text{₦}3,750$  per m<sup>2</sup>.

Rate Gen **Material Constants** and QUIV also hold labour rates per unit as defaults: for example ₦5,500 per m<sup>3</sup> of concrete, ₦600 per m<sup>2</sup> of blockwork, ₦1,300 per m<sup>2</sup> of rendering and ₦50,000 per tonne of reinforcement. Treat these as placeholders and set your own in **Material Constants**; labour rates vary widely between states and between labour-only and gang rates.

## Plant outputs

ADLM's plant allowances are **zero by default** in Rate Gen **Material Constants**. Nobody can defend a plant figure that nobody set, so plant is costed only when you enter your own.

Typical outputs for planning:

| Plant                          | Typical output                               | Assumption                 |
|--------------------------------|----------------------------------------------|----------------------------|
| Small drum mixer (one-bag)     | 10 to 15 m <sup>3</sup> per day              | Hand-fed, with a full gang |
| Backhoe loader, trench digging | 15 to 25 m <sup>3</sup> per hour             | Ordinary soil, good access |
| Plate compactor                | 100 to 200 m <sup>2</sup> per hour per layer | 150 mm layers of laterite  |
| Tipper, local haul             | 6 to 10 trips per day                        | Under 10 km, no queues     |

Plant outputs depend heavily on site conditions. Check them against your own records before you price.

## Grade and unit conversion

### Concrete grades and nominal mixes

| Grade       | Characteristic strength at 28 days | Nominal mix commonly equated | Typical use                       |
|-------------|------------------------------------|------------------------------|-----------------------------------|
| C7.5 to C10 | 7.5 to 10 N/mm <sup>2</sup>        | 1:4:8                        | Blinding, mass fill               |
| C15         | 15 N/mm <sup>2</sup>               | 1:3:6                        | Oversite, plain strip footings    |
| C20         | 20 N/mm <sup>2</sup>               | 1:2:4                        | General reinforced work in houses |

| Grade | Characteristic strength at 28 days | Nominal mix commonly equated | Typical use                              |
|-------|------------------------------------|------------------------------|------------------------------------------|
| C25   | 25 N/mm <sup>2</sup>               | 1:1.5:3                      | Suspended slabs, beams, columns          |
| C30   | 30 N/mm <sup>2</sup>               | 1:1:2                        | Heavier structural work, water-retaining |

#### IMPORTANT

These pairings are a rough guide only. A nominal mix batched by volume on site often falls short of the grade it is equated with, especially when sand is damp or the water is not controlled. Where the specification calls for a grade (C25, for example), a design mix tested by cube results governs, and its cement content comes from the mix design, not from this table.

## Reinforcement grades and notation

| Notation | Meaning                             | Typical yield strength                                                               |
|----------|-------------------------------------|--------------------------------------------------------------------------------------|
| R        | Plain round mild steel              | 250 N/mm <sup>2</sup>                                                                |
| Y or T   | High yield deformed bar (ribbed)    | 410 to 460 N/mm <sup>2</sup> under BS 8110; 500 N/mm <sup>2</sup> under BS 4449:2005 |
| H        | High yield bar in BS 8666 schedules | 500 N/mm <sup>2</sup>                                                                |

How to read the notation:

- **4Y16** means four high yield bars, 16 mm in diameter.
- **R10-200** means plain 10 mm bars (usually links) at 200 mm centres.

High yield bars carry more load per kilogram than mild steel, so never substitute mild steel without the engineer's approval.

## Block sizes

| Common name  | Size (L × H × W)   | Typical use                       |
|--------------|--------------------|-----------------------------------|
| 9 inch block | 450 × 225 × 225 mm | External and load-bearing walls   |
| 6 inch block | 450 × 225 × 150 mm | Internal partitions               |
| 5 inch block | 450 × 225 × 125 mm | Light partitions, where available |
| 4 inch block | 450 × 225 × 100 mm | Infill, fences, where available   |

Each block has a 450 × 225 mm face, so the count per m<sup>2</sup> is the same for every thickness: about 9.3 laid, 10 ordered.

## Imperial to metric

| Imperial | Metric   |
|----------|----------|
| 1 inch   | 25.4 mm  |
| 1 foot   | 0.3048 m |

| Imperial          | Metric                 |
|-------------------|------------------------|
| 1 yard            | 0.9144 m               |
| 1 ft <sup>2</sup> | 0.0929 m <sup>2</sup>  |
| 1 yd <sup>2</sup> | 0.8361 m <sup>2</sup>  |
| 1 ft <sup>3</sup> | 0.0283 m <sup>3</sup>  |
| 1 yd <sup>3</sup> | 0.7646 m <sup>3</sup>  |
| 1 acre            | 4,046.9 m <sup>2</sup> |
| 1 lb              | 0.4536 kg              |
| 1 long ton (UK)   | 1,016 kg               |
| 1 imperial gallon | 4.546 L                |
| 1 US gallon       | 3.785 L                |

A standard Nigerian plot of 60 × 120 ft is 7,200 ft<sup>2</sup>, which is **668.9 m<sup>2</sup>** (about 18.29 × 36.58 m).

To go back from metric: 1 m = 3.281 ft, 1 m<sup>2</sup> = 10.764 ft<sup>2</sup>, 1 m<sup>3</sup> = 35.315 ft<sup>3</sup>, 1 kg = 2.205 lb.

### Tonnes and trips of sand, laterite and granite

Suppliers sell sand, laterite and granite by the trip, named after the truck's nominal capacity. The volume in a trip depends on the truck and on how full it is loaded.

| Typical truck                | Typical load                  |
|------------------------------|-------------------------------|
| 5-tonne tipper               | about 3.5 m <sup>3</sup>      |
| 10-tonne tipper (6-wheeler)  | about 6 to 7 m <sup>3</sup>   |
| 20-tonne tipper (10-wheeler) | about 12 to 14 m <sup>3</sup> |
| 30-tonne trailer             | about 18 to 20 m <sup>3</sup> |

These volumes are typical, not fixed. Confirm what a trip means with your supplier before you price.

#### Conversions:

- Tonnes = m<sup>3</sup> × density (1.6 t/m<sup>3</sup> for fill, 1.44 t/m<sup>3</sup> for sharp sand, as ADLM uses).
- m<sup>3</sup> = tonnes ÷ density.
- Trips = loose m<sup>3</sup> ÷ m<sup>3</sup> per trip.

QUIV's **Material Constants** count hardcore and laterite at **5 m<sup>3</sup> per truckload trip** by default.

**Worked example.** You need 20.93 t of hardcore and the quarry sends 10-tonne trucks:

1. Volume = 20.93 ÷ 1.6 = 13.1 m<sup>3</sup>.
2. At about 6.5 m<sup>3</sup> a trip, 13.1 ÷ 6.5 = 2.0, so **2 trips**.

### IMPORTANT

Keep the three units apart: tonnes, cubic metres and trips. Most expensive filling errors come from a factor in one unit being read as another.

## Cement bag conversions

| From                                  | To                                      |
|---------------------------------------|-----------------------------------------|
| 1 bag                                 | 50 kg                                   |
| 1 tonne                               | 20 bags                                 |
| 1 m <sup>3</sup> of cement (1,440 kg) | 28.8 bags                               |
| 1 bag                                 | 0.0347 m <sup>3</sup> (34.7 L)          |
| 1 head pan of cement                  | about 0.6 bag (typical 20 L pan, level) |

A level head pan holds about 20 L, so  $20 \div 34.7 = 0.58$  bag. Head pans vary, so measure the one on your site before you rely on it.

## Percentages and slopes

### Percentages:

- Waste factor =  $1 + \text{waste \%} \div 100$ . For example, 5% gives 1.05.
- Markup and margin are different. A 25% markup on cost is a 20% margin on price:  $\text{margin} = \text{markup} \div (1 + \text{markup})$ .
- Percentage change =  $(\text{new} - \text{old}) \div \text{old} \times 100$ .

**Slopes.** Slope % =  $\text{rise} \div \text{run} \times 100 = \tan(\text{angle}) \times 100$ . A gradient of 1 in n is  $100 \div n$  percent.

| Gradient | Percent | Degrees | Typical use               |
|----------|---------|---------|---------------------------|
| 1 in 100 | 1.0%    | 0.57°   | Drain fall, minimum       |
| 1 in 80  | 1.25%   | 0.72°   | Floor and drain falls     |
| 1 in 60  | 1.67%   | 0.95°   | Flat roof falls           |
| 1 in 40  | 2.5%    | 1.43°   | Paving falls, foul drains |
| 1 in 20  | 5.0%    | 2.86°   | Gentle ramps              |
| 1 in 12  | 8.33%   | 4.76°   | Accessible ramp, maximum  |
| 1 in 10  | 10.0%   | 5.71°   | Driveways                 |
| 1 in 4   | 25.0%   | 14.04°  | Steep drive               |
| 1 in 2   | 50.0%   | 26.57°  | Roof pitch                |
| 1 in 1   | 100%    | 45.0°   | Steep roof                |

**Common mistake:** reading a 100% slope as vertical. A 100% slope is 45°: one metre up for every metre across.

This part of the handbook follows real jobs from start to finish. Each use case shows which ADLM products to use, in what order, and where one product hands over to the next. The button names are the ones you see on screen. For the full detail on any product, follow the link to its own guide.

## Which ADLM tool for which job

Start here when you are not sure which product to open. Most jobs use two or three products together, with ADLM Cloud as the place the project ends up.

| The job                                         | Use this                                                                               | Where it ends up                                                             |
|-------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Check whether a Revit model is ready to measure | <a href="#">QUIV for Revit</a> : <b>Model Checker</b>                                  | A readiness score, an issues list and a <b>QS Query Report</b>               |
| Measure the building from a Revit model         | <a href="#">QUIV for Revit</a>                                                         | A bill and a budget in Revit, saved to ADLM Cloud                            |
| Measure from PDF drawings                       | <a href="#">ADLM HERON</a> with PlanSwift 10 or 11 and the ADLM templates              | A priced bill, a budget and an Excel bill, saved to ADLM Cloud               |
| Measure MEP services from a Revit model         | <a href="#">SERVIQ for Revit MEP</a>                                                   | A priced services bill and material schedule, saved to ADLM Cloud            |
| Price a bill someone sent you in Excel          | ADLM Cloud: <b>Import Excel BoQ · HERON</b> , then <a href="#">HERON</a> to measure it | A HERON project with your measurements behind the client's lines             |
| Build and keep your own rates                   | <a href="#">ADLM Rate Gen</a>                                                          | One rate library on your account, read by QUIV, HERON, SERVIQ and ADLM Cloud |
| Plan durations and crew sizes                   | <a href="#">ADLM Time Pro</a>                                                          | An Excel record and a Microsoft Project programme                            |
| Plan the budget and purchases                   | <a href="#">ADLM Cloud</a> : <b>Budget</b>                                             | A material, labour and plant breakdown, and a buy schedule                   |
| Value work and issue payment certificates       | <a href="#">ADLM Cloud</a> : <b>Valuation</b>                                          | Interim payment applications, certificates, variations and a final account   |
| Report to a client                              | <a href="#">ADLM Cloud</a> : <b>Share dashboard, Project report, Export</b>            | A public read-only dashboard, PDF reports and Excel workbooks                |
| Work with a colleague on one project            | <a href="#">ADLM Cloud</a> : <b>Collaborators</b>                                      | Shared access at <b>View only</b> or <b>Full access</b>                      |

### NOTE

Each product is a separate subscription. QUIV does not include SERVIQ, and HERON does not include Rate Gen. Pricing from the rate library in any product needs a Rate Gen subscription on the same account.

# Use case 1: A bungalow BoQ from a Revit model

## Scenario

An architect sends you the Revit model of a three-bedroom bungalow on strip foundations. The client wants a priced bill of quantities by the end of the week.

| Who it is for                                              | Tools used                                |
|------------------------------------------------------------|-------------------------------------------|
| Quantity surveyors and estimators who receive Revit models | QUIV for Revit, ADLM Rate Gen, ADLM Cloud |

## Steps

In Revit, check the model first:

1. Open the model in Revit.
2. On the **ADLM Calculator** tab, click **Model Checker**.
3. Under **Select Model Type**, choose **Architectural Model**.
4. Tick **Check for Overlapping Elements**, then click **Run Model Check**.
5. Read the **Readiness Score** and the **Element Categories**. Click each row under **Issues** to find the element in Revit, and fix duplicates before you measure.
6. If parts of the model are missing or unclear, click **View QS Query** and send the **QS Query Report** to the architect with **Export PDF**.

Measure with QUIV:

1. On the **ADLM Calculator** tab, click **ADLM Calculator** and sign in if asked.
2. On the dashboard, under **Ready for a new project?**, click **Bungalow**.
3. On the **Take off List**, tick the items this job needs and click **Continue**.
4. Work down the sidebar from **Oversite Qty** and **Strip Qty** to **Finishes Qty**. In each module, choose the level and type, check the **Model Quantities**, enter your own inputs (bar sizes, spacing, thicknesses), then click **Save**.
5. The first time you save, type the **Project name** in the **Save Takeoff** window. Use a clear name, for example the client and the building.

### TIP

To measure everything in one go, click ♦ **Run the whole takeoff** under the take-off list, then **Let QUIV run the whole takeoff**. Items that start with a pick in Revit are listed under **Still needs you**, and you measure those yourself.

Check the bill and save it to the cloud:

1. Click **BoQ** under **Database** in the sidebar.
2. Read down the **Bill of Quantities**. Use the **Done** tick for each line you have checked.
3. Click **Save To Cloud** and confirm the **Project name**.

Price the bill on ADLM Cloud:

1. Sign in at [adlmstudio.net](https://adlmstudio.net) and open your QUIV projects ( </projects/revit> ). Click **Refresh projects** if the job is not there yet.

2. Open the project and click **Bill of Quantity** under **Views**.
3. On the ribbon, open the **Rates** tab and click **Load RateGen rates**.
4. On the **Home** tab, tick **Only fill empty rates**, then on the **Rates** tab click **Sync rates**. Rate Gen fills every line it can match.
5. For each line still without a rate, click its **Rate** cell and type part of a rate's name. Pick a rate from your library, or type a figure.
6. In the **Summary** box at the foot of the bill, set **Contingency** and **VAT**. On the **Contract** tab, set **Preliminaries %**.
7. Click **Save changes**.

Send the bill out:

1. Click **Export** in the top bar.
2. Choose **Bungalow** under **Elemental BoQ** for a bill by building element, or **Bungalow (Trade format)** for a bill by work section.
3. When the bill goes to the client, click **Mark as tendered** in the **Summary** box, and turn **Follow RateGen changes** off on the **Rates** tab.

## What you get at the end

A measured and priced bill on ADLM Cloud, an elemental or trade bill in Excel, and a budget behind every line that you can use later for purchasing and valuations.

## Tips

- You can also price inside Revit: on the QUIV Bill, click **Load RateGen Prices**, or **View Rates** and **Map** for one line. Pricing on the cloud is easier to share and to change later.
- Rates you type or pick on the cloud are kept when you re-save the project from QUIV.
- If a quantity looks double, run the **Model Checker** again with **Check for Overlapping Elements** ticked, and check that no linked model is placed twice.

## Use case 2: Pricing a tender from PDF drawings

### Scenario

A contractor sends you a tender set of PDF drawings for a two-storey office block. There is no model. You need a priced bill in Excel by the tender date.

| Who it is for                                              | Tools used                                                                                |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Estimators and quantity surveyors working from 2D drawings | ADLM HERON with PlanSwift, the ADLM templates, ADLM Rate Gen, Microsoft Excel, ADLM Cloud |

## Steps

Set up the job in PlanSwift:

1. Open **ADLM HERON** and sign in, so the ADLM templates are switched on. If this is the first sign-in on the PC, restart PlanSwift.
2. In PlanSwift, create a new job and set it to metric.

3. Import the PDF drawings, rename each page (for example "Ground Floor Plan") and group the pages into folders by drawing set.
4. Scale every page you will measure on, using a long grid dimension. Check the scale on a dimension running the other way with PlanSwift's Dimension tool.

#### IMPORTANT

The scale is set for each page, not the whole job. An unscaled page gives wrong quantities on every line measured on it.

Measure with the ADLM templates:

1. In PlanSwift's templates panel, choose **Complete ADLM TakeOff Plugin**.
2. Work trade by trade: SUBSTRUCTURE, FRAME, BLOCKWORK AND OPENINGS, FINISHES, ROOFING, then the services if they are in your scope.
3. Name each item after the room or element, for example "Offices" or "225 Wall Area". The names become the descriptions on your bill.
4. Organise the folders the way you want the bill to read, for example by storey: "Office Sub", "Office GF", "Office FF".

Read and price the takeoff in HERON:

1. In HERON, on the **Dashboard**, click **Start New**.
2. Click **Quantity Take Off**. Check the **Take Off Items** and the **Take Off Breakdown** for each folder.
3. Double-click concrete items to add reinforcement with **+ Bars** and **+ Links**, then click **Apply Reinforcement**.
4. Click **Save / Export**. The **Review Takeoff Before Saving** screen opens.
5. Type the job name in **Project Name**.
6. Click **Load rates**. HERON prices the bill from your Rate Gen library and shows how many items matched.
7. For lines with no rate, select the row, type in **Search Rate**: and double-click a result, or type a rate in the **Rate** cell.
8. Check any row shaded amber. Its rate is below the material and labour cost HERON worked out.
9. Use **Up** and **Down** to put the bill in the order you want it printed.

Export and save:

1. Click **Export All** for one workbook with every folder, or **Export Folder(s)** to choose.
2. Click **Save to Cloud** so the job is kept on your account and can be opened on the website.
3. On the **Budget** page in HERON, pick a folder, click **Load rates**, then **Export** for the linked budget workbook.

## What you get at the end

An Excel bill of quantities with one sheet per folder, QS item lettering and a **GEN SUMMARY** sheet, a linked budget workbook with a **Basic Prices** sheet, and the same job on ADLM Cloud.

## Tips

- If you rescale a page after measuring, click **Reload** in **Quantity Take Off** before you save.
- Avoid two items with the same name in one folder. HERON merges them.

- For steelwork, name items with the full designation (for example UKB 305x165x40) and run **Steel Tonnage** to get the tonnage.
- If your firm has its own tender template in Excel, use **Connect Excel** and the **Takeoff Link** pane to link HERON's figures into your own cells instead of exporting.

## Use case 3: Your firm's own rate library

### Scenario

Your firm prices excavation, concrete and blockwork its own way, with its own gang sizes, and hires an excavator for bulk digging. You want every estimator to price QUIV and HERON jobs from the same rates.

| Who it is for                                                         | Tools used                                                                                           |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Senior quantity surveyors and practice leads who set the firm's rates | ADLM Rate Gen (desktop and the RateGen pages on the website), QUIV for Revit, ADLM HERON, ADLM Cloud |

### Steps

Set up the library once:

1. On the ADLM website, open **Profile** and set your **Pricing location (State)**. Rate Gen prices from the zone your state belongs to.
2. Open **ADLM RateGen** and sign in. Click **Yes** if it asks to update your prices to your zone.
3. Click **Library** in the sidebar. Check the location line at the top.
4. On the **Material** tab, click **Edit** on each material you buy at a different price, type your price and click **Update Library**. Do the same on the **Labour** tab for day rates and plant hire.
5. Add anything missing with **Add Material +** or **Add Labour +**, then **Save to Database**. Plant goes in the **Labour** list.

Adjust the shipped rates:

1. Open a work section, for example **Concrete**.
2. Set the **Overhead** and **Profit** boxes to your firm's percentages.
3. Click a rate's **Description** to open its **Rate Composition**.
4. Double-click a **Quantity** cell and type your figure, for example your own mix or gang output.
5. Click **Save**. You will see **Rate saved successfully. Your edits will sync to QUIV and HERON**.

Build custom rates, with plant:

1. Click **Add Custom Rate +** in any section.
2. Type the **Name of Rate** and the **Description**, and set **Overhead** and **Profit**.
3. Under **Materials**, click **Add Material** and pick each material by name, with its **Quantity** per unit.
4. Under **Labour**, click **Add Labour** for the gang. Add the plant here too, for example the excavator, because plant is in the labour library.
5. Check the **Grand Total** and click **Save**.
6. Click **Sync from Cloud** in the header to send the rate to your account.

Check the rate reached your account:

1. On the website, open **RateGen** in the left-hand menu ( `/rategen` ).
2. Open **My Custom Rates**, then **Effective Rates**. Your rate shows with a **Source** of `user-custom` .

Use the rates on jobs:

1. **HERON**: on the review screen, click **Load rates**, or use **Search Rate**: to apply one rate to a row. In the **Budget**, type a name in a **Price** cell to search your library.
2. **QUIV**: save the job to the cloud, open it on ADLM Cloud, and pick your rates in the **Bill of Quantity** by typing part of the name in the **Rate** cell. Picking a rate also writes its materials, labour and plant into the **Budget**.
3. **Any ADLM Cloud bill**: use **Sync rates** on the **Rates** tab to fill empty lines from your library.

## What you get at the end

One library on your account, priced for your location, with your own build-ups and custom rates. QUIV, HERON and ADLM Cloud all price from it, so two estimators pricing the same item get the same rate.

## Tips

### IMPORTANT

Build each custom rate in one place. Rate Gen desktop keeps the cloud copy of your custom rates in step with its own **Saved Rates** list, and it does not download rates built elsewhere. So if you use the desktop app, build your custom rates there. A rate built on the website, or on another PC, can be removed from your account the next time the desktop app syncs.

- If your firm works only on the website, you can build custom rates there instead: open `/work/library` and click **Build a custom rate**. On the website a rate has a separate **Plant** group, where a plant line takes the machine's name and your price per hour.
- Change prices in the library, and quantities in the build-up. A price is shared by every rate that uses it. A quantity belongs to one rate.
- A project you have already priced keeps its rates. A library change reaches it only when the line is priced again, or when **Follow RateGen changes** is on.

## Use case 4: A budget for a client

### Scenario

A private client is building a duplex and wants to know what it will cost before committing, split into what goes on materials and what goes on labour. You have measured the building in QUIV or HERON and saved it to the cloud.

| Who it is for                                                                             | Tools used                                              |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Quantity surveyors advising a client or a developer; contractors planning their own costs | QUIV for Revit or ADLM HERON, ADLM Rate Gen, ADLM Cloud |

### Steps

Price the bill so the budget fills itself:

1. On ADLM Cloud, open the project and click **Bill of Quantity** under **Views**.
2. Click a line's **Rate** cell, type part of a rate's name and pick a Rate Gen rate. ADLM writes that line's materials, labour and plant into the **Budget** at the same time, so the bill and the budget agree.
3. Repeat for each line, or use **Sync rates** on the **Rates** tab with **Only fill empty rates** ticked.
4. Click **Save changes**.

Check the budget:

1. Click **Budget** under **Views**. Each bill item is broken down into rows of **Material**, **Labour**, **Plant**, **Equipment** or **Consumable**.
2. Price any row still at zero by typing a rate, a  $=$  formula, or a Rate Gen name in its rate cell.
3. Under **Global Overhead & Profit**, type your percentages and click **Apply to all**.
4. Click **Save changes**. The new bill rates flow up to the **Bill of Quantity**.

Plan the purchases:

1. Build the programme first on the **PM Dashboard** (click **Generate from BoQ**, or import one as in Use case 6).
2. In the **Budget**, switch from **Breakdown** to **Buy schedule**. Read **Should already be bought**, **Buy this week** and **Not yet scheduled**, and set the **Lead time** for each material.

Give the client the figures:

1. Click **Export** in the top bar and choose **Export bill & budget workbook**. The workbook has your bill plus separate Material, Labour and Plant schedules, a Schedule of Current Prices and a Material Summary.
2. Click **Project report** for a PDF progress report, then **Download PDF**.
3. To let the client follow the job without an account, open **Dashboard**, click **Share dashboard**, tick **Enable public link** and click **Copy link**.

## What you get at the end

A budget for every bill item, a workbook that shows materials, labour and plant on separate schedules, and a buy schedule tied to the programme.

## Tips

- In QUIV you can see the same split inside Revit: open **Budget** under **Database** and use the **All**, **Material** and **Labour** filters. Tick **Bought** as items are bought.
- In HERON, the **Budget** page shows **Project cost (material + labour)** against the **Take-off value (BoQ)**, with the **Margin**. **Over budget** marks an item whose cost is higher than its rate.
- A bill item counts as complete only when every line under it is marked. Buying the materials is not enough until the labour is done too.
- If a picked rate did not price the Budget, the rate has no build-up in Rate Gen. Price the budget rows yourself, or pick a rate that has one.

## Use case 5: Monthly interim valuations and payment certificates

### Scenario

The contract is signed on a priced bill. You are the QS on site and must value the work every month, issue an interim certificate, and handle variations as the client changes things.

| Who it is for                                                        | Tools used                             |
|----------------------------------------------------------------------|----------------------------------------|
| Consultant and contractor quantity surveyors running a live contract | ADLM Cloud (any QUIV or HERON project) |

### Steps

Before the first valuation, set the contract up once:

1. Open the project and click **Valuation** under **Views**. Click **Show valuation settings**.
2. Fill in **Client / Employer**, **Retention %**, **VAT %** and **Withholding tax %**. The defaults are 5, 7.5 and 2.5.
3. Choose the **Valuation basis: By Bill of Quantity** (each line valued by its own % complete) or **By Budget (Material & Labour)**.
4. Open the **Bill of Quantity**, click the **Contract** tab on the ribbon, then **Lock contract**. Choose a 4-digit PIN and type it twice.
5. Click **Save changes**.

#### IMPORTANT

Keep the PIN safe. You need the same 4 digits to unlock the contract, and a lost PIN can only be reset by ADLM support.

Each month, record progress:

1. Walk the site with the bill.
2. On the **Bill of Quantity**, tick a line's box for work fully done (**Mark as completed**), or type a percentage in its **Done** box for part-done work.
3. Tick executed PC and provisional sums on the **Provisional** tab, and tick **Done** against preliminary items as they are executed.
4. Click **Save changes**. The save is logged against that day.

Prepare the valuation:

1. Open **Valuation** and click **Show daily valuation log**.
2. Choose the day in **Valuation date**.
3. Read down **Gross value of works to date**, **Net valuation to date**, **Less previous payments**, **Subtotal before taxes** and **Total amount due for payment**.
4. Click **Print valuation** for the formal Interim Payment Application, or **Export Excel** for an editable workbook.

Issue the certificate:

1. Under **Contract administration**, open **Certificates**.
2. Click **Issue certificate**. It certifies the current value to date as the next IPC.
3. Set its status as it moves along: **Draft**, then **Approved**, then **Paid**.

4. Download the certificate as an Excel workbook for the client.

Handle variations:

1. Open **Variations** and click **Add variation**.
2. Fill in **What changed**, the **Instruction reference**, the **Type** (**Addition** or **Omission**) and the **Value (N)**.
3. When the instruction is agreed, open the variation and click **Approve**. Only approved variations change the project total.

At the end of the job:

1. Open **Final account** and check the movement against the **Contract sum**.
2. Click **Close the final account** and confirm. Then click **Download** for the final account workbook.

## What you get at the end

A dated valuation for every month, a numbered run of interim certificates with retention, VAT and withholding tax worked out the same way each time, a variations register, and a final account.

## Tips

- Lock the contract before the first certificate, so every certificate is measured against the same baseline.
- Only the latest certificate can be deleted, which keeps the sequence honest.
- If the model or drawings are re-measured after the lock, re-save from QUIV or HERON. Re-measured quantities are recorded as actuals and new lines are raised as variations, so the contract sum does not move. HERON shows a red **CONTRACT LOCKED** chip when this applies.
- Changes are kept only when you click **Save changes**. Save before you leave the page.

## Use case 6: A programme from the bill

### Scenario

The client asks how long the bungalow from Use case 1 will take, and how many workers you need to finish in 20 weeks. Your site team has been logging gang outputs on another job for a month.

| Who it is for                                                | Tools used                                   |
|--------------------------------------------------------------|----------------------------------------------|
| Site quantity surveyors, site engineers and project managers | ADLM Time Pro, Microsoft Project, ADLM Cloud |

### Steps

Build up output history in Time Pro (before you need it):

1. Open **ADLM Time Pro** and sign in.
2. Each morning, click **Current Weather**, search your town, choose it under **SELECT CITY** and click **Fetch Weather**.
3. Click **Task Log**, then **Add Task** for each gang: the **Item of Work**, the **Trade**, **Skilled Labour** and **Unskilled Labour**, **Hours Worked** and **Break Hours**, the **Output** and **Output Unit**. Click **Save Task**.
4. Record one row per gang per day, and type the item names exactly the same way every time.

### IMPORTANT

A new Time Pro has no history, so every expected duration shows 0. Start recording on site before you need the forecast. A week of honest records is enough to begin.

Turn bill quantities into durations:

1. Keep the bill open beside you, for example the Excel export from ADLM Cloud.
2. In Time Pro, click **Duration Summary**.
3. For each item, click **Click to enter BOQ Qty...** and type the quantity from the bill, in the same unit you recorded output in. Press **Enter**.
4. Read **Expected Duration (days)** and the crew under **Required Skilled / Day** and **Required Unskilled / Day**.
5. Where the programme must be shorter, type the days you want in **Planned Duration (days)**. The crew changes to match.

Export the programme:

1. Click **Excel** to keep a record of where every duration came from.
2. Click **MS Project**. Set the **Project Start Date**, choose the **Currency** and type the resource rates (or 0 to fill them in later).
3. Click **Export** and save the **.xml** file.
4. Open the file in Microsoft Project from **File**, then **Open**. Adjust the links where trades can overlap, and set a baseline.

Bring the programme into the project on ADLM Cloud (optional):

1. Save the programme from Microsoft Project.
2. On ADLM Cloud, open the project's **PM Dashboard** and click **Import MS Project**. Choose the **.xml** or **.mpp** file.
3. Track progress with each task's % and status, and read **CPI** and **SPI** on the dashboard.

## What you get at the end

Durations and crew sizes based on what your own gangs produce, a Microsoft Project programme with tasks, links, resources and costs, and (if you import it) a programme on ADLM Cloud tied to the bill and the buy schedule.

## Tips

### IMPORTANT

In Time Pro 1.1.1 the figures you type on the **Duration Summary** clear when you leave that screen, change the Task Log or close the app. Enter your quantities and export in one sitting.

- Keep the output unit the same as the bill unit. Blockwork recorded in m2 must be forecast against a quantity in m2.
- If you only need a quick programme without site records, use **Generate from BoQ** on the **PM Dashboard**, which creates one task per bill item, and set the dates yourself.
- When quantities change, export from Time Pro again and use **Update MS Project** on the **PM Dashboard**. It refreshes the schedule and keeps your progress and links.

## Use case 7: MEP services for a commercial building

### Scenario

You are pricing the mechanical and electrical package on a four-storey office building. The services engineer has a Revit MEP model. The building itself was measured separately in QUIV.

| Who it is for                                                            | Tools used                                                                                                                                           |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Services quantity surveyors and main-contract QSs pricing an M&E package | SERVIQ for Revit MEP (the <b>ADLM Revit MEP Plugin</b> in the Installer Hub, the <b>ADLM MEP &amp; HVAC</b> tab in Revit), ADLM Rate Gen, ADLM Cloud |

### Steps

Measure the services:

1. Open the services model in Revit. If the services sit in a linked file, open that file directly.
2. On the **ADLM MEP & HVAC** tab, click **Sign In** on the **Tools** panel.
3. Start with HVAC. Click **Duct Quantity**, choose a level under **Select Level**, then a type under **Select Duct Type**.
4. Check the schedule, click **Result Info**, then **Save to Take-Off**.
5. Repeat for each level, then for **Duct Fitting Quantity**, **Air Terminal Quantity** and **HVAC Equipment Quantity**.
6. Move on to **Pipe Works Quantity** and **Plumbing Fixture Quantity**, then **Lighting Quantity**, **Power Quantity** and **Cable Quantity**.

Price inside Revit:

1. Click **Show Takeoff Database** on the **Tools** panel and open **Take-Off Data**.
2. Type a **Unit Rate** on lines you price as a single figure.
3. For lines you want built up, click the **f** button. In the **Rate Build-Up** window, use **SEARCH RATEGEN LIBRARY** to add materials and labour, adjust **Qty/unit** for waste, and click **Done**.
4. Click **Apply to similar lines** to copy a build-up to other lines in the same service.
5. Type your **Overhead %** and **Profit %** at the bottom of **Take-Off Data**.
6. Open **Budget** to see the **COST COMPOSITION** (material, labour and rate-only lines), and **Material Schedule** for the procurement list.

Save and combine on ADLM Cloud:

1. Under **PROJECT**, replace "Untitled" with the job name and "Services", then click **Save to Cloud**.
2. On the website, open your Revit MEP projects ( `/projects/mep` ), or click **View Projects** on the **ADLM** panel).
3. Open the project. On its **Dashboard**, click **Price services** under **Price services from RateGen** to price the services bill from your library. The lengths, joints and fittings it allows for come from your Services Constants.
4. Open the building project (the QUIV job). On its **Dashboard**, under **Linked Services & Works**, click **+ Link a project**, choose the services project from **Select a project...** and click **Link**. The **Linked total** now shows the whole job.

## What you get at the end

A services bill measured from the model, a material schedule for procurement, a "Complete MEP BoQ" Excel workbook from **Export to Excel**, and one figure for building and services together on ADLM Cloud.

## Tips

- Set your house standards for services once at `/rategen/services-constants` : standard lengths, how joints are counted and the fitting uplift. Click **Save constants**.
- Keep the services and the building as separate projects. They come from different models and are often valued as different packages.
- A service drawn as a single-line diagram or a note is not modelled, so it cannot be measured. Price it another way and say so in the bill.
- Every Revit session starts with an empty Take-Off Database. Click **Load Latest Local** to bring back your last take-off.

## Use case 8: Importing a client's Excel bill into HERON

### Scenario

A client sends you a bill of quantities in Excel, prepared by another firm. They want it checked and priced, and they want real measurements behind the main lines.

| Who it is for                                                  | Tools used                                                                               |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Quantity surveyors re-measuring or pricing someone else's bill | ADLM Cloud ( <b>Import Excel BoQ · HERON</b> ), ADLM HERON with PlanSwift, ADLM Rate Gen |

#### NOTE

Excel bill import is switched on for your account by ADLM and needs a live HERON subscription. If you do not see **Import Excel BoQ · HERON** on your HERON projects page, contact ADLM support through </support>. There is no import button inside HERON itself.

## Steps

Import the bill on the website:

1. Sign in at [adlmstudio.net](https://adlmstudio.net) and open your HERON projects ( `/projects/planswift` ).
2. Click **Import Excel BoQ · HERON**.
3. Type a name in **Project name (optional)**, or leave it blank to use the file name.
4. Under **Excel workbook (.xlsx)**, choose the client's file, then click **Import project**.
5. Open the new project and read the bill. Where the workbook had no material and labour schedule, one has been built for you and priced from your material constants and Rate Gen.

Build the bill in PlanSwift:

1. In PlanSwift, create a job for the drawings and import and scale them.
2. In HERON, on the **Dashboard**, find the imported project under **Recent Projects** and click **Build in PlanSwift**.

3. In the **Build Bill in PlanSwift** window, check the **Template** chosen for each line and change any wrong match. Check **Measured as**, **Depth/Height**, **Width** and **Thickness** where the template needs them.
4. Click **Build in PlanSwift**. Each section becomes a take-off folder and each line a measurable item.

Measure, price and save:

1. Click **Measure** on a line in the **Build Bill in PlanSwift** window to send it to PlanSwift, then trace it on the drawing.
2. In HERON, go to **Quantity Take Off** and click **Reload**.
3. Click **Save / Export**, then **Load rates** to price from your library.
4. Click **Save to Cloud**. Your measurements land on the same lines that came from the client's file.

## What you get at the end

The client's bill, in their order and wording, with your measured quantities behind it, priced from your library and saved as a HERON project you can value and share like any other.

## Tips

- Click **Download the import template** if you want a workbook laid out the way the importer reads best.
- Upload the client's own bill, not one exported from the ADLM website. An ADLM export is refused.
- Building again is safe: lines already in the job are updated, never duplicated, and a line that already has a measurement keeps it. Use **Only lines not in the job** on a second run.
- If you only need to price the bill, not re-measure it, you can price it straight on ADLM Cloud after the import, as in Use case 4.

## Use case 9: Sharing a project with a client or collaborator

### Scenario

Your client wants to follow progress without learning the software. A structural engineer and a junior QS in another office need to work on the same project.

| Who it is for                                                                | Tools used |
|------------------------------------------------------------------------------|------------|
| Project leads and consultants who report to clients or work with other firms | ADLM Cloud |

There are two ways to share, and they are not the same. A public dashboard link is for a client with no ADLM account: they open a summary dashboard without signing in, and cannot edit anything. A collaborator signs in with their own account and the matching product licence, and sees the whole project at **View only** or **Full access**.

### Steps

Share a dashboard with your client:

1. Open the project and click **Dashboard** under **Views**.
2. Click **Share dashboard**.
3. Tick **Enable public link**, then click **Copy link** and send it to the client.

The client sees a read-only Project Dashboard: overall status, progress, the contract sum and its make-up, cost to date and forecast, interim certificate totals, and planned and actual spend. The button now reads **Shared · link on**.

Invite a collaborator:

1. Open the project and click **Collaborators**.
2. Under **Access level**, choose **View only** or **Full access**.
3. Fill in **Label (optional)**, **Restrict to emails (optional)** and **Max uses (0 = unlimited)** if you want to limit who can join.
4. Click **Generate code**.
5. Under **Active codes**, click **Copy**, or **Link & QR** for a join link and a QR code.

Join as a collaborator:

1. Open the link and sign in, or open the project list for that product, click **Add shared project**, type the **Share code** and click **Add project**.
2. The project appears marked **Shared with you**.

## What each person can see and do

|                                    | Owner | Full access                  | View only                    |
|------------------------------------|-------|------------------------------|------------------------------|
| Edit rates and progress            | Yes   | Yes                          | No                           |
| Download exports and reports       | Yes   | Yes                          | No                           |
| Invite people, delete, public link | Yes   | No                           | No                           |
| See rates and money                | Yes   | Only with a Rate Gen licence | Only with a Rate Gen licence |

A collaborator without Rate Gen sees **Rates hidden**. A **RateGen subscription is required to view rates**. Rates and money totals show as a dash, bill and budget exports are refused, and they cannot raise variations, rebuild the schedule or price services.

## What you get at the end

A client who can check progress from a phone at any time, and colleagues who can work on the same project at the level of access you chose.

## Tips

### IMPORTANT

Anyone who has the public link can open the dashboard. When you no longer want it shared, untick **Enable public link**. Ticking it again brings back the same link.

- A collaborator also needs a licence for the product the project came from, for example HERON for a HERON project. Without it they see **Subscription required**.
- Under **People with access**, change a person between **View only** and **Full access**, or click **Remove**. **Revoke** stops a code being used again.
- Give the client a **Project report** PDF each month alongside the link, so they have a dated record.

## Use case 10: Training a new estimator with the sample projects

### Scenario

A graduate joins your firm. You want them to understand what a finished, priced and valued ADLM project looks like before they touch a live job.

| Who it is for                                                       | Tools used                                                                                   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Practice leads, training managers and new users teaching themselves | ADLM Cloud learning samples, QUIV for Revit, ADLM HERON, SERVIQ for Revit MEP, ADLM Rate Gen |

### Steps

Week 1: read finished jobs:

1. Sign in at [adlmstudio.net](https://adlmstudio.net) and open a product's project list, for example your QUIV projects ( [/projects/revit](#) ) or HERON projects ( [/projects/planswift](#) ).
2. Above your own projects, find **Learning samples**. If the strip is folded away, click the button that reads **Show** and the number of samples.
3. Open a sample. The banner reads **Sample project · Read-only learning material**. Read **What to look at in this sample** first.
4. Go through every view in order: **Dashboard**, **Bill of Quantity**, **Budget**, **Valuation**, then **PM Dashboard**. On QUIV samples, open **3D Model** and click bill lines to see the elements they were measured from.
5. Try every **Export** to see what each workbook looks like.

Week 2: compare foundations and methods:

1. Open the QUIV and HERON samples for the same foundation type. One was measured from a model, the other from PDF drawings.
2. Compare their bills line by line. Note where a model gives a quantity directly and where a drawing needs a measurement.
3. Open the SERVIQ samples to see how a services bill is laid out.

Week 3: measure a small job:

1. Give the trainee a small, finished job your firm has already measured by hand.
2. Have them run the **Model Checker** (QUIV) or scale the pages (HERON), measure, and save to the cloud under a clearly marked training name.
3. Compare their bill with the sample and with your own figures. Discuss every line more than a few per cent out.

Week 4: price and value:

1. Have the trainee price their bill on ADLM Cloud from Rate Gen, check the **Budget**, and export a bill.
2. Have them record some progress, prepare a valuation in the daily valuation log, and print it with **Print valuation**.
3. Delete the training project when you are done, so it does not use a storage slot.

## What you get at the end

A new estimator who has seen every stage of an ADLM job on a realistic project, measured and priced one of their own, and can tell a good bill from a doubtful one.

## Tips

- Samples are read-only. A trainee can open every tab, filter and export, and nothing can be broken.
- Samples appear for the products your account holds an active licence for.
- **Hide samples** only hides them in that browser. Click **Show N samples** to bring them back.
- Go through the [Rate Gen guide](#) section on reading a build-up together. An estimator who can read a rate's **Rate Composition** can defend every price on a bill.

## Good habits

These habits save the most time across all the ADLM products.

### Name projects clearly

- Use the same pattern everywhere, for example client, building and stage: "Adeyemi Duplex, Lekki, Tender". The name you type in QUIV, HERON or SERVIQ is the name on the website and on every export.
- Add the discipline to the name when one job has several projects, for example "Services" for the SERVIQ project, so you can find and link them.
- Mark training and trial projects clearly, and delete them when you finish. Each project uses a storage slot.

### Check the model or the drawings first

- Run the QUIV **Model Checker** before you measure a Revit model. Duplicated elements and links placed twice are the usual reason a quantity comes out double.
- In HERON, scale every page and check the scale on a second dimension before you measure anything.
- In SERVIQ, look at the counts on the model page first. A model with far fewer fittings than the drawings show is telling you something before you measure.

### Save to the cloud often

- Save to the cloud at the end of each working session, not only at the end of the job. Each save from QUIV and HERON updates the project or adds a new version, and does not wipe the rates or purchase marks you added on the website.
- QUIV's ♦ **Run the whole takeoff** does not save to the cloud. Check the bill, then click **Save To Cloud** yourself.
- On ADLM Cloud, click **Save changes** before you leave a project. Rates, progress and settings are not kept until you do.

### Keep your rates in one place

- Keep one rate library for your firm, in Rate Gen, and price every job from it. Do not keep a second set of rates in a spreadsheet.
- Set your **Pricing location (State)** on your profile, so every product prices for the right zone.

- Build custom rates in one place only. If anyone in the firm uses Rate Gen desktop, build every custom rate there. A rate built on the website or on another PC can be removed from your account the next time the desktop app syncs.
- Turn **Follow RateGen changes** off on a bill before it goes out, so the prices you issued do not move.



Read the latest version of this guide at [adlmstudio.net/guides](https://adlmstudio.net/guides).  
Need help? [adlmstudio.net/support](https://adlmstudio.net/support)

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